COPENHAGEN BUSINESS SCHOOL

Master Thesis

MSc in Economics and Business Administration - Finance and Investments

Optimizing Portfolio Construction: Hierarchical Risk Parity (HRP) and Modified HRP Against Traditional Methods

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Abstract

Inspired by the seminal work of Marcos López De Prado (2016) and Marat Molyboga (2020), this thesis critically assesses the efficacy of Hierarchical Risk Parity (HRP) and Modified Hierarchical Risk Parity (MHRP) in contrast to a wide range of conventional portfolio creation methods. HRP constructs diversified portfolios using an algorithm that combines graph theory and machine learning techniques. The performance of HRP and MHRP is assessed against equally weighted, minimum-variance, maximum Sharpe ratio, inverse-variance, inverse-volatility, and equal risk contribution portfolios. Utilizing a comprehensive dataset from the S&P 500 index ranging from January 1993 to December 2023, we employ a quantitative methodology with a rolling walk-forward analysis to assess their performance and robustness in quasi-real market conditions. The study reveals that while HRP and MHRP exhibit promising features for risk-adjusted performance, they do not consistently outperform all the traditional methods examined based on returns or (adjusted) Sharpe ratio. They nonetheless produce portfolios with low annualized volatility, which is only surpassed by the minimum variance portfolio. This outcome may be appealing to certain investors. The findings indicate that the potential of HRP and MHRP, while important, may not entirely line with the optimistic outcomes indicated by López De Prado (2016) and Molyboga (2020). This study extends the discussions about advanced portfolio strategies by highlighting the necessity for extensive empirical evaluations to determine their real-world feasibility and performance.

Keywords: Asset Allocation, Portfolio Construction, Portfolio Optimization, Hierarchical Clustering, Machine Learning, Hierarchical Risk Parity, Modified Hierarchical Risk Parity

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1 Introduction

"Invest in seven ventures, yes, in eight; you do not know what disaster may come upon the land" - Ecclesiastes 11:2, Old Testament, The Bible.

This ancient wisdom emphasizes a timeless truth in the realm of finance and investment: market volatility and unpredictability. It elegantly hints at the notion of diversification, which is a fundamental strategy in risk management for investment portfolios. A major risk in investing is individual security's performance unpredictability. While investing in corporate securities can be rewarding, there is always a risk of major financial loss if a single investment fails. This risk arises from the variability of returns.

The process of optimally allocating capital among a preset group of assets or securities is referred to as optimal portfolio creation (Maillard et al., 2010). However, the initial theory work on investment practices has been laid out way earlier by John Burr Williams in his book "*The Theory of Investment Value*" (1938). The book introduced the dividend discount model to predict stock prices based on the present value of future dividends. This era was distinguished by difficulties in collecting financial information. The instability of markets during the Great Depression frequently led to associating investing with gambling for the wealthy. Despite this, professional investment managers such as Benjamin Graham have made significant progress by first gathering reliable information and then correctly assessing it before making any investments.

Portfolio creation has been intensively researched since the 1950s when Markowitz pioneered the mean-variance technique. His fundamental 1952 study, revised in 1959, laid the framework for modern portfolio theory, which has remained relevant for more than six decades (Hult et al., 2012; Markowitz, 1952, 1959). Markowitz's mean-variance concept is based on the notion that rational investors want to maximize expected return for a given level of risk. It offers a systematic method of portfolio diversification and optimization that has affected investment strategies and financial analysis for decades.

Reflecting its various advantages, modern portfolio theory remains centered on Markowitz's mean-variance framework for portfolio formation. The method's intuitiveness and simplicity are widely acknowledged benefits (Bruder & Roncalli, 2012). It is also thought to be

sophisticated and resilient because it can be expressed as a convex optimization problem with an analytical solution (López De Prado, 2016).

While Markowitz's discoveries from the 1950s remain essential to modern portfolio theory, Markowitz's mean-variance portfolio faced intense scrutiny following the global financial crisis. Despite optimizing many institutional portfolios using mean-variance criteria, they performed poorly, resulting in huge losses for investors worldwide. The field of portfolio construction is still under investigation as, despite its theoretical appeal, Markowitz's mean-variance portfolio can have shortcomings and produce erratic results in practice. These shortcomings stem from two fundamental issues with Markowitz's mean-variance approach to portfolio construction (Kolm et al., 2014). The issues are 1) the estimation of the expected returns and 2) the estimation of the positive-definite covariance matrix.

The first concern has resulted in a sizable percentage of investors embracing risk-focused portfolio-building strategies that ignore the expected returns over those based on forecasting expected returns (Jagannathan & Ma, 2003; Merton, 1980). Addressing the second concern, estimating the positive-definite covariance matrix presents its challenges. Risk-based allocation approaches do not rely on estimated returns but still require the inversion of a potentially problematic covariance matrix. When assets are highly correlated, this matrix becomes ill-conditioned, resulting in a high condition number. Inverting such an ill-conditioned matrix complicates intrinsic estimate problems. Such flaws eventually lead to numerically unstable portfolios. Indeed, studies have shown that the estimating errors may be substantial enough to cancel out any benefits of diversification (Ardia et al., 2015; Black & Litterman, 1992; Jorion, 1985; Kan & Zhou, 2007). Such a phenomenon is called Markowitz's curse. Therefore, numerous studies have focused on methods to improve the covariance matrix's stability and reduce its numerical instability (Kolm et al., 2014).

In 2016, López de Prado announced an alternative asset allocation strategy to overcome the inherent flaws of traditional optimization methods. The suggested hierarchical risk parity (HRP) approach combines hierarchical asset clustering with a heuristic risk-based allocation strategy to improve the allocation process. In essence, the HRP method prioritizes the allocation of risk over the allocation of capital. In this thesis, 'heuristic' refers to non-optimization-based portfolio design strategies such as equal weighting, inverse-variance, and inverse-volatility that prioritize risk considerations over exact mathematical rigor. López De

Prado's (2016) concept combines the capabilities of risk-based models and hierarchical clustering, allowing managers to generate a portfolio from a single covariance matrix. This feat is impossible with quadratic optimization-based strategies such as Markowitz's mean-variance framework (López De Prado, 2016). The proposed HRP strategy can be outlined in three primary steps:

- **Tree clustering**: Assets are grouped into a hierarchical structure based on similarities, allowing allocations to flow downstream through a tree graph, using a specific distance metric and the single linkage clustering, forming a hierarchical tree.
- **Quasi-Diagonalization**: The covariance matrix is reorganized so its largest values form a quasi-diagonal, aligning assets according to the hierarchical tree.
- **Recursive Bisection**: The reordered matrix is split through a process of recursive bisection, optimizing the allocation across the asset clusters.

Hence, López De Prado's (2016) HRP approach eliminates the necessity for expected returns and inverting the covariance matrix, addressing significant stability difficulties in portfolio formation. Although relying on simulated data, the author has demonstrated the potential for outperforming existing approaches through Monte Carlo simulations by delivering portfolios with reduced out-of-sample volatility and higher returns without depending on these frequently problematic assumptions (López De Prado, 2016).

Molyboga (2020) advanced one step further and proposed the modified hierarchical risk parity (MHRP) method, which aims to refine the original HRP strategy with strategic enhancements. A notable modification is the adoption of Ledoit-Wolf covariance shrinkage and the inverse-volatility approach instead of inverse-variance, which will be explored in greater detail in chapter 3.

1.1 Research Aim

Although López De Prado's (2016) results are impressive, they are based purely on simulated data. Additionally, using a limited set of performance indicators, the results were only compared to two different portfolio design approaches: 1) quadratic optimization, as depicted by the Critical Line Algorithm's (CLA) minimum-variance portfolio, and 2) traditional risk parity method, indicated by the inverse-variance portfolio. Additionally, the initial HRP study

employed only out-of-sample variance and the Sharpe ratio as performance metrics. Thus, it is compelling to evaluate the HRP approach utilizing historical data, compare it with a broader range of realistic portfolio strategies, and employ more comprehensive performance indicators. Therefore, this thesis seeks to answer:

How does the HRP method perform compared to traditional and heuristic allocation strategies?

The primary objective of this thesis is to conduct a comprehensive analysis of the performance of the HRP method when applied to real-world data. Additionally, it seeks to determine whether a more complex approach leads to more robust results in portfolio construction or if heuristic strategies are more effective. According to DeMiguel et al. (2009), simple strategies, such as the equally weighted (1/N) portfolio, may outperform more complex approaches, showing the potential superiority of fundamental methods. Additional sub-questions will tackle the below-mentioned points:

- How do the portfolios' performances vary across different optimization methods from both return and risk perspectives?
- What degree of concentration is observed in the portfolio weights?
- How do different linkage methods affect the results of the HRP and MHRP portfolio optimization?

Nonetheless, like every model, HRP comes with its drawbacks. Despite its innovative nature, the HRP approach has been criticized for its practical implementation. Raffinot (2017) critiques the arbitrary clustering of assets in HRP, which may disturb the portfolio's ideal structure. Pfitzinger and Katzke (2019) cast doubt on HRP's applicability for non-institutional investors, criticizing HRP's operational complexity and significant computational demands. Furthermore, a significant amount of supporting literature for HRP exists in working papers awaiting acceptance by key financial journals. This emphasizes the need for more scholarly evaluation and validation of the model's efficacy and implementation. The model's shortcomings will be presented in greater detail in section 2.2.

This thesis aims to contribute to the scientific field by empirically testing the HRP model and its modifications using real-life data. It aims to validate the theoretical benefits of HRP while also investigating its effectiveness and limits in real-world market situations. This study

examines both the basic HRP model and its modified versions and seeks to offer new insights into their efficiency, adaptability, and potential challenges. Thus, the overall goal of the paper is to provide a solid foundation for future portfolio management developments and applications.

1.2 Data, Scope, and Limitations

Data

Most of the financial data for this has been retrieved using Datastream, Refinitiv Eikon, Yahoo Finance, and Bloomberg Terminal databases. The risk-free rate has been retrieved from the Federal Reserve Economic (FRED) database. These sources are respected for their accuracy and objectivity, providing a solid foundation for the analysis. The dataset used for the analysis ranges from January 3rd, 1993, to December 29th, 2023. This timeframe's data includes several major financial events, providing a thorough analytical backdrop. This timeframe includes the dot-com bubble's collapse and depicts the market's recovery from speculative highs. Moreover, it includes the 2007–2008 Global Financial Crisis, a critical period for understanding systemic risks and assessing portfolio resilience. Finally, the extraordinary impact of the COVID-19 epidemic, leading to a significant economic slowdown, serves as a prime example of market reactions to unexpected worldwide emergencies. The subsequent economic recovery demonstrates how markets react to government measures after these disasters. By examining these numerous market situations, the thesis ensures that the financial models and strategies developed are robust and well-tested across economic cycles. An in-depth explanation of data processing can be found in chapter 4.

Academic Limitations

Given the constant improvements that characterize the field of portfolio design, particularly in the realm of HRP, it is recognized that the scope of this research may include only some available information. The 1st of March 2024 has been selected as the cutoff date for the literature review. The authors have carefully emphasized peer-reviewed articles as the foundation of their thesis. However, the inclusion of some non-peer-reviewed papers was deemed to be significant. As a result, to verify the study's integrity and accuracy, all non-academic literature was carefully cross-checked, to minimize any potential errors.

Practical Constraints and Assumptions

This thesis assumes optimal divisibility and liquidity for all assets, meaning they can be divided into the required smaller units and traded instantly at market-adjusted closing prices without delay. Utilizing adjusted prices ensures accurate stock tracking by removing distortions caused by corporate actions like dividends and stock splits. This approach provides a clearer view of a stock's value evolution, enabling decisions based on the company's actual economic performance rather than mere price fluctuations. It cuts through the noise, resulting in a more transparent analysis of stock history. Furthermore, the thesis presupposes the constant availability of a buyer and a seller at the closing price for each stock, guaranteeing seamless transactions. This analysis does not account for liquidity variations on S&P 500 rebalancing days, where trading volumes spike due to portfolio adjustments and potential arbitrage, temporarily impacting liquidity.

This thesis recognizes several practical limits and fundamental assumptions influencing its analytical approach. Firstly, the thesis does not include the tax implications of different countries. These can considerably impact on the after-tax returns of different portfolios due to differences in interest income and capital gains tax rates. Secondly, transaction costs, like brokerage fees and bid-ask spreads, while important to investment returns, are only included as a form of simplified benchmark version. These expenses can reduce total returns and influence asset allocation, particularly in scenarios involving high-frequency trading or high turnover. Finally, the analysis is based on a long-only investing strategy. Thus, it does not consider strategies that include short selling or a broader range of investment techniques, such as those utilizing derivatives and leverage. Furthermore, in practical scenarios, many portfolio managers face shorting constraints (Maillard et al., 2010). The study might overlook the intricacies by concentrating on long-only strategies to maintain simplicity and ensure comparability with other risk-based approaches. This could result in missing out on the potential benefits of employing more intricate investment methods to reduce downside risks or capitalize on market inefficiencies.

1.3 Thesis Structure

The thesis is divided into distinct sections and is outlined as follows:

Part 1: Literature review

This section introduces the notion of clustering in financial analysis. It then discusses the fundamental literature behind the HRP technique and its original flaws. Subsequently, the discussion shifts to explore modified variations of the HRP approach, presenting a structured exploration of its evolution and enhancements.

Part 2: Financial background

This section outlines the thesis' theoretical basis, starting with the necessary basics of statistics. It then explores Modern Portfolio Theory and various risk-based allocation strategies and discusses the challenges of estimating and inverting covariance matrices. Finally, it introduces clustering theory as a foundation for understanding López de Prado's HRP model, setting the stage for a detailed examination of this innovative approach.

Part 3: Data

The Data section of the thesis outlines the selection of the S&P 500 index, describes comprehensive data sourcing methods, and details a rigorous three-step data cleaning process to ensure the accuracy and integrity of the dataset used in the analysis.

Part 4: Methodology

This section describes the research methodology and the statistical measures utilized in the analysis. It describes the modifications made to the HRP model and the walk-forward approach used to assess the performance of portfolio strategies.

Part 5: Results

The empirical results of the thesis are presented in this section, which analyzes how well the HRP and MHRP models perform compared to more conventional portfolio techniques. The distribution and performance[1] metrics are used by groups to visualize the results to facilitate understanding and comparison.

[1] In this thesis, "performance" is used as a general term to indicate both return and risk.

Part 6: Discussion

In this part, the implications of the findings are covered, showing the strengths and limitations of the HRP model. It also suggests future research directions on the HRP method and related areas in hierarchical clustering to enhance the empirical validity of the proposed method.

Part 7: Conclusion

The thesis concludes by summarizing the study's contributions to the financial analysis field, mainly in portfolio management. The outcomes of diverse portfolios are compared to evaluate their efficacy and appropriateness for different purposes. The conclusion highlights the importance of continuing research in risk-based portfolio management.

2 Literature Review

This chapter of the thesis will lay out the theoretical foundation for the research. The thesis draws inspiration from papers on asset allocation, clustering, hierarchical risk parity, and its modifications. The literature review aims to present a comprehensive overview of existing methodologies and their potential benefits and drawbacks.

2.1 Clustering in Finance

Clustering has been the topic of academic interest for decades, with foundational publications from the 1960s laying the groundwork for its broad application in financial analysis. Among these pioneering efforts, Ward (1963) introduced an approach for hierarchically organizing data to optimize an objective function on large-scale datasets. This method cyclically combines sets based on the greatest similarity and particular features, thereby lowering loss as determined by an objective function. It establishes a thorough hierarchical framework and provides accurate quantitative assessments at every grouping level. Because of its adaptability, the method improves analysis and administration in various intricate data fields. This adaptability, however, raises concerns about the possibility of overfitting. It also emphasizes the difficulty of understanding clusters in a way that is consistent with economic reasoning rather than statistical convenience alone.

Subsequently, King (1966) built upon these ideas and further explored the heuristic nature of financial clustering. The author highlighted early clustering approaches' exploratory and iterative nature by recognizing the lack of specific measurements and endpoints for group clustering. Additionally, the study emphasized the significance of correlation matrix properties in determining the number of clusters, laying the groundwork for later, more sophisticated methods of financial clustering. King's (1966) emphasis on the subjective nature of financial clustering approaches reinforces the notion that cluster analysis requires some level of subjective judgment. This emphasizes the need to use thorough validation approaches to guarantee that the clusters produced are real, not just products of the analytical method used. King's work has significantly influenced the development of hierarchical clustering in financial analysis. Nonetheless, the concept of clustering in the stock market was observed even earlier. Roberts (1959) discussed how similar results tend to group, shedding light on the early

recognition of clustering possibilities in financial data. While alerting readers of the theoretical limitations of cluster-based forecasting, this insight offers first-hand knowledge of the patterns and possible predictability inside financial markets. Roberts' (1959) analysis adds another viewpoint to understanding the financial dataset clustering phenomenon.

Even though a solid theoretical framework for cluster analysis was established in the 1970s (Anderberg, 1973), it was not until much later that clustering became a standard tool in financial research. This shift was marked by the prominent study of Mantegna (1999) when financial markets became increasingly complex and globalized. Mantegna (1999) investigated the complex relationship between stocks while researching the hierarchical structure of financial markets. Using data from the Dow Jones Industrial Average and the Standard & Poor's 500 index from July 1989 to October 1995, the study discovered a valid economic classification of equities based on daily logarithmic price disparities. The main takeaway was that using correlation for clustering may highlight the economic organization of stock portfolios, providing a new viewpoint on the interconnection of financial markets. In contrast, Laloux et al. (1999) employed random matrix theory to evaluate S&P 500 empirical correlation matrices spanning 406 assets from 1991 to 1996. Their findings highlighted the challenges in differentiating actual financial correlations from the noise, indicating that a considerable portion of the patterns in these matrices could be noise-based. This paper doubts the efficacy of such matrices for financial analysis, underlining the need for better ways to differentiate true financial information from noise. While Mantegna (1999) and Laloux et al.'s (1999) research contributes significantly to understanding the financial markets, their disparate findings underscore the difficulty of discerning actual patterns from noise. This dilemma emphasizes the significance of using rigorous procedures and interpreting data cautiously, mainly as financial data grows in volume and complexity.

As the understanding of clustering has deepened over the years, it is critical to grasp its practical application in portfolio optimization, which substantially impacts capital allocation techniques. Clustering, a statistical partitioning strategy, combines similar assets based on their connection, resulting in a more robust and appealing allocation by encouraging competitive capital distribution across asset groupings (Maimon & Rokach, 2010).

Tola et al. (2008), using data from 1071 NYSE-listed stocks between 1988 and 1998, illustrated how clustering based on correlation may be used as a powerful filtering tool by condensing the correlation matrix into a more straightforward structure with fewer unique elements,

facilitating portfolio optimization. The authors focused on the average linkage clustering, demonstrating how it might reduce a portfolio's effective size and riskiness by addressing the statistical shortcomings of generating correlation matrices. Their comparison analysis revealed that clustering-based approaches outperform Random Matrix Theory (RMT) in filtering performance. However, the study acknowledges that while clustering methods, such as average linkage, improve portfolio structure, they also introduce additional complexity and may increase portfolio riskiness, especially in varying market conditions, which requires careful consideration in practical applications.

Tumminello et al. (2010) conducted an in-depth study on the hierarchical clustering of 100 top-capitalized NYSE-listed equities from 2001 to 2003, employing the Average Linkage Clustering Algorithm (ALCA) on correlation data. This study makes an essential contribution to clustering in finance by demonstrating the potential of hierarchical clustering to identify substantial economic sectors, extending its application in asset allocation and risk management beyond what RMT provides. The study demonstrates how hierarchical structures and correlation-based networks can reveal complex sectorial linkages between equities, allowing for a better understanding of market behavior and more educated investing decisions.

Recent studies, such as by Elkamhi et al. (2023), introduced an innovative approach called cluster-enhanced inverse-volatility. The paper aims to outperform classic inverse-volatility portfolios by comparing Sharpe ratios. This strategy uses machine learning clustering algorithms to combine assets with similar characteristics. Such an approach allows for portfolio formation that meets the theoretical optimality of mean-variance portfolios. The authors analyzed seven industry portfolios with asset counts ranging from 10 to 76, including data from 1926 to 2023. The data included multi-asset portfolios, equity indexes such as the S&P 500, corporate and government bond indices, and commodity indices, providing a comprehensive view for analyzing their cluster-enhanced inverse-volatility strategy. Elkamhi et al.'s (2023) study shows that this strategy routinely outperforms traditional portfolios, particularly in big asset pools. This innovative strategy effectively combines traditional portfolio theory with new computational tools, improving portfolio diversification and performance.

An even further step forward in clustering was presented by incorporating hierarchical momentum (Cirulli et al., 2023). This research combines hierarchical clustering with stock price momentum to develop portfolios aimed at maximizing returns while concurrently

managing risk and preserving stability across different economic environments. Their study analyzed the MSCI All Country World Index from 1997 to 2022. Their strategy outperformed benchmarks and reduced common risks associated with momentum investing, such as portfolio drawdowns, without complex covariance matrix inversion. Their work highlights the advanced application of clustering techniques to mitigate drawbacks in momentum-based strategies, offering a promising direction for modern financial portfolio management.

As clustering approaches get more advanced, they require significant computational power and add complexity, possibly without evident benefits. This evolution raises questions about accessibility for all investors and potential overfitting, particularly in less liquid markets. The revolutionary combination of momentum techniques and hierarchical clustering provides a new risk and return balancing approach. However, the new approaches come with obstacles, such as frequent rebalancing and continuing data analysis. This situation emphasizes the discussion about the real-world impact of complex portfolio management approaches, emphasizing the importance of striking a balance between innovation and practical consequences.

Previously mentioned papers suggested that innovative portfolio strategies could outperform traditional ones, with promising Sharpe ratios and variance levels. However, because much of this research is in the pre-peer-review stage, its credibility remains unknown. Academic working papers, typically shared in early versions, have not been subjected to top publications' rigorous peer review processes. This issue suggests that these creative approaches may lead to incorrect, false positive conclusions. As a result, a cautious approach is recommended, emphasizing the importance of careful evaluation and confirmation by the financial academic community.

In conclusion, the growth of clustering approaches mirrors a broader narrative of financial analysis innovation and adaptation. This establishes a foundation for increasingly complex and effective asset management strategies that continue to evolve in response to an ever-changing market scenario.

2.2 Initial HRP Suggestion and its Shortcomings

Hierarchical clustering divides data into sorted binary groups, resulting in a tree-like hierarchical structure. One fundamental advantage of this approach is that it recognizes that

complex systems, such as financial markets, frequently exhibit hierarchical configurations that greatly influence complicated behaviors (Simon, 1991). Hierarchical clustering contributes to formulating a more detailed and inclusive system model by quantitatively characterizing these hierarchical patterns (Raffinot, 2017).

In the 2016 paper about portfolio construction, López De Prado proposed hierarchical risk parity (HRP), a distinctive approach to asset allocation that incorporates machine learning and graph theory to deduce a hierarchical link among assets. Nobel Prize winner Herbert Simon proposed (1962) that complex systems naturally organize into hierarchies with nested substructures. It points to the significance of how elements interact within hierarchies in influencing the behavior of complex systems. This highlights the need for a quantitative framework to describe and analyze these hierarchical structures to effectively model complex systems (P. W. Anderson, 1972).

López De Prado's method uses the deduced hierarchical arrangement and inverse-variance allocation to distribute weight across clusters of correlated asset returns. Furthermore, correlation matrices often lack the concept of hierarchy (Raffinot, 2017). Moreover, one notable feature of the HRP technique is that it avoids the need to invert the covariance matrix, a typical requirement in many classic risk-based allocation methods that frequently leads to considerable estimation errors. HRP, unlike typical quadratic optimizers, can compute portfolios on ill-degenerated or singular covariance matrices (López De Prado, 2016). Empirical evidence from Monte Carlo simulations and real-world asset classes shows that HRP can produce portfolios with lower out-of-sample variance and more diversity than traditional techniques. This makes it particularly appealing to prominent asset managers and funds that use leverage, where consistent and efficient risk allocation is critical. Furthermore, HRP's adaptability allows for the insertion of multiple constraints and the customization of the tree structure, allowing for customized investment methods and objectives (López De Prado, 2016). While López De Prado's HRP model allows for extensive customization and adaptation, this adaptability may result in overfitting in some market conditions. His study utilized a simulated dataset, where returns were Gaussian simulated with a mean of 0 and a standard deviation of 10%. This controlled setup allowed for a thorough evaluation of the HRP method's effectiveness and stability in a theoretical context but lacked "real-world" implications. Given the nature of simulated data, it does not represent real-life scenarios in which unforeseen events, so-called "black swans", might significantly impact on asset correlations and market

dynamics. This limitation emphasizes the significance of testing portfolio strategies in real-world market conditions to provide resilience and adaptability to unexpected developments in the stock market. It emphasizes the need for a more comprehensive approach to assessing the efficacy of financial models such as HRP. Future research should concentrate on quantifying the influence of this customization on the model's predictive capability and stability, especially in volatile markets or with non-normal asset distributions using real asset data.

For each object to be classified, hierarchical clustering requires numerical feature measurements. A feature vector defines each object, contributing to a feature matrix in which rows and columns correspond to objects and their features, respectively. The analysis of this matrix computes the distances between the attributes of objects to construct clusters - related objects are grouped, while dissimilar ones are separated (Murtagh & Contreras, 2017). The correlation matrix is frequently used to underpin portfolio design analysis, with assets serving as objects and pairwise correlations serving as features, assisting in the definition of linkages and market interactions. The procedure is divided into two stages. The first includes arranging the data to reflect the relative distances between objects, generally accomplished by computing distances between feature vectors. Following that, it entails the actual clustering and hierarchy building. Clusters are formed based on calculated distances and a linking criterion that describes how to calculate distances between clusters. This methodical technique enables systematic and detailed asset categorization, which aids in comprehensive portfolio analysis and strategy building (Murtagh & Contreras, 2017). Finally, hierarchical clustering is classified into two types called agglomerative clustering and divisive clustering. Agglomerative clustering is considered a bottom-up approach, while divisive clustering is a top-down strategy (A. K. Jain et al., 1999). These approaches will be explained and visualized in section 3.5.

López De Prado's (2016) study introduced the HRP method, which represents an essential advancement in portfolio management. Despite its significance, the methodology faced scrutiny, particularly regarding its clustering decisions. The author's method uses clustering to reorganize assets, but it ignores the structure of the dendrogram during the bisection step, potentially dividing closely related assets.

This issue introduces a degree of arbitrariness because the initial hierarchical structure is not considered during bisection in the third step. Typically, the HRP approach uses clustering primarily to reorder assets without considering the structure of the dendrogram during bisection, which may result in the separation of highly connected assets. Following scholars

have adopted alternate techniques to address these restrictions. Notably, Raffinot (2017, 2018) pioneered a technique for dividing allocations based on the structure of the tree, effectively capturing the correlation clusters reflected by the form of the dendrogram. By analyzing data from 1995 to 2016, a multi-asset dataset, and 357 stocks from the S&P 500, Raffinot sought to optimize asset allocation. This comprehensive data utilization aimed to improve the HRP's efficacy in crafting diversified portfolios, showcasing the method's adaptability and potential in modern portfolio management. The results showed that portfolios based on hierarchical clustering were more diverse and performed statistically better risk-adjusted than classic portfolio optimization strategies. Other papers, such as Lohre et al. (2020), Huang (2020), and Reis et al. (2023), have also used and developed this technique for tree-based sectioning, demonstrating a progressive progression in the application of HRP.

Pfitzinger and Katzke's (2019) later research provides important improvements to solve the initial HRP method's shortcomings. The authors improve HRP by altering allocation to reflect asset similarities within clusters, thus avoiding the random partition of closely linked assets. This change emphasizes the dendrogram's significance in bisection decisions, protecting cluster boundaries, and leveraging hierarchical clustering's profound insights. Their research uses daily total return index data from six equities universes from 2008 to 2017, including markets such as the JSE All Share Index and the S&P 500, involving 255 to 707 stocks. Authors propose weight constraints so that portfolios align with certain criteria, introducing formerly unobtainable levels of control and personalization. Their adoption of genetic optimization for clustering differs from traditional methods, showing improved efficiency and effectiveness. The authors use a simple loss function $\mathcal{L}$ with bigger weights assigned to correlation coefficients further away from the matrix diagonal, which can be used to quantify the extent of diagonalization. The genetic permutation algorithm described in *Algorithmic Combinators* (Even, 1973) is used in this paper to reduce $\mathcal{L}$. The paper also suggests another approach, replacing the naive allocation hierarchy with an algorithm that considers asset similarity at cluster edges, allowing for more informed and nuanced allocation decisions. Finally, during the recursive bisection step, the authors impose box and group constraints, ensuring portfolios comply with specified constraints while remaining close to optimal HRP allocations. This research supports Raffinot's (2017) findings, indicating that updating the initially proposed HRP method can yield superior results. Pfitzinger and Katzke (2019) demonstrate that, while their proposed method does not achieve the mean-variance portfolio's low volatility, it delivers larger returns, lower turnover, improved diversification, and the capacity to manage singular

covariance matrices. In contrast to advances in clustering and hierarchical risk parity, DeMiguel et al. (2009) argue that even with modern portfolio management methodologies, simple strategies such as the 1/N portfolio frequently work just as well, if not better. The authors emphasize that complicated models may not provide better results unless they are based on a large amount of historical data. These findings show that one should be cautious about immediately adopting complex models without further evidence of their efficacy across various market scenarios.

To sum up, López De Prado's (2016) HRP proposal significantly advanced portfolio management by integrating hierarchical clustering into asset allocation. Despite criticism, particularly regarding asset grouping, its innovative method of avoiding covariance matrix inversion has inspired the industry to adopt and further develop the HRP approach. Raffinot's (2017, 2018) and Pfitzinger and Katzke's (2019) further revisions addressed past issues, boosting efficiency and precision. These developments illustrate HRP's progress and promise in sophisticated asset management, highlighting the importance of continual research and adaptation in the ever-changing financial market.

2.3 Current Advancements and Applications in HRP

López De Prado (2016) used simulated data to depict HRP benefits. Following this foundational research, various authors have examined HRP's effectiveness using real-world data. Lau et al. (2017), for instance, tested HRP across diverse cross-asset platforms, including alternative risk premia and standard market indices. In terms of risk-adjusted returns, their data show that HRP outperforms most conventional risk-based portfolio strategies. Even while the minimum-variance portfolio outperformed the HRP portfolio regarding risk-adjusted returns in a specific risk premia index, the HRP portfolio provided more diversification. Furthermore, their out-of-sample testing revealed that HRP has lower volatility and drawdowns than the minimum-variance portfolio (Lau et al., 2017).

However, compared to Lau et al. (2017) paper, the most recent study of HRP in the Brazilian market (Reis et al., 2023) produced ambiguous results regarding its superiority. This disparity in findings highlights the need for a more nuanced understanding of HRP's success across markets. Using monthly returns from January 2000 to June 2022 for 25 assets featured on the IBrX-100 index, the authors compared HRP against six well-known portfolio strategies and

the Bovespa index over various out-of-sample periods and window sizes. The conflicting results from Reis et al. (2023) highlight the complexity of deploying HRP in various market conditions, showing that its efficacy may depend on unique market dynamics and the model's adaptation to those dynamics. Their findings depict that while HRP competes with other strategies in terms of the Sharpe ratio, its performance across other comparative measures needs to be clarified. Such observation encourages a critical assessment of HRP's function in portfolio diversification and risk management, questioning the concept that one strategy fits all and calling for a tailored approach based on portfolio size and market circumstances. According to the research, the efficiency of HRP can vary depending on factors such as transaction costs and diversity requirements. Despite its promising results, HRP is not always the best option, emphasizing the significance of continuous innovation and empirical testing in portfolio management methods to adapt to the changing financial landscape.

On the other hand, Burggraf (2021), for example, used HRP on a cryptocurrency portfolio and discovered that it effectively managed volatility and tail risk better than other risk-based allocation strategies such as inverse-volatility, minimum-variance, and maximum diversification. Burggraf (2021) added shrinkage to the covariance matrix to further improve the method's robustness. The analysis was conducted on data for 61 cryptocurrencies from January 1st, 2015, to November 1st, 2019, with an average daily mean return of 1.25% and volatility of 27.05% across all studied cryptocurrencies. The dataset highlights the volatile environment where HRP proved effective, but it also raises questions regarding the necessity for model tweaks and refinements in diverse asset situations. Moreover, the analysis of HRP in the Iranian stock market (Nourahmadi & Sadeqi, 2022) reveals that HRP is a powerful tool for constructing portfolios that adeptly balance risk and return, outperforming various conventional approaches. Nonetheless, the paper notes that research on this subject typically encounters limits due to the complex and temporal dynamics of financial markets. As a result, while the findings seem positive, they must be interpreted considering this underlying complexity.

In parallel to HRP's evolution, Cirulli et al. (2023), in their paper on portfolio construction with hierarchical momentum, propose a novel portfolio creation approach that combines hierarchical clustering and stock price momentum. Momentum strategies are investment methods that seek to capitalize on the constant performance of assets, often known as momentum, to generate higher-than-average returns. The empirical analysis uses of a

multidimensional dataset of equities from the MSCI All Country World Index (ACWI) from June 1997 to August 2022. These equities reflect a broad asset universe, including large- and mid-cap stocks from 23 developed and 24 emerging markets. Cirulli et al.'s (2023) introduction of hierarchical momentum as a complement to HRP demonstrates that portfolio management is developing, with new techniques potentially leading to superior results. The research excludes future-oriented information and only includes assets having at least five years of historical returns and continuous trade for up to two weeks during the last five years, resulting in a comprehensive dataset free of survivorship bias. The authors use hierarchical clustering for dimensionality reduction and sparse diversification, combining it with momentum strategies to construct robust portfolios. This unique blend enhances returns and ensures risk control and portfolio stability. The paper's findings depict the strategy's superiority over conventional benchmarks, including Maximum Momentum, Time-Series Momentum (Raffinot, 2017, 2018), Mean-Variance models, and model-free strategies like equally weighted and capitalization-weighted portfolios. The Hierarchical Momentum approach by Cirulli et al. (2023) outperforms these benchmarks in key performance metrics, offering a compelling alternative that combines hierarchical clustering's methodical asset grouping with the dynamic potential of momentum investing, setting a new standard in portfolio management. This integration of techniques presents a promising option for future research, emphasizing the possibility of beneficial interactions between various investing tactics. Proper data selection with a particular focus on a "non-survivorship-biased dataset of eligible assets" (Cirulli et al., 2023) ensures the high quality and reliability of the paper's findings.

Sen and Dutta (2023) conducted research on the Indian stock market, which is comparatively underrepresented in HRP studies. Their research focused on examining the performance of Eigen portfolios against HRP within the NIFTY 50 context. Their study not only broadens the empirical base of HRP research but also provides valuable insights into its adaptability and success in emerging economies. Sen and Dutta (2023) used historical stock prices from January 1st, 2016, to December 31st, 2020, to build portfolios using two approaches. The performance of these portfolios was then analyzed using test data ranging from January 1st, 2021, to November 1st, 2021. Their backtesting results show that the HRP portfolio regularly beat its eigen counterpart across multiple sectors on training and test data sets. The research entailed calculating daily returns and annual volatility to predict stock behavior. Principal Component Analysis (PCA) was used to identify the principal components and design the best portfolio for Eigen portfolios, prioritizing the maximum Sharpe ratio. The HRP technique, on

the other hand, unfolded in three stages: tree clustering to organize stocks hierarchically, quasi-diagonalization to reorganize the covariance matrix, and recursive bisection for exact weight distribution. According to the study, HRP portfolios often beat Eigen in terms of Sharpe ratios on both training and test data, indicating improved pattern detection and increased diversity. This highlights HRP's potential for a more effective risk-return balance in portfolio structuring. Nonetheless, the optimistic results must be weighed against the financial market's intrinsic complexity and dynamism, highlighting the importance of contextualizing HRP methods (Sen & Dutta, 2023).

In a related advancement in portfolio management, Elkamhi et al. (2023) provide a novel cluster-enhanced inverse-volatility model. This strategy combines machine learning clustering and inverse-volatility strategies to improve portfolio performance. The methodology is applied across seven diverse datasets, some even dating back to 1926, highlighting its effectiveness across different asset classes and market conditions. The authors apply clustering to segregate assets by their correlations. They then utilize inverse-volatility strategies within these clusters. This approach balances asset weights and minimizes estimation errors. Their findings show the cluster-enhanced inverse-volatility method equals conventional portfolios in performance and shows adaptability across varied datasets. This adaptability imroves portfolio diversification and diminishes the potential for estimation inaccuracies. Elkamhi et al. (2023) discovered that the benefit of clustering is particularly visible in larger portfolios, with the model's performance boost being more pronounced in portfolios with 30 or more assets. The cluster-enhanced inverse-volatility model's capacity to provide substantial gains in larger portfolios supports its approach and encourages additional research into its scalability and optimization for various market sizes. This contrasts with minor improvements reported in smaller datasets, such as the 10-industry dataset, demonstrating the scalability and efficacy of the cluster-enhanced inverse-volatility model in managing varied asset scenarios. However, the efficacy of the approach may be determined by the clustering technique used and the assumption of constant asset correlations and Sharpe ratios over time. Although not a direct extension of HRP, this methodology marks a substantial advancement in the application of computational tools to improve asset allocation and portfolio optimization.

Generally, the HRP methodology has been applied on various fronts, including individual stocks (Raffinot, 2017, 2018), the Chinese (Huang, 2020) and Canadian (Jothimani & Bener,

2019) stock markets, commodities (Molyboga, 2020), and multi-asset multi-factor investment landscapes (Lohre et al., 2020).

In conclusion, numerous papers have been inspired by López De Prado's HRP method, indicating its ability to outperform conventional approaches across diverse markets, as evidenced by Lau et al. (2017). However, its efficacy varies, with some studies pointing to limitations and the need for more adaptive techniques. While research such as Burggraf (2021) and Sen and Dutta (2023) indicate promising outcomes in cryptocurrencies and the Indian stock market, the wider use of HRP necessitates careful evaluation of market conditions and algorithmic decisions. This highlights an important feature of portfolio management research: striking a balance between innovative approaches and the reliability of empirical validation. The evolution of portfolio strategies, as indicated by the numerous studies cited, not only broadens the scope of financial research but also emphasizes the importance of adaptation, empirical investigation, and the incorporation of modern technology into investment management. As previously mentioned, while innovative portfolio strategies promise better returns, more diverse portfolios, or enhanced Sharpe ratios, their true effectiveness has yet to be fully established due to a lack of rigorous peer review. Therefore, a careful attitude is encouraged when accepting their conclusions without further support from the scholarly community. Overall, HRP has the potential to improve risk-adjusted returns and diversification, but it requires greater investigation and sophisticated application.

2.4 Theoretical Background for Modified HRP and Beyond

Initially, it should be noted that there is no commonly accepted definition of Modified HRP. MHRP is a technique that improves the standard HRP model by including enhancements aimed at optimizing the risk-return equilibrium, diversifying the model, and increasing its responsiveness to market volatility. This section will examine some of the most acknowledged and implemented MHRP approaches in the field.

Many versions of the modified HRP (MHRP) method have been discussed in academic literature, but this paper will focus mostly on the MHRP method presented by Molyboga (2020). Molyboga's research provides an improved version of the MHRP approach which incorporates three key enhancements. This constant evolution of the MHRP model, as demonstrated by Molyboga's work, shows an ongoing attempt to balance risk and return more

effectively. However, the necessity for such changes may also indicate inherent difficulties in the original HRP model's assumptions regarding market behavior. Therefore, the author uses 1) Ledoit-Wolf shrinkage to replace the usual covariance matrix with an exponentially weighted one, 2) equal volatility instead of inverse-variance for greater diversification inside and across asset clusters, and 3) volatility targeting for improved temporal diversification. Molyboga then conducted a large Monte Carlo simulation on the portfolios of Barclays commodities trading advisors (CTA) from 2002 to 2016. According to the findings of his study, the MHRP technique improves out-of-sample Sharpe ratios by an average of 50% while decreasing downside risk. This study confirms the potential of MHRP to deliver a more efficient risk-return balance in portfolio construction. However, its effectiveness may vary depending on market dynamics and should be contextualized in the context of financial markets' complex, developing nature. Molyboga's MHRP technique significantly improves portfolio measures. Nonetheless, the broader application and durability of such a specific MHRP approach under different market situations merit additional investigation. Operationalizing MHRP presents obstacles, including computational demands and the requirement for adaptive rebalancing procedures to preserve portfolio efficacy. Despite these factors, Molyboga's improvements emphasize the significance of building adaptable and robust portfolio management solutions amidst market uncertainty.

Analyzing Ibovespa, the Brazilian stock market, Duarte and De Castro (2020) discovered that portfolio strategies based on partitional clustering outperformed the minimum-variance portfolio and the standard HRP in terms of average return and Sharpe ratio. Partitional clustering divides objects into discrete clusters based on similarities, optimizing for variables such as return or volatility. It varies from HRP, which employs asset covariances to create portfolios with a hierarchical structure to reduce risk. The analysis is carried out using data from the Brazilian Stock Exchange (B3) from December 2005 to April 2020. However, these strategies showed a slight increase in volatility, turnover, and drawdown. In contrast, Leon-Nieto et al. (2017) observed that when utilizing the Omega ratio as a performance indicator, a specific hierarchical clustering methodology known as Ward's method outperformed all other clustering approaches in terms of efficacy. The Omega ratio is a relative estimate of the probability of achieving a specific return, such as a minimum aim. Their research, centered on Russell 1000 Index components over a brief period, shows that machine learning technologies can greatly improve portfolio diversification and maximize the risk-return trade-off. Particularly significant is their discovery that portfolios rebalanced at shorter intervals —

approximately one hour — produced better results, implying that rebalancing frequency is critical in maximizing financial outcomes. However, a rebalancing strategy often introduces challenges due to increased monetary costs. This insight into the rebalancing technique complements Duarte and De Castro's (2020) partitional clustering approach, providing a broader perspective on how advanced algorithms in portfolio management can be used to efficiently negotiate complicated market dynamics.

Moving forward, Jothimani and Bener (2019) conducted a study on the Canadian stock market where they replaced the standard correlation used in HRP with the Gerber statistic. Employing Gerber statistic allows for a more resilient measure of predicting co-movements in covariance matrices. This innovative use of the Gerber statistic, known for its robustness in handling outliers, signifies a broader reevaluation of fundamental HRP principles considering new market data and behaviors. The Gerber statistic enhances the HRP model by providing a more stable estimation of asset price co-movements, addressing the volatility and unpredictability inherent in financial markets. The authors assessed the modified hierarchical risk parity model, termed HRP-GS, with stocks from the TSX index over a decade (2007–2016). Their findings reveal that HRP-GS not only surpasses the traditional HRP in terms of the risk-return profile and adjusted Sharpe ratio but also demonstrates the potential of integrating robust statistical measures to improve portfolio optimization techniques. This approach underscores a significant shift towards deploying advanced statistical tools to better manage estimation errors and portfolio allocation stability, suggesting a pivotal advancement in risk management strategies.

Adaptive Seriational Risk Parity (ASRP) is introduced as a modification to the standard HRP method, with 57 variations focusing on enhancing the quasi-diagonalization step (Schwendner et al., 2021). The analysis employs a complete multi-asset universe, encompassing equity indexes, sovereign bonds, and commodities futures from May 2000 to June 2020, using daily data. The research is based on data from 17 futures markets. Monthly rebalancing is performed within a rolling historical one-year timeframe, with portfolio leverage modified to attain a 5% annual volatility target. Compared to other studies, Schwendner et al. (2021) take into account transaction costs, thereby enhancing its real-world applicability and relevance. The approach tackles HRP's potential issue of segregating associated assets explicitly by introducing both seriation-based and tree-based (static and adaptive) techniques. Most static tree-based methods outperformed standard HRP in terms of risk-adjusted returns; however, adaptive tree-based and seriation-based techniques produced mixed outcomes. The success of these

improvements is heavily reliant on the quality of the tree cluster utilized for both the diagonalization and sectioning processes. A critical evaluation reveals that while the ASRP framework represents a substantial development in asset allocation strategies, its reliance on specific data sets and periods may restrict the generalizability of its conclusions. Further research is needed to confirm the consistency of these strategies across different market conditions and historical periods.

Looking at alternative codependence and distance metrics, Lohre et al. (2020) and Barziy and Chlebus (2020) investigated HRP adjustments using various distance and codependence measurements. HRP's performance was evaluated by Barziy and Chlebus (2020) using measures such as distance correlation, mutual information, and variance of information. These measures were later converted into distance matrices. Despite introducing these alternative metrics, the authors discovered that the original Pearson correlation coefficient beat all alternatives, calling into doubt the efficacy of adopting multiple metrics. The purpose of their paper was to evaluate if different measures may improve the HRP method beyond the traditional Pearson correlation in portfolio optimization. The study used a broad portfolio of ETFs encompassing equities, bonds, and commodities from January 2007 to December 2019, with data points taken before market closing to simulate daily prices. In contrast to classic HRP, this approach assesses non-linear and information-theoretic measurements. It seeks to capture complicated asset interdependencies between one another. However, the study discovered that while these metrics provide new insights into asset correlations, they do not increase portfolio performance regarding the Sharpe ratio or maximum drawdown. The study's results suggest Pearson correlation's effectiveness in portfolio optimization.

On the other hand, Lohre and colleagues (2020) investigated various distance metrics, including one based on lower tail dependency and intended for tail risk management. The authors explored HRP's performance using a multi-asset, multi-factor framework, comparing traditional risk parity and tail-dependence-based HRP methods across equity, bond, commodity, and credit markets from March 2003 to December 2017. Their findings suggest that while standard risk parity and tail-dependence-based HRP techniques can compete, the latter shows potential for managing tail risks. However, this strategy implies a trade-off in higher turnover. Therefore, the authors emphasize the complexity and potential costs of tail-risk-focused tactics. This incorporation of the tail dependency provides an alternative perspective on HRP's adaptability and potential in portfolio diversification, emphasizing its

effectiveness in the face of economic circumstances. Furthermore, P. Jain & Jain (2019) emphasized the importance of addressing covariance misspecification and its impact on out-of-sample portfolio performance, underlining HRP's relative robustness, especially when dealing with erroneous covariance estimates.

Finally, MHRP advances emphasize its potential for improved portfolio performance, with changes improving adaptability and robustness. However, the requirement for these changes and the examination of other metrics highlights the need for a serious reevaluation of the key assumptions underlying HRP. Future research must develop these models and examine the fundamental concepts of HRP in an ever-changing market scenario.

2.5 Literature Review Conclusion

The literature review highlights the critical importance of hierarchical risk parity (HRP) and clustering in evolving financial strategies while recognizing the inherent complexity of these methodologies. As Schwendner et al. (2021) point out, there is a delicate line between increasing precision and avoiding overcomplication in the modeling process. The following research should simplify HRP approaches, perhaps by addressing the high turnover and recurrent bisection issues raised by works such as Burggraf (2021).

Further research must be conducted to evaluate HRP's performance across multiple market conditions and asset classes to develop a methodology that is both robust and accessible. HRP can be improved significantly by using different clustering algorithms, covariance adjustments, and portfolio generation strategies. However, as P. Jain & Jain (2019) pointed out, resolving covariance misspecification and its impact on performance is critical for the applicability of HRP.

The academic and practical implementation of these techniques should proceed with caution, emphasizing the importance of thorough testing and validation to assure their dependability and effectiveness in various market conditions. Most of these research papers are still in the works, highlighting the dynamic interplay between innovation and empirical testing as HRP moves from a theoretical notion to a practical instrument in portfolio management, paving the way for future advances in the field.

To summarize, while HRP theoretically and sometimes empirically provides good risk-adjusted returns and diverse portfolios, its adaptation to the fast-changing financial landscape demands continual innovation and empirical validation. The continuing evolution of financial markets requires that HRP and clustering approaches not only keep pace but also remain relevant and effective.

3 Financial Background

This chapter delves deeper into the financial background necessary for this thesis. Given the widespread use of correlation, covariance, and eigenvalues in this thesis, this chapter will begin with a brief overview of these essential concepts. Subsequently, it will examine Markowitz's fundamental Modern Portfolio Theory (1952, 1959), establishing its foundational position in portfolio construction and discussing both its advantages and the criticisms that it has received. The notion of covariance instability is also introduced to explain the shortcomings of the Markowitz technique. Subsequently, the paper switches to common risk-based portfolio construction methodologies. Next, clustering theory is covered, laying the groundwork for a thorough grasp of the HRP algorithm. The thesis then critically examines López De Prado's (2016) scholarly work, 'Building Diversified Portfolios that Outperform Out-of-Sample', which provides an analytical interpretation of the original HRP model. This chapter further explores the theoretical foundations and practical applications of Molyboga's Modified Risk Parity (MHRP) with the main focus on Ledoit-Wolf covariance shrinkage (Ledoit & Wolf, 2004).

3.1 Measuring Risk

The most important characteristic of making an investment in an asset is that it is impossible to know the future returns, and therefore, the rate of return is said to be uncertain. In the realm of finance, risk measurement is a crucial tool that helps investors and analysts to measure this uncertainty. Various statistical tools and concepts are put into use to achieve this, such as probability theory, random variables, and distributions.

First, the probabilities associated with the uncertain returns are used to find the expected return, and then the deviation from this parameter, called variance or standard deviation, is derived. In addition to the mentioned statistics, to understand the specific characteristics of different datasets, distributions are assigned to them. As there are usually several assets that investors are interested in, the interrelationships between different financial assets through measures such as covariance and correlation, as well as the application of eigenvalues in matrix analysis, are essential for portfolio optimization and risk management.

In this section, we delve into the abovementioned concepts and present the primary statistical knowledge required to understand the theory of the thesis. The theoretical foundation of this section is based on Claus Munk's book "*Financial Markets and Investments*" (2022).

3.1.1 Random Variables and Key Summary Statistics

As we said, the return to obtain is uncertain, and in mathematical terms, such an uncertain quantity is represented by a *random variable* that shows the possible outcomes and their probabilities. *Discrete random variables* have a finite set of possible realizations, whereas a *continuous random variable* has infinite possible realizations usually represented in intervals.

When there are many realizations, comparing different distributions becomes a difficult task. Therefore, the key summary statistics are computed, mainly the expected value, the variance, and the standard deviation. Below, we show how these are computed for discrete and continuous random variables.

Discrete random variables

Let's denote a discrete random variable by X and assume that it has S possible outcomes, denoted by $x_1, x_2, \dots\ , x_S$. For each of these states, we denote the probability of realization by p_s:

$$p_s = Prob(X = x_s) \tag{3.1}$$

The expected value is the weighted average of the possible realizations, where the weights are the probabilities. The *expectation* of X is calculated as

$$E[X] = \sum_{s=1}^{S} p_s X_s \tag{3.2}$$

The expected value is used to predict the possible realization based on the probabilities. As it only leans towards the state with higher probability, it is impossible to state the actual outcome beforehand, therefore, it is useful to know how much the outcome can deviate from this value. To demonstrate this, *variance* and *standard deviation* are usually used. The variance of X is the sum of squared deviations from the expected value, weighted by probabilities, and it is defined as

$$Var[X] = E[(X - E[X])^2] = \sum_{s=1}^{S} p_S(x_s - E[X])^2 \tag{3.3}$$

It can also be simplified as

$$Var[X] = E[X^2] - (E[X])^2 \tag{3.4}$$

The standard deviation is the square root of the variance:

$$Std[X] = \sqrt{Var[X]} \tag{3.5}$$

Variance and standard deviation contain the same information, and the main purpose of using the standard deviation is to have a same-order statistic as the expected value and the values variables get. Both are non-negative and can only be zero if there is no uncertainty, i.e., there is only one possible outcome.

Continuous random variables

A continuous random variable can have infinitely many possible outcomes, usually represented by values in an interval. This is easier to handle than discrete random variables as it is relevant to real-life cases.

The probability distribution of continuous variables is represented by the probability density function (pdf) and the cumulative distribution function (cdf). Let's denote a continuous random variable by X and the density function by $f_X(x)$. Then for any a and b with $a < b$,

$$Prob(a < X < b) = \int_a^b f_X(x)dx \tag{3.6}$$

Since there are infinite number of possible values, the probability of the variable being equal to any value is zero. Therefore, when computing the probability of X being in some interval, it does not matter to include the boundaries or not.

Now that we have the probability distribution, we can compute the summary statistics with the same concept. The expectation of X is defined as

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)dx \tag{3.7}$$

The variance is computed as

$$Var[X] = E[(X - E[X_1])^2] = \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x)dx \tag{3.8}$$

which also can be simplified as

$$Var[X] = E[X^2] - (E[X])^2 \tag{3.9}$$

The standard deviation is again the square root of the variance:

$$Std[X] = \sqrt{Var[X]} \tag{3.10}$$

Standard deviation is usually used to measure the riskiness of a total portfolio; however, it is arguable to apply the same concept to individual assets. Investors are usually interested in a risk contribution to the total portfolio when the risk of a particular asset is involved.

3.1.2 Normal Distribution

Distributions are used to understand the characteristics of the possible realizations of random variables or sample data. However, individual data samples are usually different from each other, so using a standard distribution almost becomes impossible. There comes the normal distribution, which is proven to be approximately the general case for the real-life samples, given that they are large enough. Since it provides ease in understanding real-life data, the normal distribution is the most frequently used distribution, and in fact, many models in finance assume that the rate of return or the log-return is normally distributed. Due to the scope of this thesis, we will not go into details and only provide the main properties of normal distribution below.

The normal distribution is characterized by its two parameters – the mean and the variance – which the probability density function is based on. If X is a normally distributed random variable with mean μ and variance σ^2, then its probability density function is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}, \quad x \in \mathbb{R} \tag{3.11}$$

and generally, it is denoted as $X \sim N(\mu, \sigma^2)$. One can also show that for X,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \mu \tag{3.12}$$

$$Var[X] = \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x) dx = \sigma^2 \tag{3.13}$$

$$Std[X] = \sigma \tag{3.14}$$

If $\mu = 0$ and $\sigma = 1$, it is said that X is *standard normally distributed* and denoted as $X \sim N(0,1)$. Standard normal distribution is important because it makes computing probabilities easier. By using its cumulative distribution function, one can find the probability of the variable being smaller than any x for data distributed normally.

In many financial models, it is assumed that stock returns follow a normal distribution because of historical data demonstrating a bell-shaped curve. However, it is shown that most of those data bring more extreme outcomes than a normal distribution would, showing "fat tails". Considering this still happens with a small probability, it is not a big mistake to assume the rate of returns following a normal distribution. However, a bigger problem is that normal distribution gives a positive probability to the values smaller than -1. Since the stock cannot lose more than its value, this is usually solved by assuming logarithmic returns following a normal distribution. A logarithmic return is defined as

$$r^{log} = \ln R = \ln(1 + r) \tag{3.15}$$

As it is seen from the definition, log-returns are not bounded below by -1 or any other number. Additionally, the rate of return $r = \exp(r^{log}) - 1$ always stays above -1 due to the nature of the logarithmic function, solving the before-mentioned problem. When log-returns follow a normal distribution with mean m and variance s^2, the rate of return is said to follow a *lognormal distribution* with mean m and variance s^2.

3.1.3 Covariance, Correlation and Eigenvalues

Covariance

In situations where multiple financials assets are considered, it is important to know to what extent their returns move together. *Covariance* is a statistical metric that quantifies the degree of directional relationship between two random variables, indicating how they vary together. Covariance between X_1 and X_2 is defined by

$$Cov[X_{1,}X_2] = E[(X_1 - E[X_1])(X_2 - E[X_2])] = E[X_{1,}X_2] - E[X_1]E[X_2] \tag{3.16}$$

Covariance between the same element becomes its variance. For example, covariance between X_1 and X_1 is defined as

$$Cov[X_{1,}X_1] = E[X_1^2] - (E[X_1])^2 = Var[X_1] \tag{3.17}$$

If $\bar{X}_1$ and $\bar{X}_2$ represent the sample means (arithmetic average returns) of the two stocks, with the rates of return X_1 and X_2 on stock 1 and stock 2, respectively, a sample covariance can be used, generally calculated by

$$\widetilde{Cov}_{X_1X_2} = \frac{1}{n}\sum_{t=1}^{n}(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) \tag{3.18}$$

When X_1 equals X_2, the sample covariance is equivalent to the sample variance of X_1 (or X_2), which means

$$\widetilde{Cov}_{X_1X_2} = \frac{1}{n}\sum_{t=1}^{n}(X_1 - \bar{X}_1)^2 = \frac{1}{n}\sum_{t=1}^{n}(X_2 - \bar{X}_2)^2 \tag{3.19}$$

The portfolio variance, Var_p, can be calculated as

$$Var_p = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w} \tag{3.20}$$

where $\mathbf{w}$ depicts $n\ x\ 1$ vector of the portfolio weights (indicating the proportion of the total portfolio value invested in each element), $\mathbf{w}^\top$ is the transpose of the weights vector, and $\mathbf{\Sigma}$ denotes $n\ x\ n$ population covariance matrix. This matrix is a square matrix that provides the covariance between each pair of elements within a given random vector, with each element σ_{ij} specifically representing the covariance between the returns of elements *i* and *j*.

If we assume N dimensional random vector, $\mathbf{X}$, consisting of N individual random variables representing various dimensions, then the covariance matrix can be described as

$$\mathbf{\Sigma} = E\,[(\mathbf{X} - E[\mathbf{X}])(\mathbf{X} - E[\mathbf{X}])^\top] \tag{3.21}$$

From the definition of the covariance matrix, it has the following structure:

$$\boldsymbol{\Sigma} = E\begin{bmatrix} (\mathbf{X}_1 - E[\mathbf{X}_1])(\mathbf{X}_1 - E[\mathbf{X}_1]) & \cdots & (\mathbf{X}_1 - E[\mathbf{X}_1])(\mathbf{X}_N - E[\mathbf{X}_N]) \\ \vdots & \ddots & \vdots \\ (\mathbf{X}_N - E[\mathbf{X}_N])(\mathbf{X}_1 - E[\mathbf{X}_1]) & \cdots & (\mathbf{X}_N - E[\mathbf{X}_N])(\mathbf{X}_N - E[\mathbf{X}_N]) \end{bmatrix}$$

$$= \begin{bmatrix} E[(\mathbf{X}_1 - E[\mathbf{X}_1])^2] & \cdots & E[(\mathbf{X}_1 - E[\mathbf{X}_1])(\mathbf{X}_N - E[\mathbf{X}_N])] \\ \vdots & \ddots & \vdots \\ E[(\mathbf{X}_N - E[\mathbf{X}_N])(\mathbf{X}_1 - E[\mathbf{X}_1])] & \cdots & E[(\mathbf{X}_N - E[\mathbf{X}_N])^2] \end{bmatrix}$$

$$= \begin{bmatrix} Var[\mathbf{X}_1] & \cdots & Cov[\mathbf{X}_1, \mathbf{X}_N] \\ \vdots & \ddots & \vdots \\ Cov[\mathbf{X}_N, \mathbf{X}_1] & \cdots & Var[\mathbf{X}_N] \end{bmatrix} \tag{3.22}$$

In empirical finance, the sample covariance matrix, which is derived from historical data, is frequently used to approximate the unknown real covariance matrix of variables in a random vector. This approach is essential for studying risk management and portfolio optimization as well as for comprehending variable relationships, such as asset returns. Sample covariance matrix, thus, can be defined as

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{\mathrm{n}-1}\sum_{\mathrm{i}=1}^{\mathrm{n}}(\mathbf{x}_\mathrm{i} - \bar{\mathbf{x}})(\mathbf{x}_\mathrm{i} - \bar{\mathbf{x}})^\top \tag{3.23}$$

and expressed as

$$\widehat{\boldsymbol{\Sigma}} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \ddots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \cdots & \sigma_{nn} \end{bmatrix} \tag{3.24}$$

Correlation

The degree to which the two random variables have a linear relationship is shown by the *correlation*. The correlation between X_1 and X_2, ρ_{X_1,X_2}, is defined by

$$Corr[X_1, X_2] = \frac{Cov[X_1, X_2]}{Std[X_1] \times Std[X_2]} \tag{3.25}$$

Correlation is viewed as a normalized or standardized covariance and, therefore, by standardizing[1] the sample covariance matrix, we can construct the sample correlation matrix, $\hat{\mathbf{C}}$:

$$\hat{\mathbf{C}} = diag(\hat{\mathbf{\Sigma}})^{-\frac{1}{2}}\ \hat{\mathbf{\Sigma}}\ diag(\hat{\mathbf{\Sigma}})^{-\frac{1}{2}} \tag{3.26}$$

which gives us the following:

$$\hat{\mathbf{C}} = \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & \ddots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \cdots & \cdots & \rho_{nn} \end{bmatrix} \tag{3.27}$$

Generally, both covariance and correlation depict how two variables are related. However, correlation provides a more standardized and intuitive measure of the relationship's strength and direction, as it is measured in $[-1, 1]$ range. If two variables exhibit a correlation of $+1$, it indicates they are perfectly positively correlated.

Eigenvalues

Eigenvalues have profound implications in various fields, and it is one of the main topics in the context of linear algebra. It is also necessary knowledge for working with correlation and covariance matrices. Therefore, we will briefly describe their main properties below leaving out the mathematical derivations.

Conceptually, if a matrix is thought of as a linear transformation, a vector space that does not change its direction with the transformation is called an *eigenvector.* The scalar associated with this transformation is called an *eigenvalue.* To express more formally, consider a square matrix $\mathbf{A}$, which represents a linear transformation in a vector space. An eigenvector $\vec{\mathbf{v}}$ of $\mathbf{A}$ is a non-zero vector that only changes by a scalar factor when the linear transformation

[1] The diagonal function, $diag$, when applied to a square matrix, produces a matrix with diagonal elements equal to the corresponding diagonal elements, keeping all off-diagonal elements in the new matrix are zeros. The diagonal matrix should not be thought of as an ordinary matrix when matrix operations are applied, which are only relevant to the diagonal elements.

represented by $\mathbf{A}$ is applied to it. This scalar factor is known as the eigenvalue λ associated with the eigenvector $\vec{\boldsymbol{v}}$. Mathematically, this relationship is expressed by the equation:

$$\mathbf{A}\vec{\mathbf{v}} = \lambda\vec{\mathbf{v}} \tag{3.28}$$

The following properties will be used to simplify the calculations associated with covariance and correlation matrices. If $\mathbf{A}$ is a $n \times n$ matrix,

- The matrix $\mathbf{A}$ has n eigenvalues including their multiplicities.
- The sum of the n eigenvalues of $\mathbf{A}$ is the same as the sum of the diagonal elements of $\mathbf{A}$:

$$\sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i \tag{3.29}$$

- The product of the n eigenvalues of $\mathbf{A}$ is the same as the determinant of $\mathbf{A}$:

$$\det(\mathbf{A}) = \prod_{i=1}^{n} \lambda_i \tag{3.30}$$

3.2 Portfolio Construction

As briefly mentioned in the previous sections, Markowitz's mean-variance portfolio (Markowitz, 1952, 1959) is widely acknowledged as the foundation of the modern portfolio theory. The logic of the modern portfolio theory is that rational investors want to maximize the expected returns while maintaining risk at a specific level. Markowitz (1952) challenged the conventional belief that the primary goal of an investment portfolio should be the maximization of expected returns.

However, Markowitz's method failed during the global financial crisis as the portfolio did not provide the diversification it promised, and the high correlations between assets brought devastating results to the investors. It was found that the cause was reliance on the estimation of expected returns, and as a result, many risk-based strategies started to develop (Bruder & Roncalli, 2012).

In this section, we explain the theories and mathematical backgrounds of different portfolios, starting with the mean-variance and continuing with the heuristic portfolios that were thought to be its cure.

3.2.1 Modern Portfolio Theory and Markowitz's Mean-Variance Portfolio

Markowitz's shift in perspective recognized that solely prioritizing returns fails to consider the differing levels of risk associated with various assets, even when their returns are similar. Additionally, properly combining assets can mitigate a portion of the risks and yield greater expected returns without raising overall risk levels. In practice, this means choosing the weights in the portfolio to create a setup that maximizes expected return while minimizing risk (Maillard et al., 2010).

Markowitz's innovative ideas in optimizing returns while mitigating variance gave rise to the development of the *efficient portfolio* concept. An efficient portfolio represents a set of portfolios positioned along the *efficient frontier*, as depicted graphically in Figure 3.1. The idea is that for every given level of risk, measured by standard deviation, the most efficient portfolio is the one with the highest expected return. Then, the choice of portfolio among these is a question of how risk-averse the investor is.

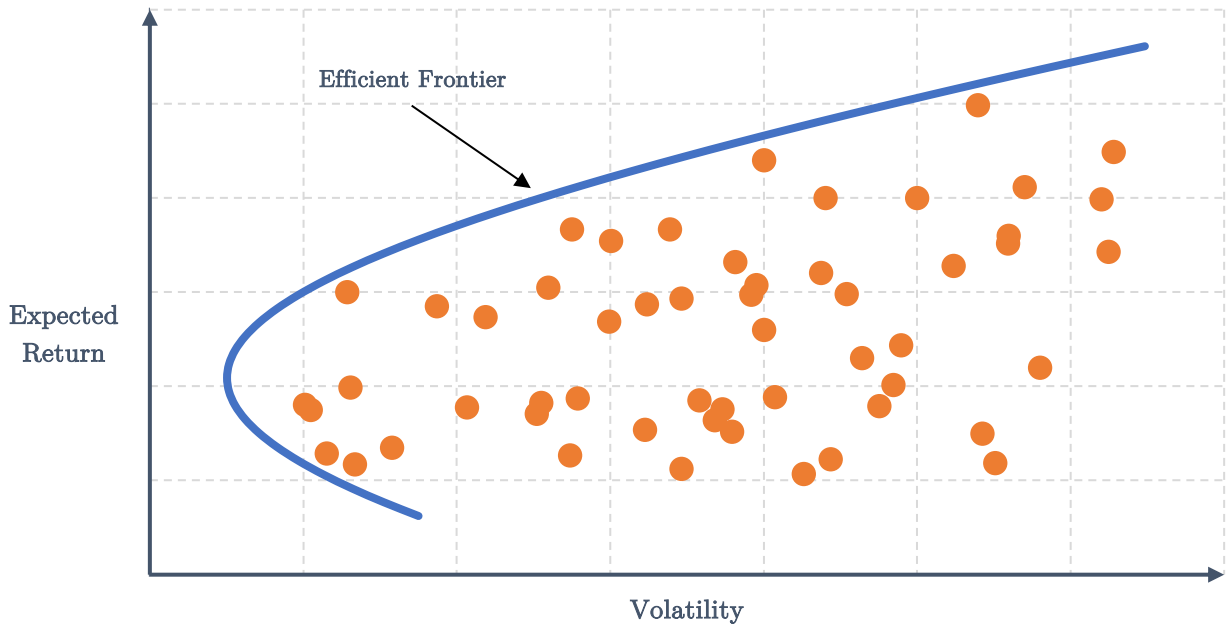

Figure 3.1: Efficient Frontier by Markowitz

As illustrated in Figure 3.1, numerous dots represent a range of portfolios, with those positioned on the efficient frontier offering the highest returns for the given levels of risk. With multiple portfolio options available, Markowitz (1959) argues that the optimal selection depends on the individual investor's risk preferences. Investors with high risk aversion require considerably higher expected returns for high risks, compared to less risk-averse investors who are willing to take more risks with less increase in expected returns. This leads the risk-averse investor to choose a portfolio on the left side of the frontier, putting the risk-seeking investor on the right. Obviously, there is no standard "coordinate" for the optimal portfolio since it depends on the shapes of indifference curves and the efficient frontier.

The mean-variance approach is a convex optimization problem predicated on the notion that the covariance matrix is positive definite[1], so that the variance can be calculated. In addition, as commonly done in portfolio optimization, constraints are set to establish both upper and lower bounds on asset weights, with a requirement that all weights must sum up to one, and short selling is not allowed on them. These constraints are applied to ensure a feasible portfolio allocation (Bailey & López De Prado, 2013).

When the convex assumption is satisfied, the optimization of the risk-return trade-off (Markowitz, 1952) can be expressed as

$$\max_{\mathbf{w}} \ \mathbf{w}^\top \mu - \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \tag{3.31}$$

$$\text{s.t. } \mathbf{w}^\top \mathbf{1} = 1 \tag{3.32}$$

$$w_i \geq 0 \tag{3.33}$$

Where $\mathbf{w}$ is used to denote the portfolio weights, μ represents the vector of expected returns, and $\boldsymbol{\Sigma}$ denotes $n \, x \, n$ covariance matrix. Therefore, $\mathbf{w}^\top \mu$ depicts portfolio's expected return while $\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$ shows the variance of the portfolio. The abovementioned constraints are denoted in the lines below the objective function. That is, $\mathbf{w}^\top \mathbf{1}$ depicts that the sum of all weights must equal one while $w_i \geq 0$ shows that all asset weights must be positive, i.e., no shorting is allowed.

[1] The matrix A is said to be positive definitive if $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is always positive for every non-zero real column vector $\mathbf{x}$. All eigenvalues of a positive definite matrix are positive (Alexander, 2012).

To expand the formula to consider the risk aversion factor, a new parameter γ can be added to the function. Here, γ indicates the investor's propensity toward maintaining a balance between risk mitigation and return maximization. An increased γ signifies a more pronounced risk aversion on the investor's behalf. Mathematically, this can be shown as

$$\max_{\mathbf{w}} \mathbf{w}^{\top}\mu - \frac{\gamma}{2}\mathbf{w}^{\top}\boldsymbol{\Sigma}\mathbf{w} \tag{3.34}$$

Both risk-return and risk aversion formulas require optimization algorithms to be solved numerically, as no analytical solution exists. This is due to the number of constraints being more than one. Markowitz (1987) proposed the Critical Line Algorithm (CLA), a specific optimization technique tailored for calculating solutions in certain circumstances. However, given the particular focus of this thesis on the HRP, neither this algorithm nor an in-depth examination of the constraint scenarios will be explained. It is worth noting that these types of problems are typically solved utilizing optimization techniques in programming languages.

Shortcomings of Markowitz's Model

As already alluded to in the introduction, there are several disadvantages associated with the mean-variance portfolio. It is known to perform poorly in practice due to its unstable nature and sensitivity to estimation errors (Michaud et al., 2015). The estimation errors arise since there are no correct estimates of expected returns or variances and covariances.

Michaud argues that this happens due to the ill-conditioned covariance matrices, where a small change in the input parameters causes the optimal weights to differ drastically. When the means are higher/lower or correlations are negative/positive, the method tends to overweight/underweight the securities (Michaud, 1989; Black & Litterman, 1992). In this situation, relying on historical data to produce input parameters, such as using the sample mean as the expected error, contributes greatly to the error-maximization of the model. Moreover, the demand for diversification grows with rising asset correlation, yet the resulting solutions become more volatile. This instability frequently undermines the benefits of diversification by magnifying the errors in estimation (López De Prado, 2016).

Moreover, market capitalization is not considered by Markowitz's model either. If an asset with low market capitalization has high expected returns and a negative correlation with the other assets in the portfolio, then the model will suggest a very high portfolio weight for the asset. Michaud (1989) also claims that the constraints applied in the MV optimization model

turn it into, rather than finding an optimal portfolio, an exercise of finding an "acceptable" portfolio.

Another significant flaw in the Markowitz's model stems from its approach to smaller principal components within the covariance matrix, leading to unrealistic asset weights (Pedersen et al., 2021). This occurs because the model often misinterprets noise as signal, assigning overly optimistic weights based on inaccurate estimations of risk and return. Such overemphasis on these components can result in portfolios with undue exposure to specific risks, deviating markedly from optimal performance (Pedersen et al., 2021). This highlights the model's limitations in distinguishing genuine asset correlations from fluctuations in the noisy financial market data.

Several papers were published, and models were developed to cope with the estimation errors related to both input parameters – expected returns and covariance matrix. The most popular model for estimating expected returns was developed shortly after Markowitz's publications, called the Capital Asset Pricing Model (CAPM), developed by William Sharpe (1964). The model is built on simple linear regression and suggests that there is a relationship between expected returns and systematic risk.

Similarly, due to the inconsistency and poor performance of both Markowitz's mean-variance portfolio and other risk-centered optimization techniques, many models were developed to make the covariance matrix more robust. Some examples of these strategies can be using Bayesian priors (Black & Litterman, 1992), including new restrictions in the quadratic optimization problem (Thorley, 2001), or taking into consideration tax effects and transaction costs (Litterman, 2003; Stein, 2001). Due to the scope of this thesis, only the most famous one suggested by Ledoit and Wolf (2004) will be covered in this thesis. They applied a transformation on the sample covariance matrix called *shrinkage* that pulled the most extreme coefficients towards more central values, thereby systematically reducing estimation error.

Although the parameters both have considerable estimation errors, studies have shown that the estimation of expected returns has a greater impact on portfolio instability than the calculation of the covariance matrix (Bruder & Roncalli, 2012). As a result, a considerable amount of research has been performed to assess the usefulness of the various portfolio construction approaches that neglect the expected returns, such as risk budgeting and risk

parity (Bruder & Roncalli, 2012; Maillard et al., 2010). Starting from the next subsection, we will briefly analyze some risk-based portfolio construction strategies.

3.2.2 Minimum-Variance Portfolio

Among the collection of portfolios located on the efficient frontier, the minimum-variance (MinVar) portfolio is the only one that can be constructed without the need for expected returns, that is, by disregarding any return requirement. As it has the lowest variance among the myriad of possible portfolios, the MinVar portfolio can be determined by minimizing the variance of the portfolio with a full investment constraint:

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \tag{3.35}$$

$$\text{s.t.} \ \mathbf{w}^\top \mathbf{1} = 1 \tag{3.36}$$

Since the global minimum-variance portfolio is subject to only one constraint, the convex optimization problem above can be solved analytically by using the Lagrangian method, and it gives the following solution for the portfolio weights:

$$\mathbf{w}_{MinVar} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}} \tag{3.37}$$

Note that if all the assets have the same expected return, the minimum-variance portfolio will equal the mean-variance portfolio. This is also in accordance with the theory of Markowitz's model: as all the portfolios yield the same expected return, the one with the lowest variance will be the efficient portfolio.

As we described above, the minimum-variance portfolio can be found on the efficient portfolio, as depicted in Figure 3.2 below:

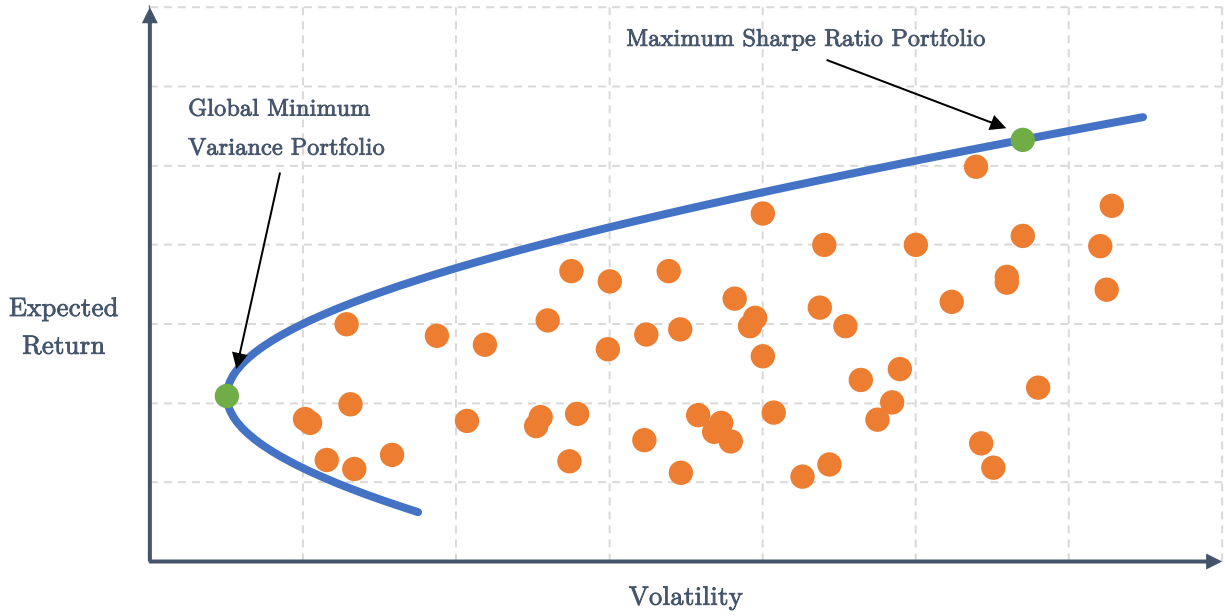

Figure 3.2: Global Minimum-Variance Portfolio on the Efficient Frontier

As we have defined a long-only strategy to compare different portfolios in this thesis, the optimization problem must take the following form:

$$\mathbf{w}_{MV} = \underset{w}{\operatorname{argmin}}\ \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \tag{3.38}$$

$$s.t.\ \mathbf{w}^\top \mathbf{1} = 1 \tag{3.39}$$

$$w_i \geq 0 \tag{3.40}$$

In this case, the minimization problem does not have an analytical solution and needs to be solved numerically.

Minimum-variance portfolios are praised by many authors for achieving higher Sharpe ratios with lower volatility; however, Maillard et al. (2010) states that in terms of portfolio turnover, they are less efficient due to high concentration, and they may also demonstrate higher drawdowns in short-term.

3.2.3 Maximum Sharpe Ratio Portfolio

Another critical portfolio on the efficient frontier in Figure 3.2 is the maximum Sharpe ratio (MaxSharpe) portfolio. This portfolio is designed to achieve the highest possible Sharpe ratio,

which is the ratio of the excess return to the standard deviation of the portfolio, thereby optimizing the risk-adjusted return. The Sharpe ratio is calculated as

$$Sharpe\ Ratio = \frac{\mathbf{w}^\top(\mu - r_f)}{\sqrt{\mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}}} \tag{3.41}$$

where μ represents the vector of expected returns, r_f is the risk-free rate, and $\mathbf{\Sigma}$ is the covariance matrix of asset returns.

To find the MaxSharpe portfolio, one maximizes the Sharpe ratio subject to the full investment constraint. Considering a long-only strategy, the optimization is formulated as below, and again numerical solutions are needed:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top(\mu - r_f)}{\sqrt{\mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}}} \tag{3.42}$$

$$s.t.\ \mathbf{w}^\top \mathbf{1} = 1 \tag{3.43}$$

$$w_i \geq 0 \tag{3.44}$$

The MaxSharpe portfolio, while theoretically optimal in terms of the Sharpe ratio, may also be susceptible to estimation errors in expected returns and the covariance matrix, leading to potential instability and significant turnover. As highlighted by many researchers, the reliance on accurate estimates of μ makes it more sensitive to model assumptions compared to the MinVar portfolio.

3.2.4 Equally Weighted Portfolio

To reduce the riskiness of the asset, some methods, such as the equally weighted portfolio, have been introduced, focusing on diversification. In an equally weighted (EW) portfolio, also known as $\frac{1}{N}$ or "naive diversification" approach, all assets are given the same weight. Therefore, the portfolio weights are:

$$w_i^{EW} = \frac{1}{N} \quad \text{for all } i = 1, \dots, N \tag{3.45}$$

where N is the number of assets. Although it is constructed in a simple way without addressing the return and risk estimations, the EW portfolio has proven to perform better than many

risk-based and traditional portfolio methods. In a popular study by DeMiguel et al. (2009), 14 different models were evaluated for enhancing the mean-variance portfolio. However, none of them outperformed the naïve diversification in out-of-sample tests in terms of Sharpe ratio.

Although many studies cited this study to prove the strong performance of the equally weighted portfolio, many authors also argue that the study's design was the main reason for the underperformance of optimized portfolios, proposing several reasons. Lohre et al. (2022) present some of these explanations and reasons the strong performance of the EW portfolio as a combination of loadings on low-size, short-term reversal, as well as seasonality effects.

Cirulli and Walker (2023), however, applied 3 different modifications to the EW portfolio (removing the stocks with the 1) lowest mean return, 2) highest standard deviation, 3) lowest Sharpe ratio, and equally weighing the remaining assets), and the resulting strategies outperformed the original equally weighted portfolio in terms of returns, risk-adjusted returns, and downside-risk measures. In summary, as Cirulli and Walker (2023) stated themselves, there is still a debate about the performance of the equally weighted portfolio and the conditions that make it outperform the optimized portfolios.

3.2.5 Risk Parity

The risk parity approach, being one of the beta portfolios, focuses on the allocation of risks rather than the allocation of capital, stating that if the asset allocations are determined by the same risk level, the portfolio can achieve a higher Sharpe ratio. Although its theoretical background started to develop in the 1950s, the first risk parity fund was introduced in 1996 by Bridgewater Associates, LP, an American investment management firm. The term, however, was first used by Qian (2005), where he claimed that by reaching true diversification with risk parity, the impact of losses coming from individual assets could be limited. In his paper, he mostly discourages the use of the traditional approach – the 60/40 portfolio that many investors prefer. He argues that this "balanced" portfolio is highly concentrated in terms of risk allocation, where the variance of stocks is roughly nine times big as that of bonds.

Maillard et al. (2010), in pursuit of finding an alternative to minimum-variance and equally weighted portfolios due to their limitations, proposed an approach called *Equal Risk Contribution Portfolio* (ERC), where the "1/N" filter is applied in terms of risk. They show that the volatility of this portfolio is located between those of minimum variance and equally

weighted portfolios, and it is potentially a good middle-point between the two approaches in terms of absolute level of risk: risk budgeting and diversification.

Bruder and Roncalli (2012), however, described the ERC approach as a special form of *risk budgeting*, which does not require the risk contributions to be the same, and considered the term "risk parity" equal to "risk budgeting". Although risk budgeting was in the literature for years, there were few results about the theoretical results of those portfolios until Bruder and Roncalli presented its main applications.

As different financial managers have different methods of categorizing assets, defining risk and the allocation of volatility, the principles of risk parity are defined differently in literature. Therefore, in this subsection, we will explore the equal risk contribution and its general form – risk budgeting portfolios. For the formulas, we derive the simplest explanation from two studies that contributed to the risk parity to a large extent, Bruder and Roncalli (2012) and Maillard et al. (2010).

Risk Budgeting (RB) Portfolio

Risk budgeting portfolios are constructed by setting the risk contribution of each asset in the portfolio to a specific risk budget, usually determined by the portfolio manager. Risk contribution of an asset is the part of total portfolio risk that is caused by that asset (Bruder & Roncalli, 2012).

To start with, every risky component of the portfolio contributes to the total risk determined by its own risk and the covariance with other assets. The marginal risk contribution, $\delta_{w_i}\sigma(\mathbf{w})$, shows how much the portfolio risk, $\sigma(\mathbf{w})$, increases by one unit increase in the asset weight, $\mathbf{w}_i$. Using differentiation formulas, we get the following:

$$\delta_{w_i}\sigma(\mathbf{w}) = \frac{\delta\sigma(\mathbf{w})}{\delta_{w_i}} = \frac{(\boldsymbol{\Sigma}\mathbf{w})_i}{\sqrt{\mathbf{w}^\top\boldsymbol{\Sigma}\mathbf{w}}} = \frac{\sum_{j=1}^{N} w_j\sigma_i\sigma_j\rho_{ij}}{\sigma(\mathbf{w})} \tag{3.46}$$

To determine how much risk has been contributed by the asset, i.e., risk contribution[1], the marginal risk contribution should be scaled by the asset weight:

[1] One can show that the sum of risk contributions by all components of the portfolio is equal to the total risk.

$$RC_i = w_i * \delta_{w_i}\sigma(\mathbf{w}) \tag{3.47}$$

Again, with a binding full investment constraint and long-only strategy, the weights for the risk budgeting portfolio can be defined as below:

$$\mathbf{w} = \{\mathbf{w} \in [0,1]^N : \mathbf{w}^\top \mathbf{1} = 1, w_i * (\mathbf{\Sigma w})_i = b_i * (\mathbf{w}^\top \mathbf{\Sigma w})\} \tag{3.48}$$

where b_i denotes the risk budgeting for asset i.

An easier demonstration of the risk budgeting portfolio is by using the asset betas. By definition, β for asset i is defined as

$$\beta_i = \frac{Cov[R_i, R_p]}{Var[R_p]} = \frac{Cov[R_i, \sum_{j=1}^{N} w_j R_j]}{\sigma^2(\mathbf{w})} = \frac{\sum_{j=1}^{N} w_j \sigma_i \sigma_j \rho_{ij}}{\sigma^2(\mathbf{w})} \tag{3.49}$$

Hence, β of the asset i can be used to show its risk contribution:

$$RC_i = w_i \beta_i \sigma(\mathbf{w}) \tag{3.50}$$

As $\frac{RC_i}{\sigma(\mathbf{w})} = b_i$, the portfolio weights can then be shown as

$$w_i = \frac{\frac{b_i}{\beta_i}}{\sum_{j=1}^{N} \frac{b_j}{\beta_j}}$$

This is still not an analytical solution since to find beta, one needs to know the portfolios weights. Obviously, when the risk budget dedicated to the asset increases, the weight of the asset increases as well. The opposite is true for its beta as the rising correlation with the other assets makes it non-preferred. As alluded in the introduction to the risk parity, risk budgeting is seen as a general case to the other popular approach – the ERC portfolio (Bruder & Roncalli, 2012).

Equal Risk Contribution (ERC) Portfolio

The equal risk contribution portfolio, being a specific case of risk budgeting, implies that the risk contributions of the portfolio components must be the same. In terms of volatility hierarchy, this portfolio offers a case where no asset dominates the portfolio risk (Maillard et al., 2010).

The portfolio is usually constructed by solving the risk budgeting formula and setting $b_i = \frac{1}{N}$. However, the complete formula for the ERC portfolio, with the budget and no shorting constraints, can still be shown as follows:

$$\mathbf{w} = \left\{\mathbf{w} \in [0,1]^N : \mathbf{w}^\top \mathbf{1} = 1, w_i * (\mathbf{\Sigma w})_i = w_j * (\mathbf{\Sigma w})_j \quad \forall\, i,j = 1, \dots, N\right\} \tag{3.51}$$

Many authors have described the equal risk contribution portfolio as a simple method to ensure and even maximize risk diversification through portfolio components. As we shortly mentioned before, Maillard et al. (2010) state that in a situation where equally weighted portfolios are inferior in terms of performance and minimum-variance portfolios are not efficient because of the high concentration, the ERC portfolio is a useful alternative for investors.

3.2.6 Inverse-Variance Portfolio

Being one of the heuristic approaches, the inverse-variance portfolio is an easy way to reduce the instability and concentration that optimized portfolios face. The inverse-variance approach allocates the weights to assets based on the inverse of their variances, that is, to avoid allocating large weights to assets with large variances. The proportion of the inverse variance to the sum of inverse variances determines the weight of the asset, denoted as follows:

$$w_i^{IVAR} = \frac{\frac{1}{\sigma_i^2}}{\sum_{j=1}^{n}\left(\frac{1}{\sigma_j^2}\right)} \tag{3.52}$$

where σ_i^2 is the variance of the asset. Inverse-variance is also known as a special case of the minimum-variance portfolio, where the assets are fully independent of each other, meaning that the covariance matrix is diagonal. If the expected returns are the same, it coincides with the mean-variance portfolio as well.

3.2.7 Inverse-Volatility Portfolio

The inverse-volatility portfolio implies the same idea of diversification as the inverse-variance portfolio; however, it focuses on the volatility (standard deviation) rather than the variance. The proportion of the inverse volatility to the sum of all inverse volatilities determines the weight of the asset, denoted as follows:

$$w_i^{IVOL} = \frac{\frac{1}{\sigma_i}}{\sum_{j=1}^{n}\left(\frac{1}{\sigma_j}\right)} \tag{3.53}$$

Inverse-volatility is a special case of the mean-variance portfolio under the assumption that all assets are independent of each other, meaning no correlation, and have equal risk-adjusted returns.

In conclusion, all heuristic portfolios, including risk-based optimized portfolios, take the covariance matrix as a main input parameter, being exposed to its numerical instability. Therefore, in the next section, we describe the problem in greater detail and then a popular method to cope with this problem.

3.3 Covariance Instability

We have shown that many risk-based and other heuristic portfolio optimization methods, by fully ignoring the expected returns, solve the estimation problem of the returns. However, these portfolio optimization methods still use the sample covariance matrix for asset allocation. Hence, the matrix's quality greatly influences the resulting weight distribution's alignment with the optimization goal. Therefore, this section explains the inherent instability of estimating sample covariance and correlation matrices, setting the stage by exploring various root causes of this issue. This initial knowledge is followed by an in-depth exploration and discussion about the possible strategies to overcome these obstacles.

The accuracy of the covariance estimations is critical, as more precise computations provide weight allocations that are closer to the optimal solution. Ledoit and Wolf (2004), in their paper focusing on shrinking the covariance matrix, emphatically argue against using the sample covariance matrix for optimizing portfolios. The core issue is that the sampled covariance matrix has a propensity to show outlier values caused by significant errors. This anomaly is exploited by mean-variance optimization, which causes it to overinvest in the least dependable coefficients and is known in financial literature as the "error maximization" phenomenon (Michaud, 1989).

The stability of portfolio optimization, which relies on a covariance matrix, hinges on the condition number (López De Prado, 2016). This is calculated as the absolute value of the ratio

of the largest to the smallest eigenvalues. As the portfolio incorporates more correlated assets, this condition number increases, leading to a deterioration in the matrix's condition. Consequently, while diversification through the addition of more assets aims to mitigate risk, the accompanying estimation errors frequently negate these advantages. Furthermore, it is critical to distinguish between estimated and actual population matrices. The estimated matrices aim to approximate the true yet unobservable population matrices, but they are not precise copies. This attempt at estimation inevitably leads to the introduction of additional noise.

To summarize, the instability in computing covariance and correlation matrices is mostly due to *noise* from sampling errors and *signal*, particularly given the hierarchical nature of financial cross-correlations (López De Prado, 2020). The two elements that contribute to estimating inaccuracies will be discussed further in the next two subsections.

3.3.1 Noise-Induced Instability in Covariance

According to the law of large numbers, as the number of observations, T, increases indefinitely with a fixed number of variables, N, the covariance estimator $\widehat{\boldsymbol{\Sigma}}$ must match the true population covariance, $\boldsymbol{\Sigma}$. In a similar manner, Anderson (1963) has shown that when the sample size grows indefinitely while the number of variables stays constant, the eigenvalues from a sample correlation matrix tend to match the eigenvalues of the real population matrix. The almost equal finding states that when the number of variables, N, is not fixed and variables to observations ratio ($q = N/T$) approaches 1, sample eigenvalues lose accuracy in reflecting population eigenvalues, regardless of sample size T (Bun et al., 2017). Hence, in high-dimensional settings, such as real-life financial applications, where both N and T are large, the sample estimator $\widehat{\boldsymbol{C}}$ is susceptible to deviation from true values, a phenomenon called *noise*. This is known as the curse of dimensionality, and it emphasizes the complexity and obstacles that arise in such settings (Bun et al., 2017).

To address the challenge of the curse of dimensionality, Ukrainian mathematicians Vladimir Marčenko and Leonid Pastur (1967) developed the Marčenko-Pastur distribution. This distribution is essential in Random Matrix Theory for examining the eigenvalue spectrum of large correlation matrices in high-dimensional environments. This occurs when both the number of assets and the number of observations are large, yet their ratio q remains substantial, rather than approaching zero (Bun et al., 2017). The Marčenko-Pastur distribution

facilitates discerning the relationship between the empirical correlation matrix $\widehat{\boldsymbol{C}}$ and the true correlation matrix $\boldsymbol{C}$, focusing on the eigenvalues' distribution. Therefore, $\boldsymbol{C}$ is shown as

$$\rho_{\boldsymbol{C}}(\lambda) = \frac{T}{N}\frac{\sqrt{(\lambda_{max} - \lambda)(\lambda - \lambda_{min})}}{2\pi\lambda\sigma^2} \tag{3.54}$$

where

$$\lambda_{max} = \sigma^2\left(1 + \sqrt{N/T}\right)^2 \tag{3.55}$$

$$\lambda_{min} = \sigma^2\left(1 - \sqrt{N/T}\right)^2 \tag{3.56}$$

This proposed framework enables the deduction of the connection between the empirical correlation matrix $\widehat{\mathbf{C}}$ and the true correlation matrix $\mathbf{C}$ through their eigenvalues. It also explains why empirical correlation matrices are significantly noisy. These results depict how the ratio between assets and periods of observations affects the distribution of eigenvalues in a null hypothesis. Results allow for distinguishing the significant eigenvalues, that are consistent with noise (Laloux et al., 1999) and are found within the range of $\lambda \in [\lambda_{min}, \lambda_{max}]$, from those considered as noise, lying outside this range. It is worth mentioning that the eigenvalue distribution from real financial data, marked by interdependent returns, does not perfectly match the Marčenko-Pastur model. Nonetheless, Pedersen et al. (2021) argue that the model nonetheless plays an important role in determining what constitutes random noise within eigenvalues.

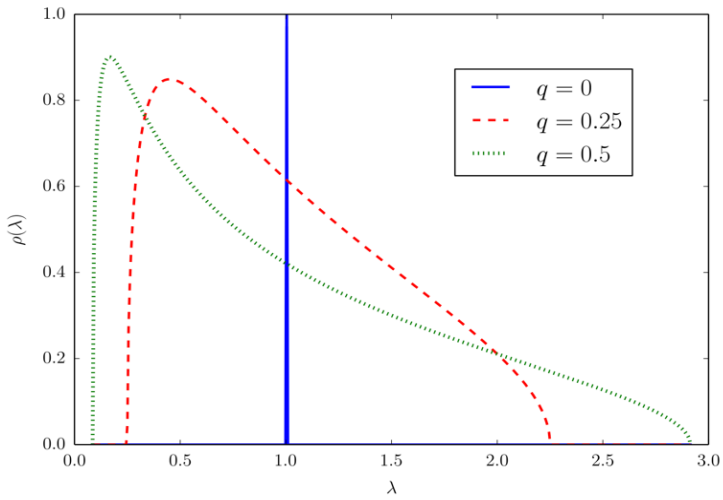

Figure 3.3 below depicts that when the number of variables N is large and close to T, the sample estimator $\widehat{\boldsymbol{C}}$ becomes unreliable due to significant systematic errors in eigenvalue

calculation caused by numerous noisy variables, assuming the null hypothesis where $C = I_N$ and $N = 500$ (Bun et al., 2017).

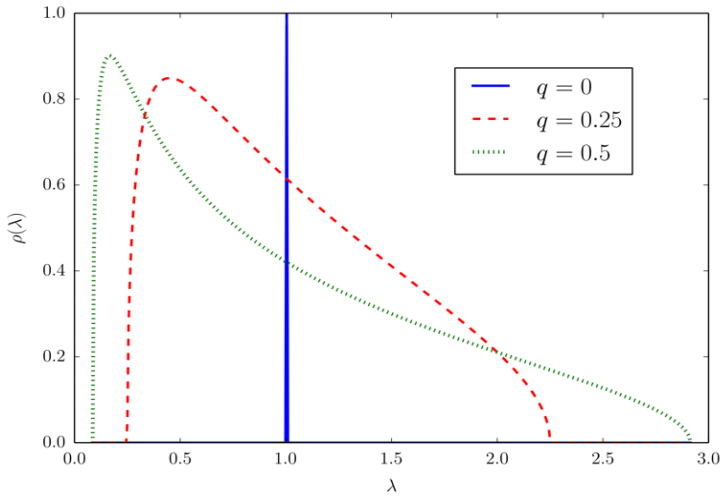

Figure 3.3: Density of sample eigenvalues (Bun et al., 2017)

The growth in portfolio size without an equivalent increase in observations introduces more noise into covariance-based allocation methods, highlighting the need for effective solutions. Various techniques exist; however, given the scope of this thesis, the Ledoit-Wolf covariance shrinkage methodology (Ledoit & Wolf, 2004) will be the primary approach examined and elaborated upon in subsequent chapters. Investigating matrix inversion reveals how noise impacts covariance matrix stability. As q approaches 1, the determinant of the $N \, x \, N$ matrix and its inverse become unstable, underlining the necessity for stable portfolio optimization in high-dimensional situations (López De Prado, 2020).

3.3.2 Signal-Induced Instability in Covariance

Signal instability arises from the data structure and the interconnections among assets, leading to fluctuations. This phenomenon challenges conventional analysis by opposing the objective of extracting meaningful signals (López De Prado, 2020). However, signals generate complications, and because they are an intrinsic element of the structure, signal-induced instability cannot be easily addressed by increasing the number of observations (López De Prado, 2020). Beyond the theoretical scenario where asset correlations are zero, $\boldsymbol{C} = \boldsymbol{I_N}$, resulting in a condition number greater than one, real-world data typically exhibits non-zero correlations among assets and between clusters. This empirical reality, highlighting prevalent correlations, contradicts the idealized assumption of independence (Mantegna, 1999).

Signal instability in covariance matrices is caused by clusters of tightly correlated variables. A cluster is formed when a collection of assets is more closely tied to one another than to the larger asset pool. Such clustering increases the matrix's condition number, compromising its stability. To address this instability, it is proposed to use clustering algorithms that manage the impact of these dominating clusters without depending on matrix inversion.

Figure 3.4 shows the comparison between the actual eigenvalue distribution from the U.S. stock data ranging from 1991 to 1996 and a theoretical distribution based on an identity matrix with equivalent q, highlighting how estimation errors can distort the eigenvalue spectrum. Assets in the same cluster share a common eigenvector, leading to a single eigenvalue explaining a significant portion of their total variance. Since the sum of the correlation matrix's diagonal values $\boldsymbol{tr(C)}$ equals N (with each value being 1), an increase in one eigenvalue results in a decrease in others, given the constant sum of eigenvalues. This illustrates the trade-off among eigenvalues within the correlation matrix. It can be shown as

$$tr(C) = N = \sum_{i=1}^{N} \rho_{ii} = \sum_{i=1}^{N} \lambda_i \tag{3.57}$$

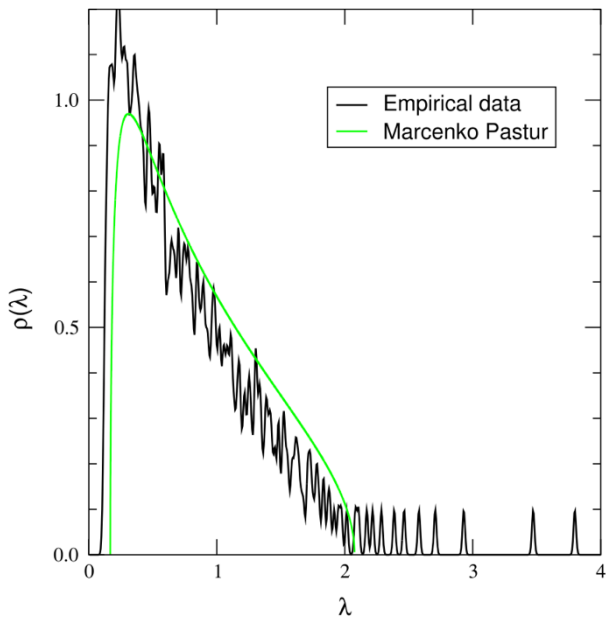

Figure 3.4: Empirical spectral density of the correlation matrix corresponding to N = 406 and T = 1300 so that q ≈ 0.31, compared to Marčenko–Pastur spectrum (Bun et al., 2017)

The condition number increases as the difference between maximum and minimum eigenvalues grows, worsened by strong correlations within clusters. High signal levels, like noise-induced instability, drive most eigenvalues to zero, resulting in instability in optimization that includes inverting the covariance matrix. The extreme eigenvalue of the most correlated cluster has a considerable influence on the condition number, highlighting the importance of dense clusters in stabilizing and optimizing processes.

3.3.3 Conclusion of Covariance Instability

López De Prado's (2016) research demonstrates the major influence of covariance instability on the efficacy of typical Markowitz-style portfolios. As a result, the author uses simulated data to demonstrate that the mean-variance portfolio is the most sensitive strategy to errors within the sample covariance matrix. To approximate real market conditions, the author produces ten series of random Gaussian returns, each consisting of 520 observations, to simulate two years of daily trading activity with a zero mean and a standard deviation of 10%. Random shocks and random correlations are added to mimic the properties of real prices (Merton, 1976). The author computes HR), critical line algorithm (CLA), and inverse-volatility portfolio (IVP) using a year's worth of data (260 observations), rebalancing each portfolio monthly. This entire process is repeated 10,000 times to determine the portfolios' out-of-sample performance. López De Prado's (2016) results show that HRP beats CLA substantially, with a 31.3% increase in its out-of-sample Sharpe ratio, demonstrating its greater ability to negotiate complicated market dynamics.

Although these findings are encouraging, and the HRP approach theoretically mitigates the impact of covariance instability, López De Prado's (2016) simulated data faces inherent challenges. Simulating and modeling complex market dynamics accurately, particularly in the face of considerable economic change, remains challenging (Chen & Huang, 2008). These problems highlight the fundamental necessity for empirical validation based on actual historical financial data. This validation is required to prove the resilience of HRP portfolios in dealing with real-world market volatility.

3.4 Ledoit-Wolf Shrinkage

This section of the thesis will introduce the sample covariance shrinkage approach presented in the paper *Honey, I Shrunk the Sample Covariance Matrix* (Ledoit & Wolf, 2004). It is a refined strategy for overcoming the limitations associated with using the sample covariance matrix in portfolio optimization. This method employs a shrinkage process to mitigate estimation errors that greatly impact mean-variance optimization, effectively reducing tracking errors and improving the information ratio for portfolio managers. It achieves these results using real stock market data without necessitating changes to other portfolio management procedures.

Due to the scope of this thesis, this paragraph provides only a general overview of the Ledoit-Wolf shrinkage method. For a detailed numerical analysis and comprehensive calculations, please consult the original paper[1]. Theoretically, the shrinkage works in three main steps.

Shrinkage Estimation

The first phase of the Ledoit-Wolf shrinkage approach introduces the notion of shrinkage estimation for the covariance matrix, with the goal of addressing the sample covariance matrix's limits in portfolio optimization. The authors suggest a convex linear combination of the sample covariance matrix and highly structured estimators, which is controlled by a shrinkage constant.

Shrinkage Target

The shrinkage target is essentially a structured estimator that aims to offer a more stable alternative to the sample covariance matrix. This target fulfills two crucial criteria: 1) it incorporates a significant amount of structure by limiting the number of free parameters, and 2) it accurately reflects key characteristics of the true covariance matrix it seeks to estimate. Ledoit and Wolf favor the continuous correlation model to the single-factor model for the shrinkage due to its simplicity and efficacy. The authors advocate for the constant correlation model over the single-factor model for the shrinkage target due to its straightforwardness and effectiveness. The latter model simplifies the estimation process by assuming a uniform

[1] The original paper for Ledoit-Wolf shrinkage can be accessed at www.ledoit.net/honey.pdf.

correlation among all assets, facilitating the calculation of average correlations and individual asset variances (Ledoit & Wolf, 2004).

Shrinkage Constant

The shrinkage constant acts as the shrinkage intensity. This constant quantifies the extent to which the sample covariance matrix is adjusted towards the shrinkage target, aiming to optimize the balance between reducing estimation error and maintaining the empirical data's relevance. The determination of shrinkage intensity involves a methodical calculation, which assesses the trade-off between the potential bias introduced by the shrinkage target and the variance of the sample covariance. This process aims to enhance the accuracy and stability of the resulting covariance matrix, particularly for portfolio optimization purposes.

3.5 Hierarchical Clustering

Hastie, Tibshirani, and Friedman (2009) roughly categorize learning problems as *supervised* or *unsupervised.* In contrast to supervised learning, where the outcome is predicted based on several input variables, namely the sample data of previous learnings, in unsupervised learning, the outcome measure does not exist, and the goal is to describe the patterns of the input measures. Being an unsupervised learning method, clustering analysis helps to form descriptive statistics to understand different groups of data that are characterized by distinct features.

Clustering analysis, also called data segmentation, relates to grouping a collection of objects into "clusters" by making sure that the objects within the same cluster are closer to each other than those in other clusters. For all clustering techniques, grouping is based on the distance or dissimilarity measure selected (Hastie et al., 2009). The differences, however, are many: the metrics used in each step, the relationship between different clusters, the specification of the number of clusters, or the computational cost to build them.

Jain (2010) broadly divides the clustering algorithms into two groups: *hierarchical* and *partitional.* Hierarchical clustering algorithms build nested clusters, whereas partitional clustering algorithms assume the clusters to be simultaneous parts of the data without demonstrating a hierarchical relationship. For example, the most popular partitional clustering method, K-means clustering, divides the data into non-overlapping subsets without any internal structure, specifies the number of clusters beforehand, and is computationally cheap.

In contrast, hierarchical clustering has an internal structure (clusters inside clusters), does not require the number of clusters determined beforehand, and is computationally expensive. As hierarchical risk parity is built on it, we only cover hierarchical clustering in this thesis.

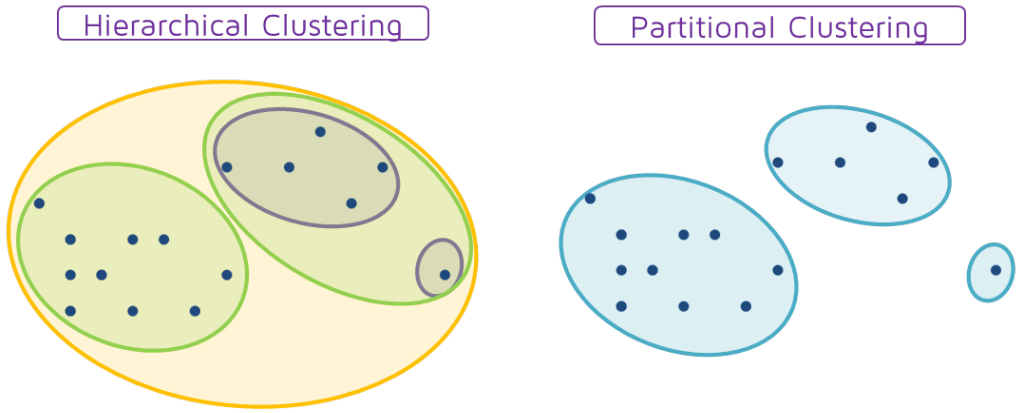

Figure 3.5: Hierarchical and Partitional Clustering

Hierarchical structuring algorithms build the nested clusters in two ways: agglomerative mode and divisive mode. Both algorithms have their own advantages and disadvantages; however, due to its academic base, agglomerative mode is preferred more (Hastie et al., 2009). Overall, the divisive mode (top-down) starts with one cluster that has all assets inside and divides until every asset has its own cluster. Agglomerative mode (bottom-up) starts with clusters that have single assets inside and build up to eventually have all assets in one cluster (A. K. Jain, 2010). From this point on, we will only focus on agglomerative hierarchical clustering.

Once the hierarchical clustering method is chosen, there are two main steps in the process (Papenbrock, 2011):

1. Choice of a distant measure
2. Choice of the clustering algorithm

In the first step, using the correlation matrix, dissimilarity or distant measures are derived, and the clustering process is based on the numerical features of the asset relationships. In the later step, the method of how to further group the clusters, i.e., the linkage method, is chosen to eventually form the clustering.

It is important to understand the decisions made in each step and how different datasets can require different approaches. Therefore, the next subsections will focus on the choice and calculation of the distant measure and how the clusters are "linked" together.

3.5.1 Distance Measure

To group the data, there must be a way to measure the similarity between the elements to decide which elements belong to the same group. This is usually achieved by introducing a *dissimilarity measure*[1], which is used interchangeably with *distance metric.* The goal is to convert the correlation matrix into another matrix so that every asset can be represented with a vector of distances.

The distances are usually measured in Euclidean space, which is the fundamental space of geometry. A distance metric in the Euclidean space between i and j must fulfill the following requirements (Murtagh & Contreras, 2017):

- Symmetry: $d(i,j) = d(j,i)$.
- Non-negativity: $d(i,j) \geq 0$.
- Identity of indiscernibles: $d(i,j) = 0$ if and only if $i = j$.
- Triangular inequality[2]: $d(i,j) \leq d(i,k) + d(k,j)$,

In order to satisfy these, López De Prado (2016) uses the distance metric proposed by Gower et al. (2016):

$$d_{i,j} = \sqrt{\frac{1}{2}\left(1 - \rho_{i,j}\right)} \tag{3.58}$$

[1] A dissimilarity measure, while closely related to distance measures, is a broader concept that quantifies how different two objects are without necessarily adhering to the strict mathematical properties of distance measures. Dissimilarity measures can be applied to both quantitative and qualitative (categorical) data and are not restricted to geometric spaces.

[2] Triangular inequality is only applicable to distance metric as it is one of the main features of triangles in geometric spaces. This is the only axiom that separates distance measure from a dissimilarity measure.

From the characteristics of the correlation coefficient and the basic algebra, it is obvious that the first three requirements are fulfilled, and the last axiom has been numerically verified by Mantegna (1999) in all portfolios he investigated.

Using the distant metric, the correlation matrix can be transformed to a distance matrix **D**:

$$\mathbf{D} = \begin{bmatrix} 0 & d_{12} & \cdots & d_{1n} \\ d_{21} & \ddots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & \cdots & \cdots & 0 \end{bmatrix} \tag{3.59}$$

Nonetheless, this metric only assesses items in pairs, yet it's also crucial to consider each item's role within the entire universe of items. For instance, consider two securities with a low correlation, but their interactions with the rest of the securities might be similar. For example, Asset 1 and Asset 2 have a low correlation, but both Asset 1's and Asset 2's correlations with Asset 3 are similar. In this case, it would be insightful to view Asset 1 and Asset 2 as more similar than first assumed. Thereby, the Euclidian distance matrix $\widetilde{\mathbf{D}}$, computed as the Euclidean distance among the columns of the primary distance matrix **D**, is also defined as distance of distances. This is also the main distance metric in the original HRP approach (López De Prado, 2016):

$$\tilde{d}_{ij} = \tilde{d}[D_i, D_j] = \sqrt{\sum_{n=1}^{N} (d_{ni} - d_{nj})^2} \tag{3.60}$$

The final distance matrix, denoted as $\widetilde{\mathbf{D}}$, is used to determine the clusters in the agglomerative hierarchical clustering algorithm.

3.5.2 Linkage Criteria

After finding the distances between potential portfolio assets, further distances between clusters must be found to determine the clusters in the next steps. This is a relatively simple task in the first level of the hierarchy as the clusters consist of single assets, and the intergroup distance between clusters is equal to the asset-based distances. However, there are different methods of using the distance matrix to determine the distances between multi-asset clusters, called *linkage methods*. Below, four common linkage methods are explained.

Single Linkage

According to the single linkage method, the distance between two clusters is the minimum of the distances between any two points in the clusters. If C_i and C_j are two clusters, the distance between them is calculated as

$$d_{C_i,C_j} = \min_{x,y} \{D(x,y) \mid x \in C_i, y \in C_j\} \tag{3.61}$$

Single linkage is mathematically cheap and a straightforward method; however, it is susceptible to mistakes as it only needs one pair of close assets, i and i', $i \in x, i' \in y$, to state that the clusters x and y are to be grouped together. This can lead to the "chaining" phenomenon and violate the "compactness" property that all observations within each cluster tend to be similar to one another (Hastie et al., 2009).

Complete Linkage

According to the complete linkage method, the distance between two clusters is the maximum of the distances between any two points in the clusters. If C_i and C_j are two clusters, the distance between them is calculated as

$$d_{C_i,C_j} = \max_{x,y} \{D(x,y) \mid x \in C_i, y \in C_j\} \tag{3.62}$$

Complete linkage works in the opposite way to the single linkage method. This method usually produces compact clusters of similar size; however, it is sensitive to the outliers, as one distant pair of assets will not allow the clusters to be grouped together (Raffinot, 2017). It can produce clusters that violate the "closeness" property; that is, observations in a cluster must be closer to members of their own cluster than they are to any member of other clusters (Hastie et al., 2009).

Average Linkage

According to the average linkage method, the distance between two clusters is the average distance between all pairs of observations in the clusters. If C_i and C_j are two clusters, the distance between them is calculated as

$$d_{C_i,C_j} = \operatorname*{mean}_{x,y} \{D(x,y) \mid x \in C_i, y \in C_j\} \tag{3.63}$$

Average linkage can produce compact clusters and make them relatively far apart. However, it is variant to monotone transformations and, therefore, is not preferred over complete or single linkage methods (Hastie et al., 2009).

Ward's Method

According to Ward's Method, the grouped clusters should be the ones that lead to minimum increase in the variance of the resulting cluster (Ward, 1963). Thus, the distance between two clusters is calculated as the increase in the squared error that results from merging two clusters. If C_i and C_j are two clusters, and m_i and m_j are the number of observations, respectively, the distance between them is calculated as

$$d_{C_i,C_j} = \frac{m_i m_j}{m_i + m_j} \left\| c_i - c_j \right\|^2 \tag{3.64}$$

where c_i and c_j are the centroids of the clusters, defined by Ward as

$$c_i = \frac{1}{m_i} \sum_{x \in C_i} x \tag{3.65}$$

Ward's method, although stated by many authors as the most reliable method because of its robustness to noise and outliers, is biased in creating large clusters (Raffinot, 2017).

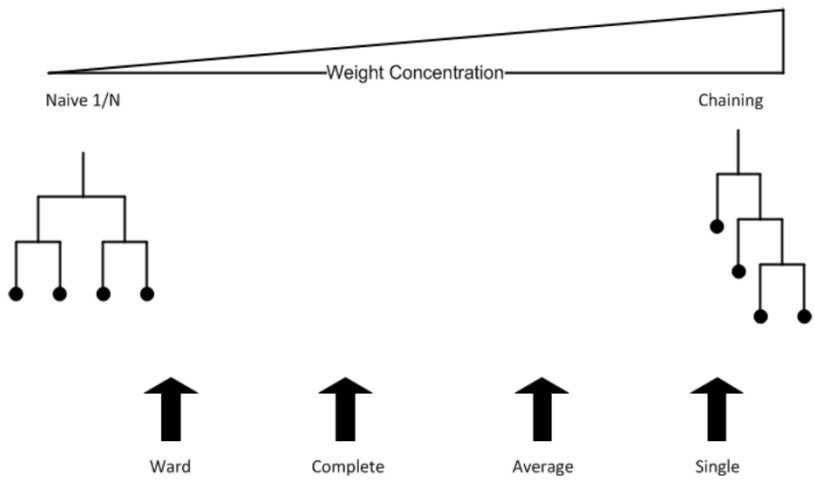

Figure 3.6: Weight concentration of linkages by Papenbrock (2011)

In the study conducted by Papenbrock (2011), single linkage and Ward's method were found to yield better results in terms of performance. Among these two methods, the single linkage method was seen to produce but a low correlation between clusters, whereas Ward's method built balanced clusters. However, as Papenbrock (2011) has explained in his study, and it is depicted in Figure 3.6, weight concentration is highest when constructed by single linkage, in contrast to Ward's method which provides low concentrations.

3.5.3 Dendrogram

An agglomerative hierarchical clustering algorithm forms a hierarchical structure in the shape of a binary tree, usually demonstrated using a *dendrogram*. Dendrogram is a useful visualization tool that shows the grouping of clusters and the order of groupings. It also gives information about the dissimilarity between clusters. An example of a dendrogram is illustrated in Figure 3.7.

In the dendrogram below, every level represents a merging point for two clusters, starting with the ones at the bottom. The numbers at the bottom show single assets (or singleton clusters), and the most similar ones are clustered first at the lowest branch. In the final step, all assets are merged into one cluster that is shown at the top. Moreover, the y-axis shows the distance between two clusters defined by the linkage criterion. This is also called the *cophenetic distance* (Raffinot, 2018). Given that the single and ward linkages are two extremes shown in the picture before, we depict them here, and the visual presentation for average and complete linkages can be found in Appendix A.

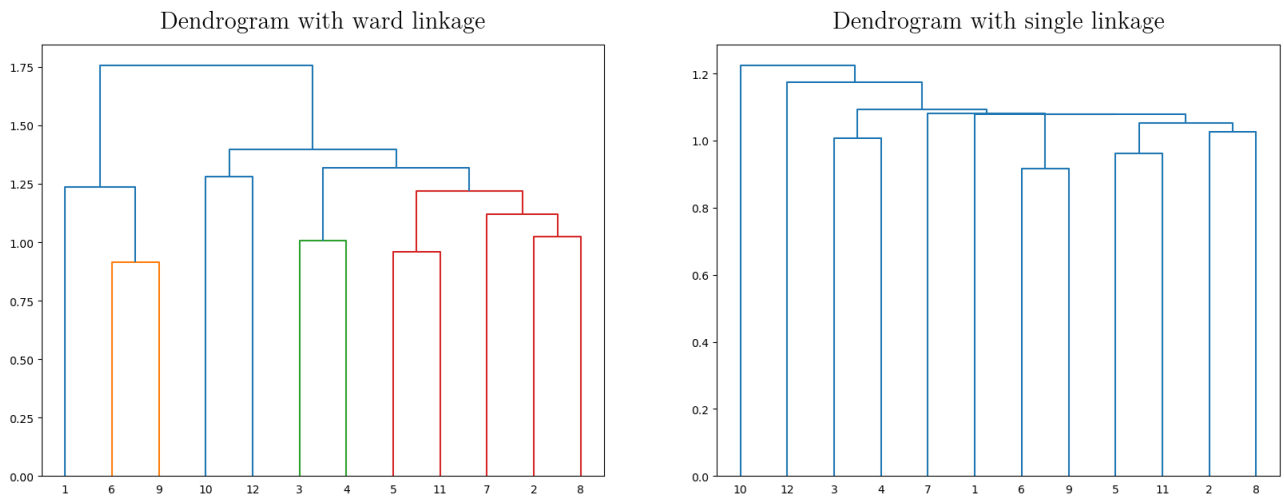

Figure 3.7: Dendrogram example using Ward's method (left) and single linkage (right)

3.5.4 Hierarchical Structure in Portfolio Optimization

The main reason for choosing hierarchical clustering to optimize the portfolio is its usefulness for financial data, enhancing the parameters, and filtering techniques (Hastie et al., 2009). Continuing this exploration, more recent studies have further highlighted the crucial role of clustering in financial portfolio management.

According to López De Prado (2016), the correlation matrix lacks the necessary hierarchical structure, and correspondingly, all assets are assumed to be interchangeable with each other. This unintuitive approach is consistent with traditional portfolio methods, where all substitution opportunities, that is, $\frac{1}{2}N(N-1)$ relationships between N assets, are evaluated. This can be visually shown in Figure 3.8, where nodes of 50 assets create 1,225 relationships, called *the complete graph* by López De Prado (2016).

However, in a myriad of stocks to include in the portfolio, if one needs to group the investment opportunities so that they can be substitutable, in the first stage, the portfolio components that are similar in almost every feature will be clustered together. López De Prado (2016) explains it, *"In deciding the allocation to a large publicly-traded U.S. financial stock like J.P. Morgan, we will consider adding or reducing the allocation to another large publicly-traded U.S. bank like Goldman Sachs, rather than a small community bank in Switzerland, …"*. Here, he also infers that the cluster that holds two U.S. banks will eventually be clustered with a small community bank in Switzerland in higher levels of hierarchy.

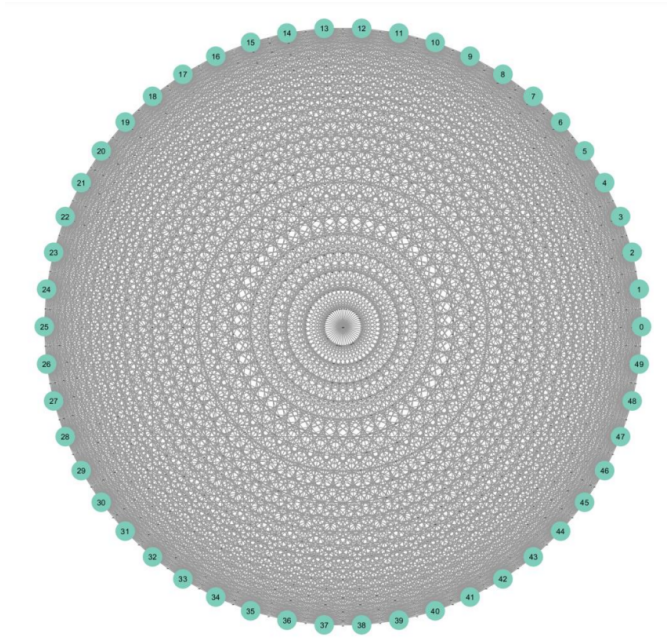

Figure 3.8: The complete graph structure by López De Prado (2016)

Consistent with this view, one does not need to give the assets substitution chances with all other portfolio components. To prevent this, portfolio construction must prefer a tree structure, as demonstrated in Figure 3.9, where there are only $N - 1$ relationships for N assets, and only the assets that make up a cluster together can be substituted with each other.

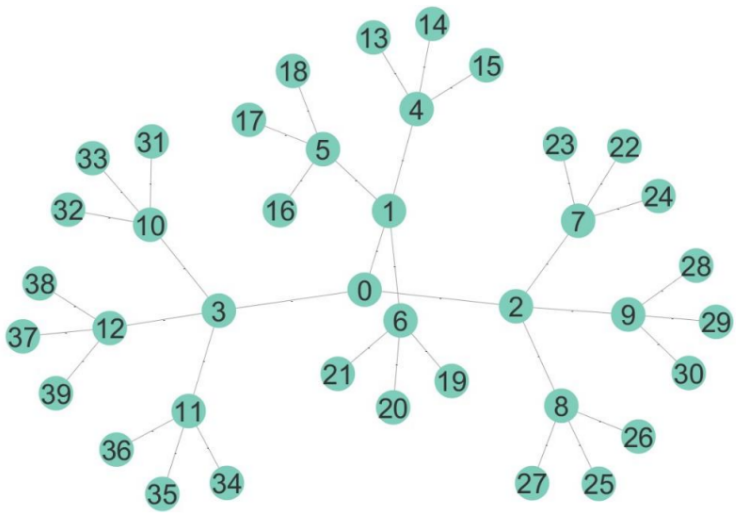

Figure 3.9: Tree graph structure by López De Prado (2016)

3.6 Hierarchical Risk Parity

This section outlines López de Prado's (2016) hierarchical risk parity method from his seminal 2016 work, providing a rationale for its inception and an overview of its structured three-phase approach. It then delves into each phase in detail across subsequent sections. It is important to acknowledge that the initial HRP methodology also presented certain limitations, however, this section will present the model as it was originally formulated.

3.6.1 General Overview

The HRP method targets portfolio optimization issues like instability, concentration, and underperformance. Distinct from conventional approaches, the HRP directly mitigates instability issues by sidestepping the need to predict returns or invert the covariance matrix. It also tackles the concentration problem by moving away from the minimum-variance strategy's tendency towards extreme concentration. Moreover, HRP has been shown to potentially outperform simple 1/N portfolios, contrary to DeMiguel et al.'s (2009) findings, addressing the underperformance issue. The HRP approach is intuitive, recognizing assets as complements or substitutes based on more than just quantitative metrics. Thus, the approach incorporates a notion of hierarchy. The algorithm operates through three stages:

1) tree clustering
2) quasi-diagonalization
3) (naïve) recursive bisection

The three-step approach offers a structured yet flexible framework for portfolio construction that enhances stability and performance. However, it should be noted that the recursive bisection method only considers leaf-node order, ignoring the hierarchical clustering's depth or relationships (López De Prado, 2016). Consequently, the binary tree structure depends on asset count rather than the hierarchy, leading to its characterization as naïve recursive bisection.

Step 1 – Tree Clustering

At the algorithm's initial phase, assets are grouped into a hierarchical cluster structure through agglomerative clustering. Each cluster is formed by combining two elements, which could either be a single asset or a pre-existing cluster of assets. In this process, correlation is used as the

primary measure for assessing similarities among assets in this clustering process (López De Prado, 2016). López De Prado suggests converting the correlation matrix into a distance matrix $\mathbf{D}$, and then to the Euclidian distance matrix $\tilde{\mathbf{D}}$, as presented in subsection 3.5.1. This can be shown as

$$\tilde{\mathbf{D}} = \{\tilde{d}_{ij}\}_{i.j=1,\dots,N} \text{ where } \tilde{d}_{ij} = \sqrt{\sum_{n=1}^{N} (d_{ni} - d_{nj})^2} \tag{3.66}$$

In the original paper (López De Prado, 2016), the author has chosen to use the single linkage approach, therefore, this approach will be explained in detail. The practical example below is the authors' creation to visualize the single linkage clustering.

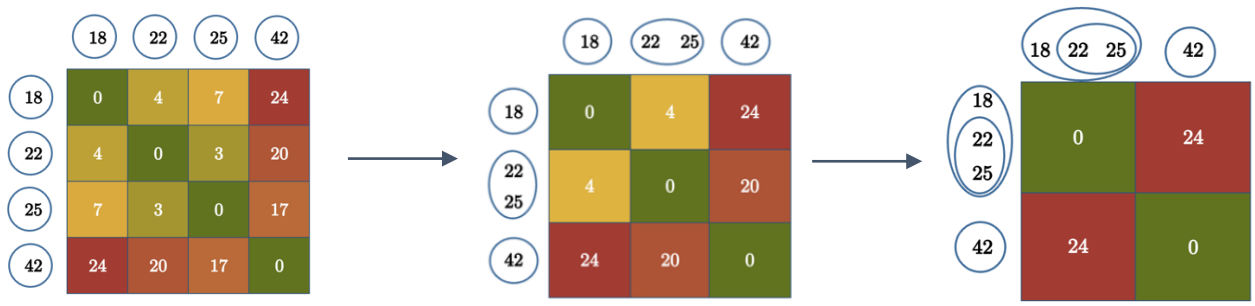

Figure 3.10: Authors' creation of the single linkage clustering

Assume we have four different numbers (*18, 22, 25, 42)* displayed in the distance matrix as depicted in Figure 3.10. The minimum distance can be found between numbers *22* and *25* with the value of 3. Hence, step 1 indicates clustering values *22* and *25* together. Now, it is seen that the minimum distance between *18*, *(22,25),* and *42* is 4. Therefore, in step 2, *18* and *(22,25)* are clustered together into a cluster of (*18, (22, 25)).* Furthermore, a dendrogram can be utilized to visually represent the concept of hierarchical clustering. Figure 3.11 depicts a dendrogram that is based on a hierarchical structure shown in Figure 3.10.

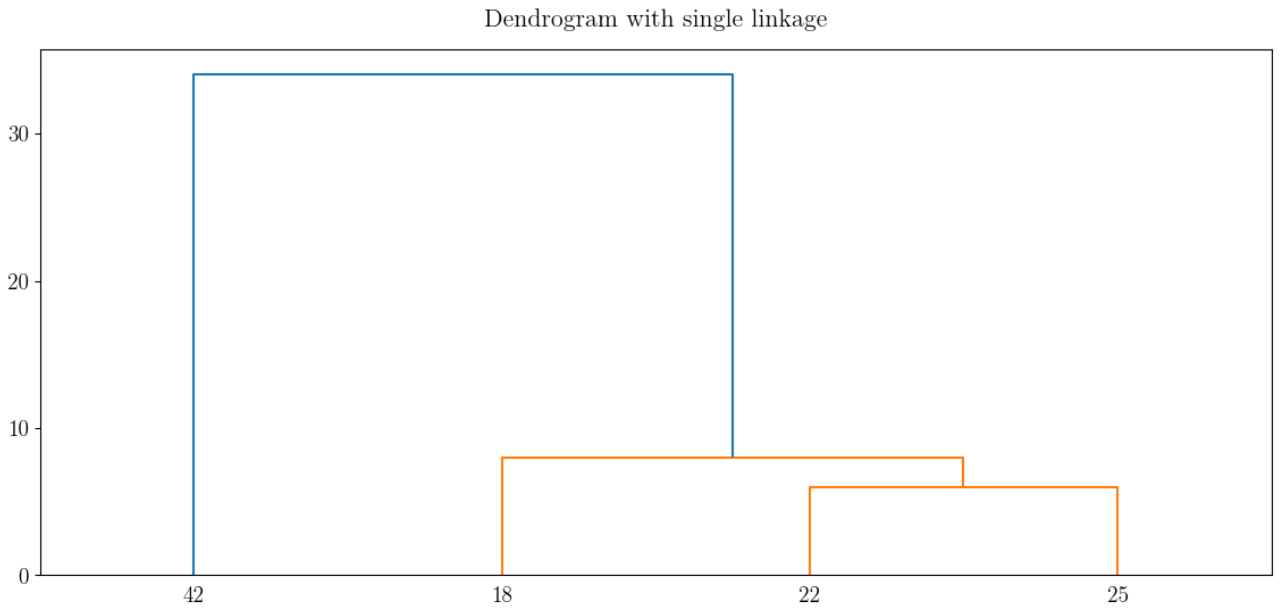

Figure 3.11: Dendrogram based on the single linkage hierarchical clustering

It's important to recognize that while the example provided in this subsection is demonstrated using the single linkage clustering method, other linkage methods can be applied in a similar manner. The algorithm for the tree clustering step of the HRP can be found in Appendix B.

Step 2 – Quasi-diagonalization

The second phase of the HRP approach focuses on quasi-diagonalization, also known as matrix seriation, which is a statistical technique for reordering the covariance matrix to enhance portfolio optimization. As outlined by Eisen et al. (1998) and López De Prado (2016), this method efficiently groups assets based on their similarities derived from hierarchical clustering. Inverse-variance allocation, which is essential to this phase, uses the rearranged matrix to distribute investments proportionately to the inverse of their variances, prioritizing assets with lower volatility for a more stable portfolio (López De Prado, 2016). By arranging similar assets closely and distancing dissimilar ones, it optimizes the allocation process, minimizing the variance among uncorrelated assets and thus justifying its application for improved portfolio performance. Figure 3.12 depicts the effects of matrix seriation for S&P 500 data.

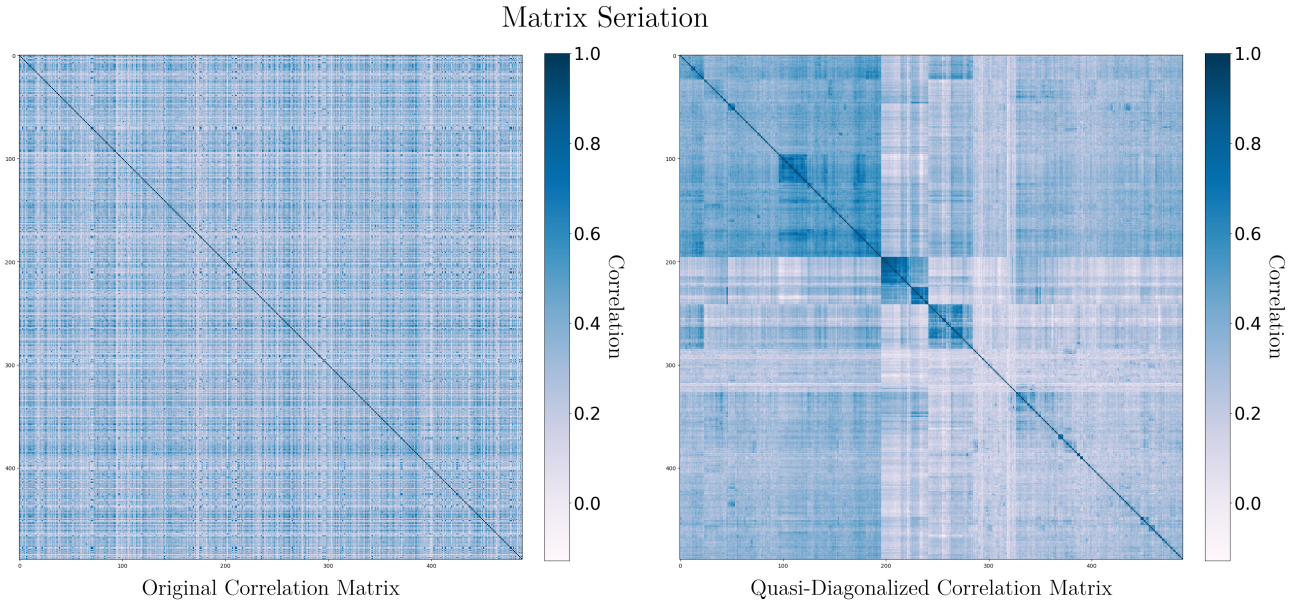

Figure 3.12: Correlation matrix before and after matrix seriation

The above image illustrates the correlation matrix after seriation, highlighting a clearer clustering structure of assets. Initially, clusters are scattered in small groups, but after quasi-diagonalization, their organization and grouping become significantly more defined.

Step 3 – (Naïve) Recursive Bisection

Recursive bisection marks the final, pivotal stage of our algorithm, where the algorithm systematically assigns the actual weights to assets in a hierarchical, top-down fashion. This process is essential for determining the precise allocation within the portfolio. The process begins with dividing each cluster into sub-clusters by beginning with the largest cluster and working it down the tree in a top-down manner. Subsequently, recursive bisection uses the quasi-diagonalized covariance matrix to recurse into clusters.

Setting the list of items, L, and assigning unit weights, $W = 1$, to all items are the first steps for initializing the algorithm:

$$L = \{L_0\} \ with \ L_0 = \{n\}_{n=1,\dots N} \tag{3.67}$$

$$W_n = 1, \forall_n = \{1, \dots N\} \tag{3.68}$$

The naïve recursive bisection approach divides clusters into two sub-clusters, L_1 and L_2, and uses a binary tree from top to bottom to allocate weight. It uses the asset order obtained in the second step, rather than the initial tree structure, to distribute weights uniformly among

sub-clusters during the portfolio construction process. Multiple authors (Huang, 2020; Lau et al., 2017; Raffinot, 2018) have all pointed out that the hierarchical structure created in the first stage of the process is not utilized in the third stage.

When the algorithm divides a cluster into two new clusters, it first allocates the intra-cluster asset weights using a top-down inverse-variance method. This requires calculating the variance of the clusters and can be shown as

$$Var\left(L_i^{(j)}\right) = \mathbf{w}_i^{(j)\top}\boldsymbol{\Sigma}_i^S\mathbf{w}_i^{(j)} \tag{3.69}$$

where $L_i^{(j)}$ refers to sub-cluster j of the cluster i, $\mathbf{w}_i^{(j)}$ represents the allocation weights assigned to the components within a particular sub-cluster, and $\boldsymbol{\Sigma}_i^S$ depicts the covariance matrix amongst the securities of the sub-cluster.

The allocation weights for assets within a cluster, $\mathbf{w}_i^{(j)}$, is calculated using the intra-cluster inverse-variance allocation method in a bottom-up manner:

$$w_i = \frac{1/\sigma_i^2}{\sum_{j=1}^{N} 1/\sigma_j^2} \tag{3.70}$$

The inverse-variance allocation method is explained in greater detail in the subsection 3.2.6. Subsequently, final weights for each sub-cluster are adjusted with a split factor, α_i, and calculated through the inverse-variance allocation method applied between the clusters. In this case, let's take two sub-clusters L_1 and L_2, therefore, their split factors can be shown as

$$\alpha_1 = 1 - \frac{Var_1}{Var_1 + Var_2} \tag{3.71}$$

$$\alpha_2 = 1 - \alpha_1 \tag{3.72}$$

where Var_1 and Var_2 indicate clusters L_1 and L_2 variance, respectively. Using the split factor, the asset weights inside the sub-clusters, W_1 and W_2, which were assigned the same weights ($W_n = 1$), are updated:

$$W_1 := W_1\,\alpha_1 \tag{3.73}$$

$$W_2 := W_2\,\alpha_2 \tag{3.74}$$

The algorithm sequentially breaks down the main cluster containing all assets into individual clusters per asset, applying weights based on intra- and inter-cluster risks. This recursive approach continues until the entire hierarchy is segmented, successfully assigning weights to all assets.

In a hypothetical example with a constant factor $\alpha = 0.4$ for a four-asset portfolio, the weights would be updated this way in each stage of clustering:

$$[1, 1, 1, 1] \rightarrow [0.4, 0.4, 0.6, 0.6] \rightarrow [0.16, 0.24, 0.24, 0.36]$$

The algorithm for the recursive bisection step of HRP can be found in Appendix B.

4 Data

The Data chapter is organized into three main components. We start by explaining the rationale behind our selection of the constituents of the S&P 500 index. Following this, we describe the sources from which the data utilized in this thesis was acquired. Finally, we elaborate on the principal stages undertaken during our data-cleaning procedure.

4.1 Data Selection

The data used in this thesis is the price series of S&P 500 constituents updated constantly with joiners and leavers. Acknowledging the S&P 500's significant focus on the technology sector, its selection was driven by the index's comprehensive representation of the current stock market environment. The S&P 500 exemplifies the world's largest and most fluid capital market, thereby eliminating currency risk and streamlining data analysis with its singular, impactful scope. With the U.S. market capitalization reaching $49 trillion, of which $40 trillion is accounted for by the S&P 500 firms, this index represents approximately 37% of the global equity market capitalization as of the end of 2023 (Securities Industry and Financial Markets Association, 2024).

The dataset covers from January 3rd, 1993, to December 29th, 2023, and includes key financial events that provide a solid background for the study. This period includes the dot-com bubble crash, the 2007-2008 Global Financial Crisis, and the economic consequences of the COVID-19 epidemic. Furthermore, by covering every previous member of the S&P 500, the analysis eliminates survivorship bias, which might skew results by focusing just on the 'winners' and eliminating the 'losers'. This comprehensive method not only evaluates the robustness of financial models and tactics over economic cycles but also ensures an unbiased study that encompasses the whole range of market variables, preventing success from being overstated.

4.2 Data Sourcing

The dataset encompasses all stocks included in the S&P 500 index from January 3rd, 1993, until December 29th, 2023, drawing from several primary sources:

1. Primary and Master Data Acquisition: Daily price data, along with the master data detailing each stock's origin, operating country, currency, first price date, delisting date, and identifiers (e.g., security ID, ISIN number), were retrieved using Datastream Financial Database.
2. S&P 500 Constituents List: A comprehensive list of stocks included in the S&P 500 from January 3rd, 1995, until December 29th, 2023, complete with their ric_id (Refinitiv Identification Code), security, and ISIN numbers, was obtained from Refinitiv Eikon database. The analysis assumes that stocks are added to or removed from the index at the start of the trading day. This assumption simplifies the study, aligning with the common industry practice of updating indices at the day's beginning, which ensures clarity and consistency in the evaluation of constituent changes.
3. Supplementary Price Data: For securities retrieved from Datastream, which lack price data, additional price information was retrieved from Bloomberg Terminal.
4. Risk-Free Rate Data: In this thesis, the risk-free rate benchmark was established using the daily rates of 3-month U.S. Treasury bills, as recorded in the Federal Reserve Economic Data (FRED) database. The short duration of these securities minimizes price volatility and interest rate risk, aligning them closely with the theoretical concept of a risk-free asset, which presupposes no default or price volatility risk. Furthermore, the rates are determined in the secondary market, where securities are traded among investors after being issued by the government in the primary market. This ensures the rate accurately reflects current market conditions and investor sentiment, making it an apt measure for risk-free returns.

The selection of price data aligns with the seminal HRP analysis (López De Prado, 2016) and is well-suited to the scope of this thesis.

4.3 Data Cleaning

The preparatory phase involved a meticulous data-cleaning process to ensure the integrity and usability of the dataset for portfolio construction. For clarity and ease of understanding, we have divided this procedure into three main parts in this section. While these processes are described chronologically, it is crucial to highlight that in fact, they frequently happened

concurrently and iteratively, reflecting the dynamic and complex aspects of data management. The lists of stocks excluded at each step are detailed in Appendix C.

First Step – Data Aggregation

The initial stage of data integration involved consolidating all existing information about securities into comprehensive master data. This dataset identified each security and indexed them according to their dates of inclusion and exclusion from the S&P 500 index. Subsequently, there arose a need to refine this dataset to produce a net list of assets. Although the Refinitiv Eikon database already accounts for companies changing their ticker symbols over time by representing these variations with a single issue, there were still remaining duplicates, and the initial step was to merge them under a single issue. Secondly, redundant entries for companies with multiple issues were eliminated, focusing on those that showed longer activity in the S&P 500 index and those with substantial liquidity. This elimination was crucial to avoid high correlations among multiple issues of the same company. Following this, the master dataset was utilized to filter the extensive price data to include only the relevant stocks, thereby optimizing the dataset for further analysis.

From the original investment universe, which comprised 1,216 securities listed in the index at any time from 1995 to 2023, the process merged the duplicate entries of 8 securities and removed 16 securities that represented non-primary issues of companies.

Second Step – Data Retrieval

In this phase, we enhanced data integrity by identifying and excluding securities with incomplete data records. Of the original securities, those lacking complete data entries for required periods were omitted from further analysis to maintain consistency and reliability. Among the remaining 1,192 securities from the first step, 13 were found to have incomplete data, resulting in their full or partial exclusion from subsequent analysis. Specifically, data could not be located for 10 of these securities, and 3 did not have available data for certain periods of their inclusion in the S&P 500. To preserve the integrity and consistency of the thesis, these securities, or their incomplete data segments, were omitted from further analysis. Of the 1,182 securities advanced to the next phase, data for 1,170 were sourced from Datastream and the remaining 12 from Bloomberg.

Third Step – Data Gap Analysis

The third step of the data cleaning process focused on data availability verification as we investigated the dates with missing stock values. The 49 dataset's instances of absent data can largely be traced to the observance of U.S. public holidays and significant events that caused market closures, shedding light on periods of trading inactivity. The data notably excludes:

- Presidents' Day, Memorial Day, and Independence Day, each accounting for 10 missing days throughout the observed period (30 in total),
- Thanksgiving Day closures contribute to another 9 days of missing data,
- Martin Luther King Jr. Day saw the markets closed on 5 different occasions,
- A unique closure occurred on Christmas Eve of 1999,
- A single-day closure for the funeral of President Nixon (April 27th, 1994),
- A 3-day hiatus during and following the events of 9/11.

These data gaps underscore how holidays and significant events play a role in market dynamics, a key factor considered in our study's data assessment. However, because some stock exchanges work half-day or are not affected by a holiday on these days, some stocks still have price data. Nevertheless, they were excluded for a fair comparison.

Additionally, after the exclusion of the above-mentioned dates, to refine our dataset for accuracy, securities exhibiting prolonged inactivity were excluded. This approach resulted in the exclusion of securities with missing price data for over two consecutive years or for more than half of the required days. This resulted in the removal of 59 securities, including instances where the absence extended beyond a thousand days.

To ensure data integrity, the data cleaning process was finalized by specifically addressing securities with incomplete historical records. Following the established criteria, we implemented a strategy to address securities with minor gaps in their historical records. Specifically, for every security with fewer than 7 consecutive days of missing data, we utilized a substitution approach by filling these gaps with the previous day's data. This approach ensured continuity in the dataset and minimized the impact of sporadic missing values on our analysis. For those securities that have missing data on more than the specified number of consecutive days, we excluded those exact days from the analysis as they would violate the criteria of having at least two years of price data available prior to their inclusion in the S&P

500[1]. The list of excluded stocks in this step, as well as in the previous steps, can be found in Appendix C.

Outcome

The careful data cleaning technique used resulted in the construction of a large dataset of 1123 securities measuring roughly 370 MB in size, featuring 4,098,220 data points across 5 key fields, including the following: 'sec_id' for security identification in the price data, 'Issue' for the unique security name, 'date' of the recorded price data, 'daily_return' of the stock and 'IntervalStatus' indicating whether the stock was active in S&P 500 index on that date.

The dataset includes 7,300 distinct dates ranging from January 3rd, 1995, to and including December 29th, 2023, thereby yielding an average of 251,7 trading days each year, which is nearly a common estimate in the finance world. It is worth noting that, while our dataset begins in 1993, the first two years serve only as historical data necessary to calculate portfolio parameters. Our data cleaning steps reveal that the average number of securities observed on the first day of each month over a 29-year period is 481.

However, despite our systematic approach to include all historical and current S&P 500 constituents, we acknowledge the presence of inherent flaws in our data methodology. These flaws include inconsistencies caused by using several data sources, as the data-providing companies may use different methodologies for data gathering. Furthermore, our assumptions during the data-filling process for stocks with short gaps, as well as the removal of inactive securities, may have a subtle impact on how market dynamics are represented, thereby favoring the "better" doing stocks. Recognizing these possible biases is critical for a fair interpretation of our findings, emphasizing the importance of continued attention and modification of our data management technique.

[1] A further explanation of our walk-forward methodology can be found in section 5.2.

5 Methodology

The methodology chapter outlines the approaches utilized in this thesis. It begins with an overview of our research methodology, followed by an explanation of our approach to walk-forward analysis. Subsequently, we discuss the implementation of the Modified HRP approach. The chapter concludes with an analysis of the statistical measures used to evaluate and compare the performance of various portfolio strategies.

5.1 Research Method and Perspective

Given the analytical requirements of our research, a quantitative approach was chosen to address the research topic effectively. This method emphasizes data quantification for analyzing and testing hypotheses that are formulated based on current literature, aligning with the research questions and their subcomponents (Bryman et al., 2022). Therefore, the thesis employs a deductive strategy where a theory-driven hypothesis is developed and subsequently empirically tested.

This thesis focuses on the question of portfolio optimization from the perspective of institutional investors. Institutional investors, including entities such as pension funds, insurance companies, and mutual funds aggregate capital from various individuals or organizations to invest in a range of financial instruments, representing their clients. Due to the scope of this thesis, multiple restrictions have been implemented, which were detailed in section 1.2.

5.2 Walk-Forward Analysis

In our research, we use a walk-forward optimization analysis to backtest portfolio performance, resulting in a realistic and dynamic approach that is consistent with real-world investing situations. A walk-forward analysis consists of two main stages, namely optimization (training) and testing. During optimization, portfolio weights are established using a subset of the data known as training data, which is used to build the sample covariance matrix and expected return. Subsequently, during the testing phase, the performance of alternative methods is evaluated using a distinct set of data called evaluation data (Coqueret & Guida, 2019; Pardo,

2008). The walk-forward analysis typically involves examining a series of consecutive periods, and its visual representation can be seen in Figure 5.1.

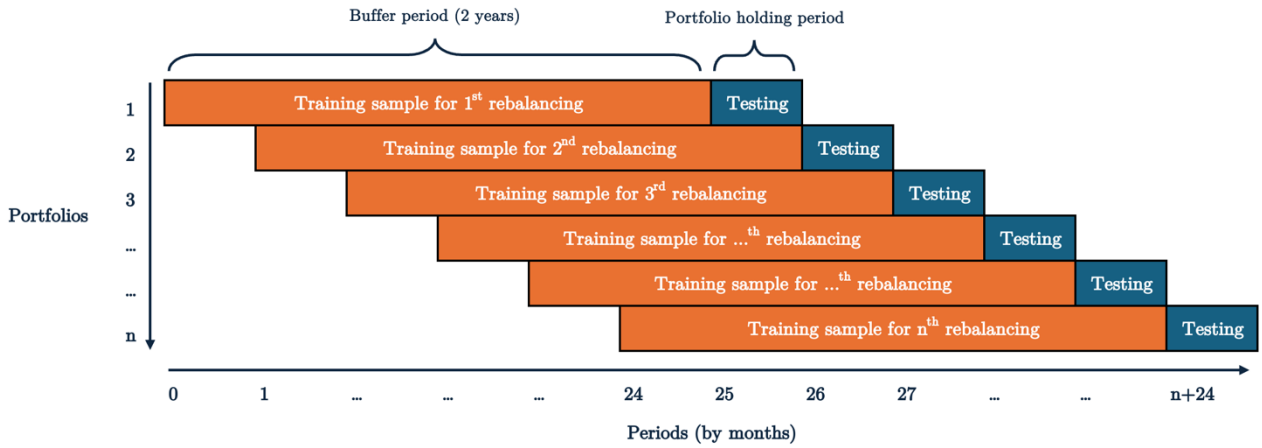

Figure 5.1: Historical walk-forward backtesting with rolling rebalancing windows

To account for the impacts of leap years, we use a span of 2 calendar years instead of a constant 730-day interval during the optimization phase. The testing phase is conducted monthly. This more regular rebalancing schedule is motivated by the belief that more granular rebalancing gives a more accurate assessment of portfolio performance, thereby allowing us to fine-tune the portfolio in response to changing market conditions. This approach leads to obtaining rolling out-of-sample results. Though our dataset began in 1993, we only used the first two years to determine the initial portfolio weights, with no testing data obtained from that period.

However, we must acknowledge that our approach to the amount of out-of-sample data differs from that of some scholars. Pardo (2008) suggests that the testing data should account for a significant amount of the in-sample data, namely 25-35%. Tokat and Wicas (2007), on the other hand, advocate for a more cautious approach, recommending quarterly rebalancing to account for market movements. However, the dynamic nature of the stock market compels several stocks to leave and join the S&P 500 index within a single month. Therefore, it has been determined that quarterly rebalancing may not provide the required accuracy to effectively manage these changes. Our methodology strikes a balance between these views by opting for a monthly testing phase, allowing for responsive portfolio adjustments while managing computational demands. This tailored methodology for walk-forward analysis

ensures that it is both applicable and optimal for dynamic real-world portfolio management situations.

5.3 Modified HRP Approach

In this thesis, we also implement the MHRP approach, outlined by Marat Molyboga (2020). MHRP differs from the original HRP method in three ways:

1. Covariance matrix estimation: MHRP introduces a sophisticated technique, combining exponentially weighted approaches with Ledoit-Wolf shrinking. This modification better accounts for the heteroskedastic structure of financial returns, resulting in more precise covariance matrix estimates (Molyboga, 2020). The covariance matrix's shrinkage estimation is based on the Ledoit and Wolf (2004) theory and is executed in Python using the PyPortfolioOpt package.
2. Asset allocation approach: Unlike the original HRP's inverse-variance method for intra-cluster weight allocation, the MHRP uses an equal volatility (inverse-volatility) allocation technique. This strategy seeks to spread risk more equally across portfolio assets as the variance enhances the risk differences compared to the standard deviation. Therefore, the approach allows for improving diversification and lowering the risk of concentration.
3. Volatility Targeting: The MHRP introduces volatility targeting to achieve constant expected portfolio volatility. Volatility targeting, while useful in managing portfolio risk, includes a dynamic aspect that can make direct comparisons between allocation methods difficult. By excluding it, this thesis more clearly and consistently compares the outcomes of different allocation strategies.

As a result, in this thesis, we implement the first two of Molyboga's (2020) three proposed modifications.

5.4 Statistical Measures

This section presents the statistical measures used in this thesis to compare the outcomes of the portfolio creation. All the calculations have been made in Python using relevant packages or libraries, such as Pandas and Microsoft Excel. These measures are grouped into return, risk, and distribution shape metrics.

5.4.1 Return Measures

The return metrics include Annualized Returns, Cumulative Returns, Sharpe Ratio, Adjusted Sharpe Ratio (ASR), and Average Turnover per Rebalancing which is the necessary component of ASR.

Annualized Returns

Annualized returns are computed to offer a consistent measure of returns throughout different time periods, making them comparable after accounting for the duration of the investment period. The formula for calculating geometric annualized returns is shown below:

$$r_{annual} = \left(\prod_{t=1}^{T}\left(1 + r_{i,t}\right)\right)^{\frac{252}{T}} - 1 \tag{5.1}$$

where $r_{i,t}$ is the daily return of the asset i at time t, and T defines number of periods. In the financial world, 252 is commonly used as an approximation for the average number of trading days each year.

Cumulative Returns

The cumulative return reflects the combined impact of price fluctuations on the value of the investment and is shown as

$$Cumulative\ returns\ = (1 + r_{t+1}) \cdot (1 + r_{t+2}) \cdot (1 + r_{t+3})\ \ldots (1 + r_{t+n}) - 1 \tag{5.2}$$

where r is a linear return for each day in the year of a given stock and is calculated as

$$r_t = \frac{P_t}{P_{t-1}} - 1 \tag{5.3}$$

Sharpe Ratio

The Sharpe ratio is a commonly used measure for evaluating investment performance while accounting for its risk. It signifies the extra return gained by an investor for each unit of additional risk undertaken. The formula for calculating the Sharpe ratio is as follows:

$$SR = \frac{r_{annual}^{pf} - r_f}{\sigma_{annual}^{pf}} \tag{5.4}$$

where r_{annual}^{pf} is the annualized return of the portfolio, r_f is the annualized risk-free rate, and σ_{annual}^{pf} is the annualized standard deviation of the portfolio's excess returns. This formula suggests that both the returns and the volatility should be annualized to maintain consistency in measurement, especially when comparing investments over different time frames. This approach differs from the original approach by Sharpe (1964) but is used in our thesis for comparison.

Given that the true population mean and the standard deviation are unknown, they must be estimated from the sample data (Lo, 2002) as

$$SR = \frac{\hat{r}_{annual}^{pf} - r_f}{\hat{\sigma}_{annual}^{pf}} \tag{5.5}$$

A higher Sharpe ratio is desirable, ceteris paribus.

Average Turnover per Rebalancing

Turnover quantifies the extent of changes in a portfolio's allocations. In light of Raffinot's comprehensive exploration of HRP (2017, 2018), we have chosen to employ the methods outlined by the author for calculating average turnover per rebalancing. Calculations for ATR are as follows[1]

$$ATR = \frac{1}{F}\sum_{t=2}^{F}\sum_{i=1}^{N}\left|w_{i,t} - w_{i,t-1}\right| \tag{5.6}$$

[1] We amended Raffinot's (2018) formula for determining portfolio turnover to include a summing of all assets, which the formula initially lacked.

where F represents the total count of rebalances undertaken, N represents the number of securities in rebalancing periods, and $w_{i,t}$ denotes the asset i's weight at time t. ATR, though not a return metric, helps to calculate transaction costs, which are important since they can considerably impact a strategy's profitability. Naturally, accurate transaction cost comparisons are difficult due to varied implementation techniques and investment sizes. However, the ATR gives insight into the relative transaction costs among strategies, with more turnover signifying higher costs (DeMiguel et al., 2009).

Adjusted Sharpe Ratio

Given that this thesis does not delve extensively into transaction costs, a simplified estimation is employed to recognize their impact on the Sharpe ratio. Following Lassance (2021), transaction costs are approximated to be 20 basis points proportional to the ATR, which measures the portfolio's turnover rate. With monthly portfolio rebalancing, the annual transaction cost is derived from the ATR, providing a basic understanding of transaction costs' effect on returns. Therefore, costs can be estimated as

$$TC = ATR \cdot 12 \cdot 0.2\% \tag{5.7}$$

indicating that the adjusted Sharpe ratio (ASR) can be shown as

$$ASR = \frac{\hat{r}^{pf}_{annual} - r_f - TC}{\hat{\sigma}^{pf}_{annual}} \tag{5.8}$$

where TC is the annualized transactions costs and the other parameters are the same as in the Sharpe ratio description.

5.4.2 Risk Measures

This subsection focuses on the risk metrics, such as Annualized Volatility, Maximum Drawdown, Value-at-Risk, Conditional Value-at-Risk, and Sum of Squared Portfolio Weights, as it provides insights into the concentration or diversification of a portfolio.

Annualized Volatility

Volatility is a statistical measure of the spread of returns for a particular securities or market index. It is typically used as a risk indicator and is frequently computed as the standard

deviation of returns. The formula for computing the variance (squared volatility) of daily returns from a sample is

$$\sigma_i^2 = Var(r_i) = \frac{1}{T-1}\sum_{t=1}^{T}\left(r_{i,t} - \overline{r_i}\right)^2 \tag{5.9}$$

where $\overline{r_i}$ indicates average return over period T, and the other notations remain the same as above. Volatility must be annualized to provide a more thorough picture of the risk over a longer period. This is particularly useful for comparing long-term investment risk profiles, and this is the approach used in this thesis. Therefore, the formula for annualized volatility becomes

$$\sigma_{annual} = \sigma_i \cdot \sqrt{252} \tag{5.10}$$

This conversion implies that daily returns are independently and identically distributed, such that annualized volatility represents the expected variability in annual returns based on daily movements. More detailed information on risk measures can be found in section 3.1.

Maximum Drawdown

The concept of maximum drawdown (MDD) serves as a metric for assessing the risk within a portfolio by measuring the largest single drop from peak to trough over a certain period (Chekhlov et al., 2005). Intuitively, the MDD is the greatest loss from a peak before a new peak is achieved, offering a clear picture of the worst-case scenario for an investment. Hence, MDD can be shown as

$$MDD = \left|\frac{Trough\,Value - Peak\,Value}{Peak\,Value}\right| = \max_{\tau\in[0,t]}\frac{(W_\tau - W_t)}{W_\tau} \tag{5.11}$$

Where W_τ represents the peak value of the portfolio before the largest drop within the interval, W_t is the value of the portfolio at the lowest point or trough following the peak value W_τ, and the term $\tau \in [0, t]$ indicates that we are looking for the maximum difference between the peak W_τ and the trough W_t over the interval period $[0, t]$.

While the MDD provides a simple measure of possible downside risk, it does not account for the frequency or duration of drawdowns, which proves crucial for evaluating the investment's entire risk profile. When comparing the two strategies, Strategy **A**'s maximum drawdown of -25% with repeated 15% dips may look less dangerous than Strategy **B**'s single -30% drop.

Despite the lower maximum loss, **A**'s repeated dips may indicate more risk due to ongoing market exposure and investors' stress. While Strategy **B** shows a greater fall, it may represent a more consistent, perhaps less worrying path for long-term investors who are unaffected by short-term swings. Moreover, if two strategies share the same MDD, uncertainty remains regarding the duration required for each strategy to recover to its initial point. This demonstrates that maximum drawdown does not properly represent a strategy's risk and, therefore, should not be used as a single measure of risk.

Value-at-Risk

Value-at-Risk (VaR) is a popular risk indicator that estimates the greatest probable loss that an investment or portfolio may sustain over a certain time horizon with a specific confidence level. In mathematical terms, if the returns are normally distributed, VaR is calculated as follows:

$$VaR = \mu + \sigma N^{-1}(\alpha) \tag{5.12}$$

where α is a confidence level between 0 and 1, and N^{-1} is the inverse of a cumulative normal distribution. Given that portfolio returns typically deviate from a normal distribution, the standard VaR calculation does not provide an accurate risk assessment. Consequently, the historical simulation method is the preferred technique for determining VaR, offering a more effective measure of risk by directly analyzing past return data (Föllmer & Schied, 2011; Hull, 2023). This method accounts for the skewed nature of return distributions, capturing the risk more effectively than the normal-based VaR calculation and be shown as

$$VaR_{\alpha}(X) = -\inf\{x \in \mathbb{R} : F_X(x) > \alpha\} \tag{5.13}$$

where $F_X(x)$ represents the cumulative distribution of X. We have chosen to proceed with this method as it better reflects the real implications of real-world financial data.

Nonetheless, it's important to acknowledge that VaR has limitations. Despite its simplicity, VaR overlooks loss magnitudes over its threshold and lacks coherence as a risk measure since it fails to meet subadditivity, neglecting diversification advantages (Hull, 2023). This feature challenges the coherence of VaR as a risk measure and necessitates careful interpretation within the context of broader risk management strategies.

Expected Shortfall

Conditional Value-at-Risk (CVaR), also known as expected shortfall (ES), improves the Value-at-Risk model by not only estimating the risk of a loss surpassing a specific threshold but also quantifying the expected amount of such severe losses. When normal distribution assumptions hold, CVaR at confidence level α is determined as

$$CVaR = \mu + \sigma \frac{e^{\frac{-Y^2}{2}}}{\sqrt{2\pi} \cdot (1-\alpha)} \tag{5.14}$$

where μ is the mean, σ is the standard deviation, and Y is the α-percentile of a standard normal percentile. Despite this, the historical simulation approach is still preferred because it better depicts the risk profile of financial returns, particularly when they exhibit non-normal behavior (Hull, 2023). In this thesis, CVaR is calculated as the average loss beyond the determined VaR (Munk, 2022) and stated mathematically as

$$CVaR_{(\alpha)} = \frac{1}{\alpha} \int_0^{\alpha} VaR(\alpha, X)\, dx \tag{5.15}$$

CVaR provides a sophisticated risk assessment method that considers all potential losses beyond VaR. Therefore, resulting in a more complete and cohesive risk picture. While more complex, it provides useful insights into severe risk scenarios but requires careful interpretation and a deeper understanding of its computation.

Sum of Squared Portfolio Weights

The sum of squared portfolio weights (SSPW) measures portfolio diversity by tracking how investment weights are dispersed throughout a portfolio. Raffinot (2017, 2018) defined SSPW as the average of squared portfolio weights over a set number of rebalancing periods, and it is shown as

$$SSPW = \frac{1}{F} \sum_{t=2}^{F} \sum_{i=1}^{N} w_{i,t}^2 \tag{5.16}$$

where F represents the total count of rebalances undertaken, N represents the number of securities for that day, and $w_{i,t}$ denotes the asset i's weight at time t. This metric ranges between 0 and 1, with lower values denoting a more evenly distributed, hence diversified,

portfolio. As Blume and Friend (1975) and Goetzmann and Kumar (2008) noted, a lower SSPW suggests a deviation from market portfolio weights, indicating less concentration in particular stocks or sectors.

5.4.3 Distribution Shape Measures

This subsection provides an overview of two significant distribution shape metrics, skewness and kurtosis, to facilitate the understanding of the third (asymmetry) and fourth (tail-heaviness) moments within the return distribution.

Skewness

Skewness provides insight into the asymmetry in the return distribution for financial assets. Technically, skewness is computed using the distribution's third standardized moment. A distribution with no skewness is symmetric and has a skewness of 0. For positively skewed distributions with a longer right tail, the mean and median are both higher than the mode. In contrast, a negative skewness suggests a longer or fatter left tail, implying a higher likelihood of substantial negative returns (Munk, 2022). The return distributions of most financial assets are skewed. Daily stock returns, in particular, frequently exhibit negative skewness, as substantial price falls are generally followed by further declines as investors attempt to reduce losses by closing their positions (Alexander, 2012). Skewness can be defined as

$$Skewness = \sum \frac{(x_i - \bar{x})^3}{ns^3} \tag{5.17}$$

where n is the number of observations, s is the standard deviation, $\bar{x}$ is the mean and x_i represents each individual asset in the same data set. Adjusting for non-asymptotic sample sizes involves using a correction factor to make the skewness calculation more accurate for smaller sample sizes (Joanes & Gill, 1998). This adjustment helps to correct for the bias that smaller samples inherently have and is shown as

$$Skewness = \frac{n}{(n-1)(n-2)} \cdot \sum \frac{(x_i - \bar{x})^3}{ns^3} \tag{5.18}$$

The adapted formula for skewness offers a more accurate reflection of the population's skewness, especially for smaller data sets. However, with larger sample sizes, such as in our dataset, the disparity between these two formulas becomes negligible.

Kurtosis

Kurtosis quantifies the thickness of the tails in a distribution relative to a normal distribution. A distribution with positive excess kurtosis is known as leptokurtic, suggesting a higher possibility of extreme events due to thicker tails. A distribution with negative excess kurtosis, on the other hand, is platykurtic, meaning it has thinner tails and fewer extreme events (Munk, 2022). Financial datasets frequently exhibit leptokurtic behavior; however, kurtosis may vary over time (Albuquerque, 2012), emphasizing the need to address kurtosis in risk assessment. Kurtosis can be shown as

$$Kurtosis = \sum \frac{(x_i - \bar{x})^4}{ns^4} \tag{5.19}$$

where n is the number of observations, s is the standard deviation, $\bar{x}$ is the mean and x_i represents each individual asset in the same data set. Excess kurtosis subtracts three from the conventional kurtosis metric to match it with the normal distribution, allowing for easier comparison of tail risks:

$$Excess\ kurtosis = \sum \frac{(x_i - \bar{x})^4}{ns^4} - 3 \tag{5.20}$$

A positive value then implies a leptokurtic distribution with many extreme outliers, whereas a negative value indicates a platykurtic distribution with fewer outliers. Again, adjusting for the biases (Joanes & Gill, 1998), the kurtosis formula, which is used in this thesis, can be shown as

$$Kurtosis = \frac{n(n-1)}{(n-1)(n-2)(n-3)} \cdot \sum \frac{(x_i - \bar{x})^4}{ns^4} - \frac{3(n-1)^2}{(n-2)(n-3)} \tag{5.21}$$

The Adjusted kurtosis formula takes into account the size of the sample size, akin to adjusted skewness.

6 Results

This section's focus is to present the results—both visual and quantitative—of the analyzed portfolios. For simplicity and clarity, we will employ the following abbreviations: EW (equally weighted), MinVar (minimum-variance), MaxSharpe (maximum Sharpe ratio), IVAR (inverse-variance), IVOL (inverse-volatility), ERC (equal risk contribution), HRP (hierarchical risk parity), and MHRP (modified hierarchical risk parity). Unless specified otherwise, both HRP and MHRP portfolios utilize Ward's linkage. We will first outline the characteristics of the data set. Then, we will explore the composition of weights within each portfolio, followed by an evaluation of return measures and risk metrics.

6.1 Characteristics of the Data Set

With a monthly rebalancing approach, the rolling condition number offers insights into the stability of the sample covariance matrix across time. As illustrated in Figure 6.1, this ratio of the largest to smallest eigenvalues of the correlation matrix is computed for each rebalancing time. A high condition number often points to potential instabilities, especially during market tumult when correlations tend to rise, a phenomenon reflective of "Markowitz's curse". Notably, when the condition number increases, as it does follow significant economic events, it indicates assets exhibit more correlated behavior in response to market dynamics. High condition numbers coincide with periods where diversification is key. Yet, they also highlight times when the dependability of the optimization model may be compromised. The peaks, especially in the aftermath of the 2008 crisis and during the COVID-19 pandemic, indicate times of heightened interconnections among asset returns. These incidents emphasize the likelihood of increased systemic risk, as well as the difficulties of attaining enough diversity during critical times when investors seek to mitigate risk.

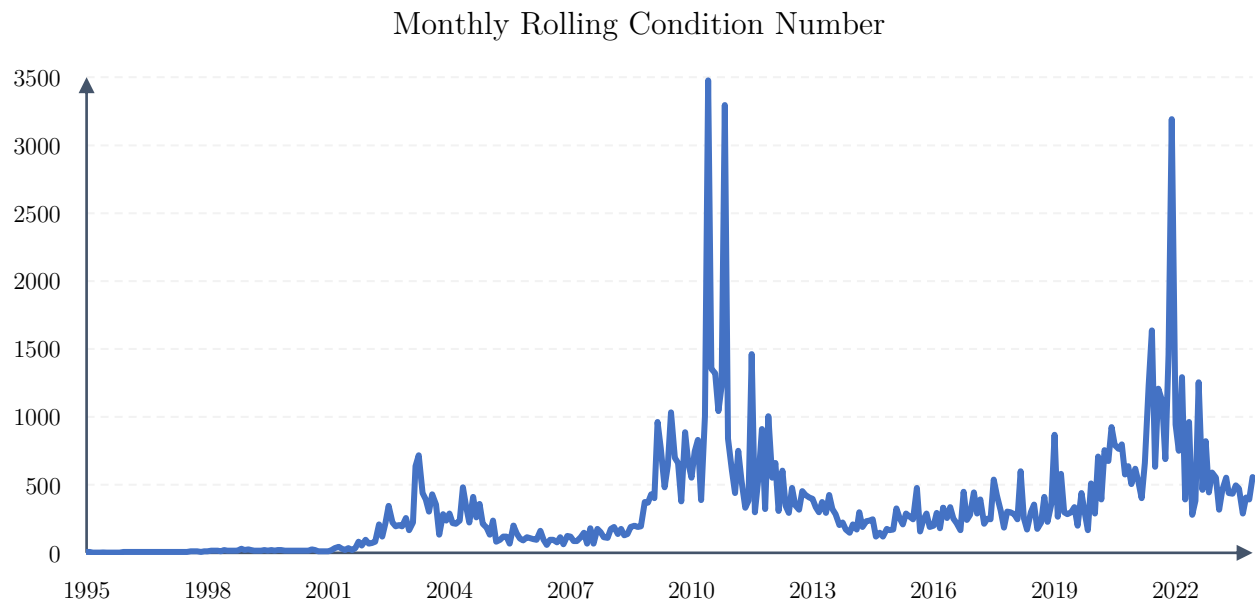

Figure 6.1: Monthly rolling condition number (scaled down by 10,000)

The HRP method, which is central to this thesis, relies on the covariance matrix — a fundamental element for determining asset weights. Given the ever-changing composition of the S&P 500, with over 1200 stocks cycling through since 1995, it is impossible to display a covariance matrix for the entire period. Therefore, the correlation matrix presented herein is selected for a single, random day, specifically May 1st, 2015, to reflect a snapshot of the index at a point in time. The sample correlation matrix for this date offers a more interpretable and nearly equivalent set of insights since correlation is a normalized measure of covariance. Figure 6.2 illustrates the correlation matrix on the given day. On May 1st, 2015, the correlation matrix for S&P 500 components revealed a spectrum of correlation coefficients ranging from negative 0.12 to 0.9, indicating the variety of links between asset pairs. Despite the large number of assets, which makes visually determining pairwise correlation values difficult, we have chosen to identify pairings with extreme correlation values. Hudson City Bancorp and M&T Bank, both in the Financials sector, show a correlation close to 0.9. Similarly, pairs like Xcel Energy and CMS Energy, as well as WEC Energy Group and CMS Energy, exhibit high correlations between 0.85 and 0.9, as they are all within the Utilities sector (MSCI, 2023). On the other hand, expectably low correlations are seen between businesses across diverse sectors and industries. Diamond Offshore Drilling within the Energy sector's Oil and Gas Drilling industry shows negative correlation with Southwest Airlines from the Industrials sector's Airlines industry, reflecting their divergent responses to economic variables. In another pairing,

Genworth Financial of the Financials sector diverges from Whole Foods Market in the Consumer Discretionary sector, thus highlighting the distinct influences on insurance and retail industries. Lastly, Express Scripts Holding from the healthcare sector has little correlation with Pepco Holdings in the Utilities sector's Electric Utilities industry (MSCI, 2023).

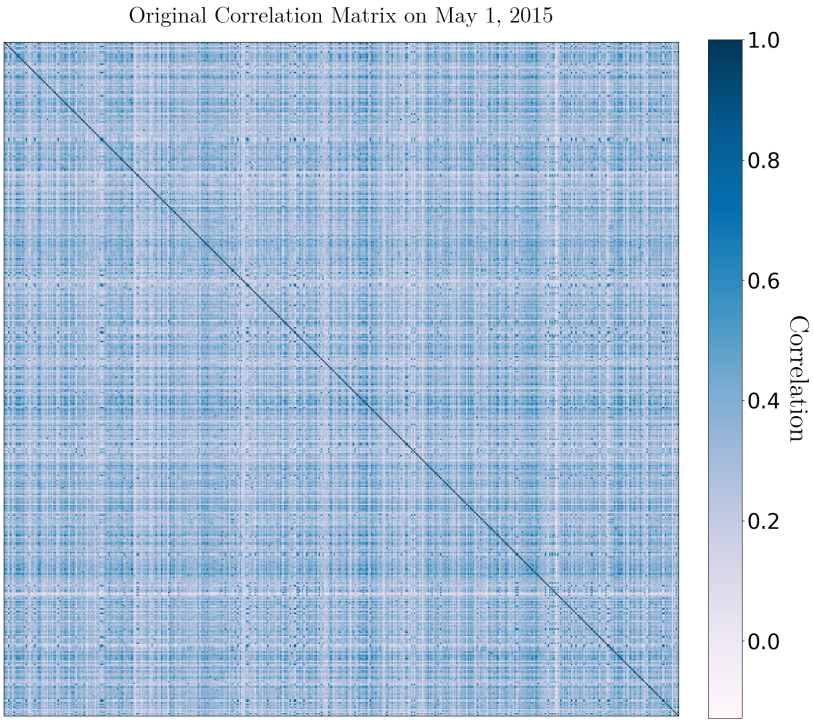

Figure 6.2: Correlation matrix of S&P 500 constituents on May 1st, 2015

In adherence to the HRP strategy (López De Prado, 2016), assets are clustered using single linkage. However, as discussed in subsection 3.5.2, this method can lead to a chaining effect. Raffinot (2018) delves into the chaining effect, where the single linkage method can erroneously group assets due to their slight proximity in risk space, not actual similarity. This results in extended clusters that compromise the method's intent to form groups based on strong financial commonalities, which is essential for creating a diversified portfolio. Therefore, the single linkage approach contradicts López De Prado's (2016) claim that the HRP approach is intuitive and straightforward. In refining the HRP method, Ward's method is utilized for asset clustering to address the chaining issue of single linkage. This method promotes a balanced dendrogram by minimizing the variance within each cluster, leading to groups with genuine financial similarities. Papenbrock (2011) found that single linkage and Ward's method outperformed other methods, with single linkage leading to denser but less correlated clusters and Ward's method creating more balanced groupings. Our analysis primarily promotes

Ward's method. For the S&P 500 on May 1st, 2015, Ward's method aligns with López De Prado's (2016) intention for an intuitive HRP approach, forming clusters that truly reflect the assets' relationships. Figure 6.3 depicts the dendrogram using Ward's method. We have chosen to omit names on the x-axis to ensure a clearer and more comprehensible visual presentation, as there were 489 stocks included on this day.

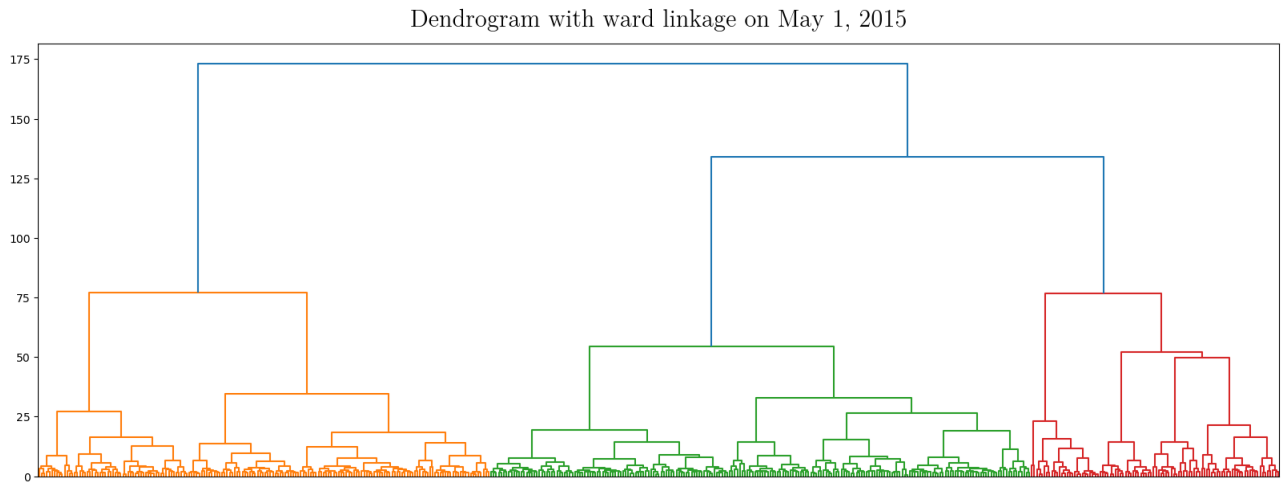

Figure 6.3: Dendrogram using Ward's method for May 1st, 2015

6.2 Weights

After briefly introducing the properties of the data in the previous part, it is appropriate to explore the composition of the built portfolios before proceeding with the statistical measure analysis. On May 1st, 2015, a weight distribution analysis of the HRP and the minimum-variance portfolios revealed significant differences. While the minimum-variance portfolio has concentrated weights in fewer assets, the HRP method uses asset clustering to provide more equitable weights across the asset spectrum. This pattern of distribution is visible in HRP regardless of the linking method used, whether single, complete, average linkage, or Ward's method[1]. In our analysis, as shown in Figure 6.4, HRP with Ward's method demonstrates a broader allocation, which differs from the tight concentration of the minimum-variance portfolio. The inverse-variance (IVAR) technique allocates weights inversely proportionate to the variances of individual assets, resulting in a more balanced weight distribution, comparable

[1] Linkage and method might be used interchangeably in this thesis. However, the correct usage of the term is "method", as in Ward's method, and "linkage" otherwise.

to HRP. To maintain clarity of the visuals, the individual names of the stocks have once more been excluded.

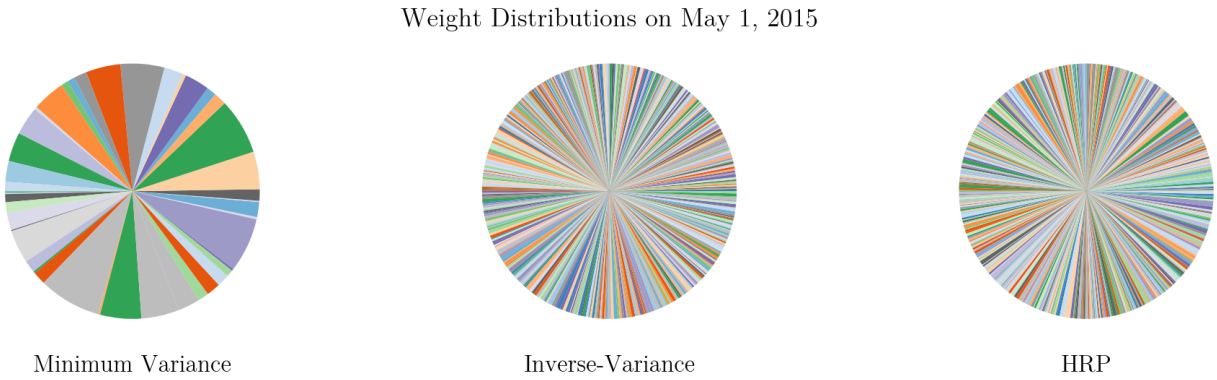

Figure 6.4: Example of weight allocations for three portfolios

Looking closer into each portfolio's weights, we have chosen to illustrate the composition by visualizing the top 15 stocks, ranked by their average weight within each portfolio. These weights are calculated as the mean of all monthly rebalanced weights assigned to each constituent over the study period. Given the lengthy duration of our study, spanning over 29 years, it is expected that the weights of many stocks have varied within their respective portfolios. However, it remains intriguing to see the weight distribution of the largest assets for each portfolio. Figure 6.5 depicts the weight distributions within each portfolio construction method. In the analysis of portfolio weights, we compare assets within their own portfolio (intra-portfolio) rather than across different portfolios (inter-portfolio). This comparison method is justified by first anchoring the greatest weight in each portfolio to a fixed value and then modifying the following weights of the other assets within that portfolio in a downward proportional way. Such an approach ensures that the weight distribution within each portfolio can be examined on a consistent scale. For example, in an EW portfolio, each of the (roughly) 500 assets would generally have a weight of around 0.2%. In contrast, the MinVar or MaxSharpe portfolios may devote more than 5% or 10% of its resources to singular assets. By using this proportional scaling, we maintain a meaningful frame of reference when evaluating the weight evolution of individual assets throughout our study period.

Therefore, in Figure 6.5, the size of a marker indicates how much weight the portfolio has allocated to the asset, thus the bigger the marker, the higher the allocation. As expected, the EW portfolio has consistent allocations across the top 15 assets. In contrast, the MinVar and MaxSharpe portfolios have a high concentration of certain assets, indicating more

concentrated strategies. The IVAR, IVOL, ERC, HRP, and MHRP portfolios have relatively balanced assets of these top assets.

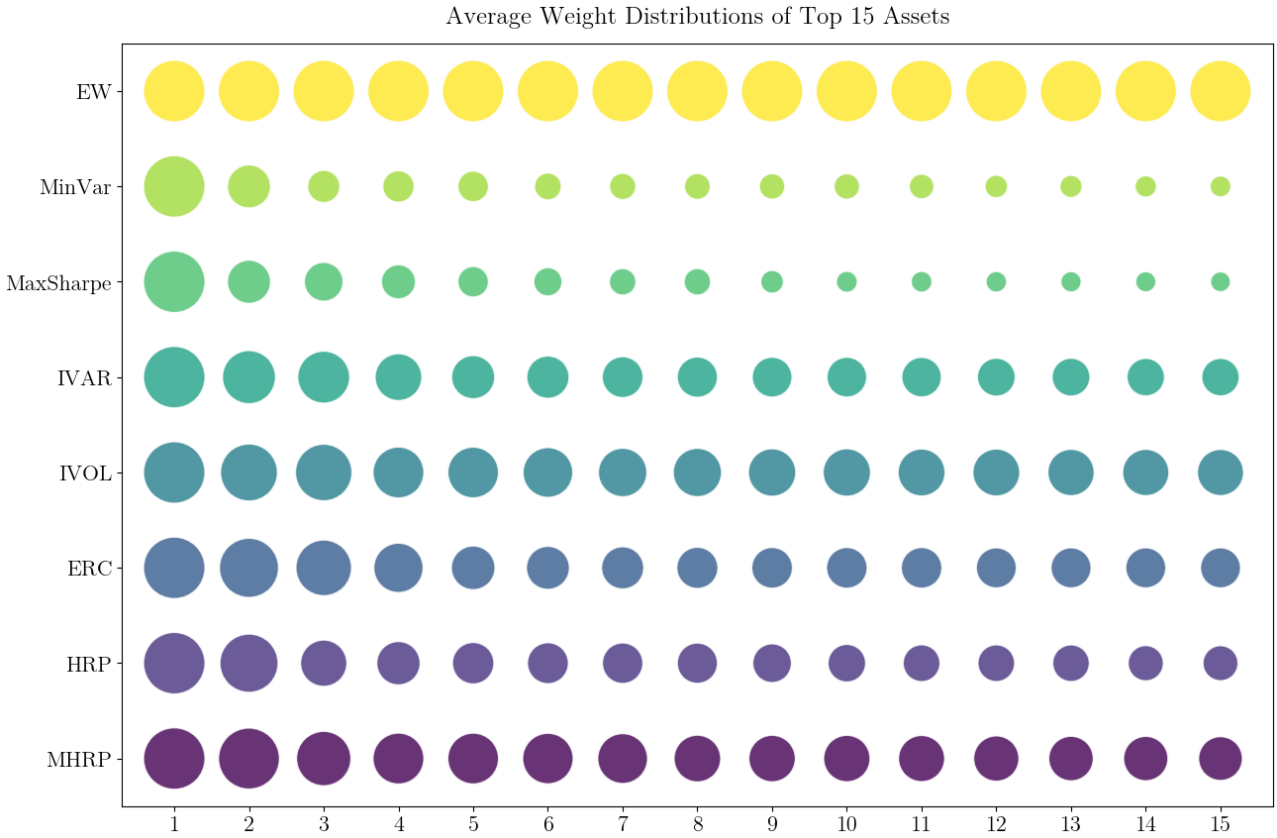

Figure 6.5: Average weight distribution of Top 15 assets for each portfolio

Furthermore, Table 6.1 depicts numerical minimum and maximum values of average asset weights, as well as SSPW values for every portfolio. The table highlights the most severe weight allocations and overall diversification across each approach. Examining the minimum values, the EW portfolio stands at 0.2%, indicating that a portfolio of 500 assets would result in a total weight of 100%. The minimum weight across many portfolios (MinVar, MaxSharpe, IVAR) is 0.00%, which is reasonable given that each rebalancing portfolio may contain up to 500 assets. The IVOL, MHRP, ERC, and HRP portfolios had the lowest average weights, at 0.01%, 0.02%, 0.05%, and 0.08%, respectively.

The MaxSharpe portfolio has the greatest average weight value of 23.86%, indicating its emphasis on a singular goal – maximizing the Sharpe ratio. Second, the MinVar portfolio is at 9.57%, emphasizing risk reduction through variance minimization. The remaining portfolios exhibit maximum weight values of less than 1%, indicating a larger diversification

strategy. HRP leads with 0.92%, followed by IVAR at 0.77%. MHRP displays a broader diversification with a maximum value of 0.69% compared to the original HRP approach, while ERC and IVOL yield 0.57% and 0.42%, respectively. The EW portfolio's minimum average weight value of 0.20% does not differ significantly from its maximum value.

Regarding SSPW, a higher value indicates lower diversification. Notably, the MaxSharpe portfolio stands out with an SSPW of 46.54%, emphasizing its pronounced focus on a limited number of stocks. Similarly, the MinVar portfolio exhibits an SSPW of 6.65%, suggesting a relatively high degree of concentration within its holdings. The SSPW values of the remaining portfolios are below 0.5%, highlighting their well-diversified nature. This reflects a more even distribution of investment weights across assets, in line with the objective of achieving broader diversification. SSPW results align with the conclusions derived from Figure 6.5.

Table 6.1: Minimum, Maximum and SSPW values for different portfolio construction strategies

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
Min	0.20%	0.00%	0.00%	0.01%	0.05%	0.08%	0.02%	0.01%
Max	0.22%	9.57%	23.86%	0.77%	0.42%	0.57%	0.92%	0.69%
SSPW	0.21%	6.65%	46.54%	0.28%	0.23%	0.24%	0.33%	0.31%

Table 6.2 illustrates the impact of different linkages on the minimum and maximum average weights, as well as SSPW values for both HRP and MHRP for all linkages. Notably, all portfolios consistently yield minimum value for weights very close to zero. HRP portfolios average at 0.02%, while MHRP shows lower minimum values at 0.01%. Regarding maximum weights, HRP reaches a peak of 0.96% using single linkage, while the single linkage approach for MHRP results in 0.84%. This trend suggests that HRP generally produces higher maximum and minimum weights. This disparity can be attributed to the methodology used by each approach, with HRP relying on variance and MHRP employing volatility, as highlighted by Molyboga (2020). Regarding SSPW, MHRP yields slightly higher diversification, potentially highlighting the impact of Ledoit-Wolf Shrinkage. All values fluctuate between 0.30% and 0.33%.

Table 6.2: Minimum, Maximum and SSPW values for HRP and MHRP with various linkages

	HRP				MHRP			
	Single	Ward	Average	Complete	Single	Ward	Average	Complete
Min	0.02%	0.02%	0.02%	0.02%	0.01%	0.01%	0.01%	0.01%
Max	0.96%	0.92%	0.92%	0.86%	0.84%	0.69%	0.70%	0.67%
SSPW	0.33%	0.33%	0.32%	0.32%	0.31%	0.31%	0.30%	0.31%

Table 6.3 illustrates the Average Turnover per Rebalancing (ATR) of all allocation strategies. It emphasizes the potential transaction costs associated with portfolio turnover during rebalancing periods. The EW portfolio demonstrates the lowest ATR of 0.2%. This outcome aligns with the portfolio's construction, as the weights of S&P 500 securities remain unchanged regardless of whether there are joiners or leavers. IVOL has an ATR of 1.5% while IVAR has an ATR of 2.9%, as IVAR operates with variances and considers higher-order factors, which amplifies the differences between portfolio weights. Therefore, a higher ATR for IVAR is expected. Furthermore, the ERC portfolio's low 2.7% ATR is attributed to its methodology. By adjusting weights to evenly distribute risk across assets, ERC ensures minimal changes in asset allocations, maintaining stability. Unlike IVOL, ERC accounts for both standard deviation and covariances, leading to a slightly higher ATR. Both the HRP and MHRP portfolios, with ATRs of 22.3% and 20.1%, modify asset allocations on a regular basis to account for changes in asset correlation structures. Table 6.3 depicts HRP and MHRP results using Ward's linkage. This methodology rebalances weights within clusters rather than across individual assets, resulting in an effective yet frequent risk management approach. As a result, their dynamic risk management approach produces greater ATRs than in previously mentioned portfolios. Next, the MinVar portfolio has the second-highest ATR of 27.9% of the portfolios examined. This shows that, despite trying to minimize total portfolio risk, the minimum-variance portfolio may need frequent adjustments to maintain its optimal allocation. Finally, the MaxSharpe portfolio considers both risk and return, making it the only portfolio that considers expected returns. This dynamic optimization approach yields the maximum ATR of 57.8%. The MaxSharpe portfolio is very concentrated because of its strict focus, potentially resulting in drastic changes in weights.

Table 6.3: Portfolios' average turnover per rebalancing

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
ATR	0.002	0.279	0.578	0.029	0.015	0.027	0.223	0.201

Finally, Table 6.4 shows that MHRP consistently has lower ATRs than HRP across all linkages, with HRP ATRs ranging from 0.200 to 0.225 and MHRP from 0.184 to 0.205. This, again, can be attributed to the impact of inverse-volatility allocation instead of inverse-variance.

Table 6.4: Portfolios' average turnover per rebalancing for HRP and MHRP with various linkages

	HRP				**MHRP**			
	Single	Ward	Average	Complete	Single	Ward	Average	Complete
ATR	0.200	0.223	0.217	0.225	0.184	0.201	0.196	0.205

6.3 Return Measures

Figure 6.6 illustrates the cumulative returns of 8 different portfolios from 1995 through the end of 2023. The cumulative returns are normalized, with each portfolio starting at the identical baseline for ease of comparison.

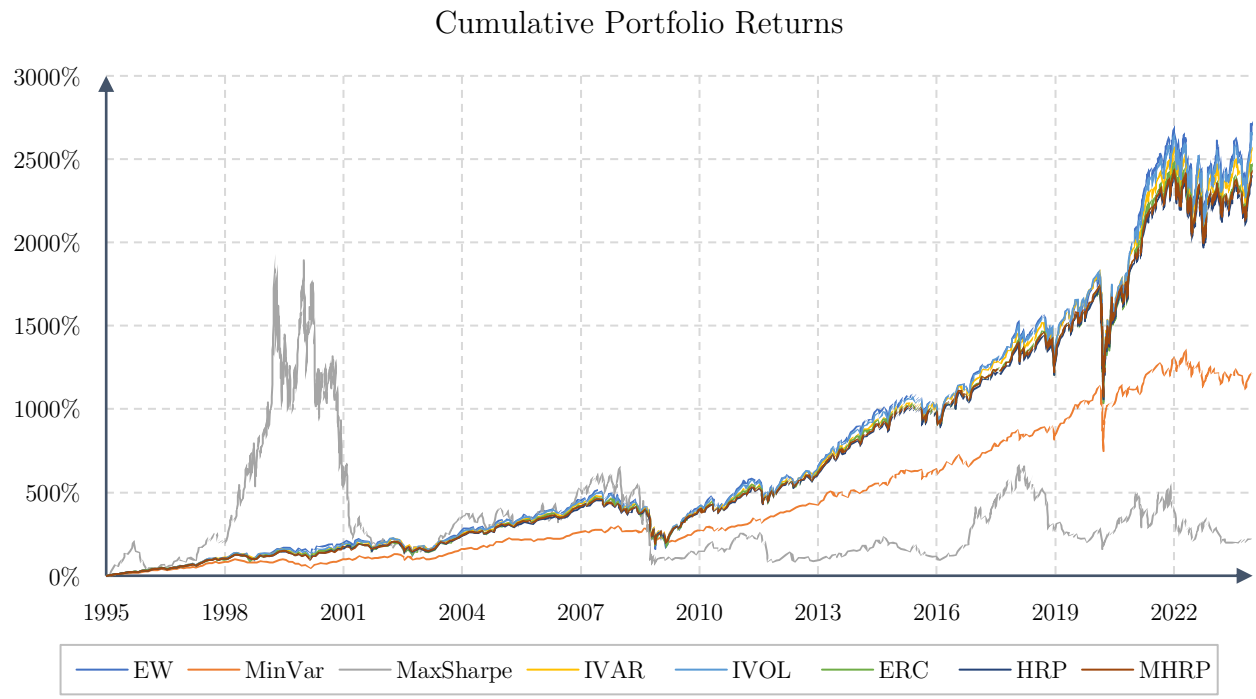

Figure 6.6: Cumulative portfolio returns for 1995 to 2023 period

The graph clearly shows the MaxSharpe portfolio as a significant outlier, with very strong cumulative returns between 1998 and 2000. Such results can possibly be attributed to well-

performing returns performing even better return and risk-wise. However, after the 2008 financial crisis, the MaxSharpe portfolio performance significantly lags compared to other portfolios. The MinVar portfolio also underperforms in comparison to the other strategies. The EW portfolio emerges as the top performer, with cumulative returns over a 29-year horizon reaching 2730%. These results align with the findings of DeMiguel et al.'s research (2009), where the authors underscore the simplicity and effectiveness of the 1/N portfolio. For the IVAR, IVOL, ERC, HRP, and MHRP portfolios, the results are comparatively consistent.

Table 6.5 depicts the comparative returns of all investment portfolios over distant time frames. Initially, the MaxSharpe portfolio yielded the highest returns over 5 and 10-year periods. Its performance, however, deteriorated after 15 years, delivering the lowest returns among the portfolios studied, while the EW portfolio emerged as the leader. This trend persisted through the 20-year mark. At the 25-year interval, the IVOL portfolio achieved the highest cumulative return at 1780%, narrowly surpassing the returns of the IVAR and EW portfolios. Over the full study period, the EW portfolio consistently outperformed its counterparts, culminating in the most substantial cumulative return of 2730%.

Table 6.5: Comparative analysis of cumulative portfolio returns across various time periods

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
5 years	166%	66%	1795%	139%	151%	138%	128%	127%
10 years	339%	212%	416%	320%	330%	308%	303%	308%
15 years	412%	271%	139%	398%	408%	388%	384%	388%
20 years	1046%	610%	201%	1006%	1033%	992%	971%	984%
25 years	1762%	1091%	252%	1767%	1780%	1683%	1667%	1688%
29 years	2730%	1224%	226%	2581%	2674%	2482%	2417%	2447%

Table 6.6 shows the mean, minimum and maximum values of cumulative monthly returns over the full study period. When examining average returns, the EW portfolio exhibited the highest mean monthly return at 1.09%, while the MinVar portfolio showed the lowest at 0.79%. In assessing the minimum monthly returns, the MaxSharpe portfolio recorded the most significant decrease of -50.32%, in contrast to the MinVar portfolio, which saw its value decline by -10.73% during its worst-performing month. On the other hand, looking into maximum monthly gains, the MaxSharpe portfolio rose by 64.14%, substantially surpassing the MinVar portfolio's increase of 10.45%. These outcomes are reflective of the strategies of each portfolio, with the MinVar approach centered on variance minimization and the MaxSharpe strategy aimed at optimizing the Sharpe ratio. Meanwhile, the remaining portfolios' results were

rather homogenous, with minimum and maximum values almost equal distance from 0%, all of them slightly negatively skewed.

Table 6.6: Cumulative monthly returns' characteristics over a 29-year period

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
Mean	1.09%	0.79%	1.05%	1.04%	1.06%	1.04%	1.01%	1.02%
Min	-19.82%	-10.73%	-50.32%	-17.65%	-18.91%	-18.78%	-17.28%	-17.40%
Max	18.30%	10.45%	64.14%	12.61%	14.67%	14.35%	12.91%	12.81%

Figure 6.7 displays the distribution of daily returns for a range of investment portfolios, each of which follows almost the normal distribution. However, the skewness of all these distributions is negative, suggesting a tendency for the returns to trend towards the left side. This implies a tendency for returns to be negative, which might imply a higher frequency of negative returns or larger negative outliers. Notably, the HRP and MHRP portfolios exhibit the most extreme negative skewness, each with a score of -0.27, showing a stronger bias toward negative returns. In contrast, the MaxSharpe portfolio has the lowest negative skewness, with a score of -0.07. The MaxSharpe portfolio's least negative skewness result is quite counterintuitive, considering that it had the worst-performing month, as seen in Table 6.6.

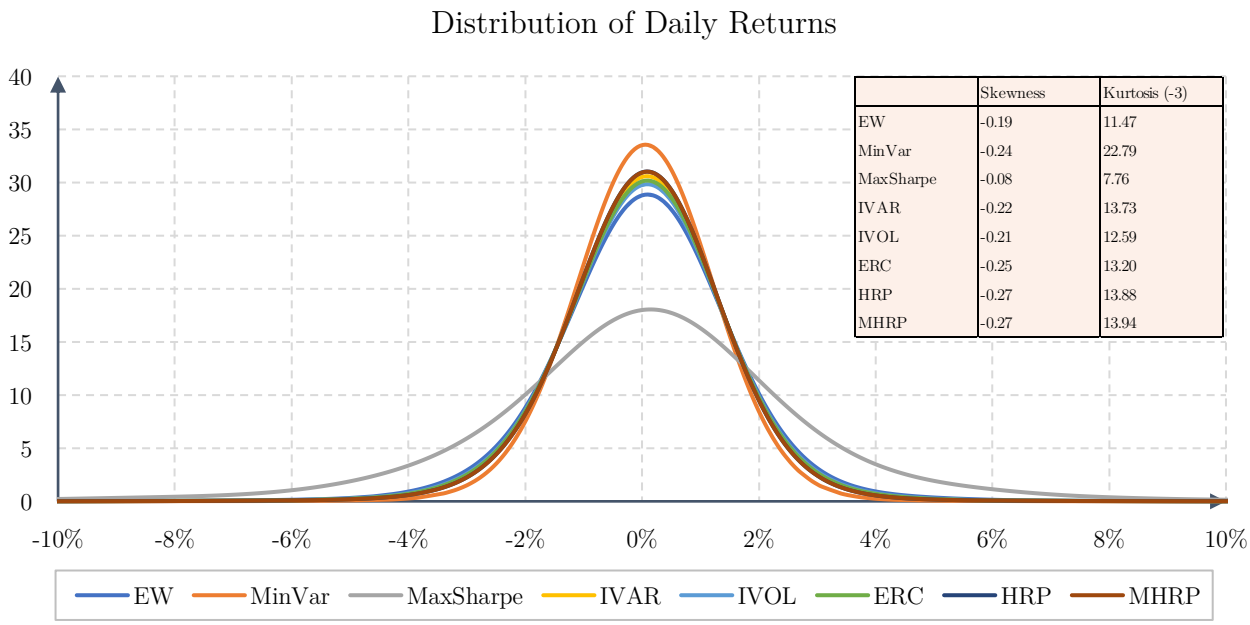

Figure 6.7: Gaussian Kernel Distribution of daily returns across multiple portfolio strategies

All portfolios have leptokurtic distributions; the MinVar portfolio has the greatest excess kurtosis at 22.79, indicating that the tails are substantially fatter than a normal distribution. Following closely are MHRP (13.94) and HRP (13.88), IVAR (13.73), ERC (13.20), IVOL (12.59), and EW (11.47). The MaxSharpe portfolio has the lowest excess kurtosis of 7.76, indicating a lower potential for extreme returns than the others. However, this is mainly because of the fatter tails, as the more uniform distribution reduces the frequency around the mean and pulls the distribution towards the normal distribution.

HRP and MHRP Across Linkages

Table 6.7 depicts that the MHRP portfolio with average linkage generates the highest cumulative returns at 2490%, whereas the HRP portfolio with single linkage has the lowest result at 2365%. This observable difference in performance highlights the impact of linkage selection on portfolio diversification and, subsequently, returns. Whether the magnitude of the 125% difference in extreme values is significant is influenced by factors such as investment objectives, risk tolerance, and the scale of capital investment. Investors' strategy, risk profile, and investment scale dictate whether a 125% difference between the extreme value is significant. The findings show a substantial homogeneity between the HRP and MHRP portfolios across all linkages and throughout all time frames.

Table 6.7: Cumulative returns for HRP and MHRP portfolios using various linkages

	HRP				**MHRP**			
	Single	Ward	Average	Complete	Single	Ward	Average	Complete
5 years	126%	128%	132%	130%	123%	127%	132%	129%
10 years	304%	303%	314%	310%	304%	308%	313%	304%
15 years	378%	384%	390%	387%	380%	388%	392%	381%
20 years	966%	971%	985%	986%	974%	984%	994%	979%
25 years	1664%	1667%	1709%	1691%	1671%	1688%	1714%	1682%
29 years	2365%	2417%	2484%	2377%	2371%	2447%	2490%	2374%

Table 6.8 presents the characteristics of the cumulative monthly returns for the HRP and MHRP portfolios over a 29-year period, categorized by different linkage methods. Uniformity among links is noticeable, with mean returns ranging from 1.01% to 1.02%. Both portfolios have modest variability in their minimum and maximum returns, with minimums ranging from -17.17% to -17.54% and maximums from 12.69% to 12.93%. This consistency shows that the choice of connection mechanism has little effect on the overall return characteristics of these portfolios.

Table 6.8: Cumulative HRP and MHRP monthly returns' characteristics over a 29-year period

	HRP				MHRP			
	Single	Ward	Average	Complete	Single	Ward	Average	Complete
Mean	1.01%	1.01%	1.02%	1.01%	1.01%	1.02%	1.02%	1.01%
Min	-17.44%	-17.28%	-17.39%	-17.17%	-17.54%	-17.40%	-17.38%	-17.43%
Max	12.73%	12.91%	12.92%	12.80%	12.69%	12.81%	12.85%	12.72%

Figure 6.8 shows consistent skewness and excess kurtosis across all links in the HRP and MHRP portfolios. Skewness scores are uniformly negative, ranging narrowly between -0.26 and -0.28, indicating a similar degree of leftward tilt in the return distributions. Excess kurtosis (13.88 to 14.09) shows the existence of heavy tails throughout these distributions.

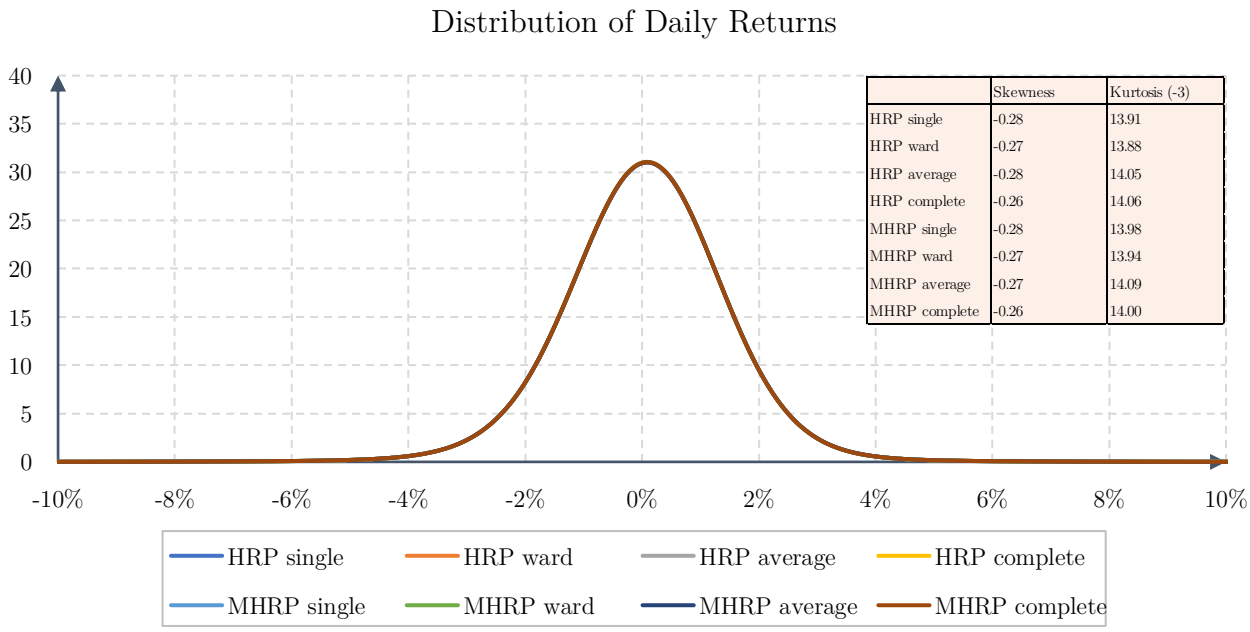

Figure 6.8: Gaussian Kernel Distribution of daily returns across HRP and MHRP portfolios

6.4 Risk Measures

To illustrate the volatility of daily returns, we use the standard deviation and annualize it for comparison purposes, as explained in the methodology. Figure 6.9 displays rolling annualized volatilities for each day through the years, each with a 252-day window. The MaxSharpe portfolio has the highest volatility throughout the whole period, as expected, due to the high concentration to achieve the maximum SR possible. Meanwhile, the MinVar portfolio shows

the lowest volatility aligned with its purpose of achieving the minimum variance. Among the other portfolios that seem to follow each other, the EW portfolio has slightly higher volatility than the others since it is constructed without any performance input and includes every asset possible; hence, it is expected to have higher variance.

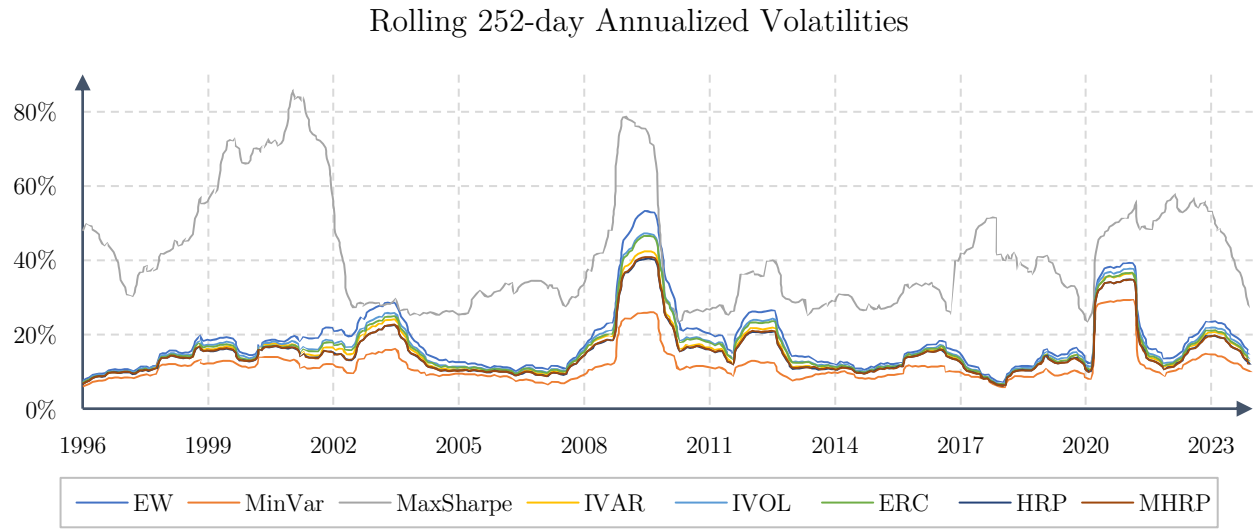

Figure 6.9: Rolling 252-day annualized volatilities for all portfolios

Table 6.9 illustrates the annualized volatility of each portfolio. MaxSharpe exhibits the highest volatility at 43.79%, while MinVar, designed to minimize variance, displays the lowest volatility at 12.43%. The HRP and MHRP portfolios, despite not prioritizing volatility minimization, rank second and third lowest in volatility, at 16.44% and 16.50%, respectively. IVAR follows at 17.19%, then ERC at 17.97%, IVOL at 18.48%, and EW at 20.16%. Across all linkages for HRP and MHRP portfolios, results vary between 16.46% and 16.56%.

Table 6.9: Annualized volatility of each portfolio throughout the study time

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
Ann. Volatility	20.16%	12.43%	43.79%	17.19%	18.48%	17.97%	16.44%	16.50%

To display the financial risk in a more extensive way, we have also included tail-risk measures. Table 6.10 shows three tail-risk metrics together with risk-adjusted return metrics to measure the return and risk together. Overall, Table 6.10 indicates that the MinVar portfolio has the highest Sharpe ratio. After adjusting for transaction costs, the portfolio outperformed only the EW and MaxSharpe portfolios, as these portfolios yield less robust results compared to others due to their aim not being purely risk minimization. With regards to the tail-risk measures,

the MinVar and MaxSharpe portfolios again show the lowest and highest risk, respectively, while the EW portfolio shows slightly higher risk compared to similar portfolios. Apart from the MinVar portfolio, HRP and MHRP portfolios give more robust results than the others.

Table 6.10: Risk measures

	EW	MinVar	MaxSharpe	IVAR	IVOL	ERC	HRP	MHRP
Sharpe Ratio	0.525	0.619	0.058	0.604	0.569	0.570	0.617	0.617
Adjusted SR	0.525	0.565	0.026	0.600	0.567	0.566	0.584	0.588
VaR (5%)	-1.84%	-1.09%	-4.24%	-1.58%	-1.69%	-1.63%	-1.51%	-1.51%
ES (5%)	-3.01%	-1.77%	-6.57%	-2.56%	-2.76%	-2.69%	-2.45%	-2.46%
Max Drawdown	70.47%	55.23%	96.90%	61.82%	65.84%	66.01%	60.77%	61.06%

Table 6.11 shows the same information across different linkages of HRP and MHRP. A closer look at the Sharpe ratio across all links in both the HRP and MHRP portfolios reveals that the HRP with average linkage has the greatest Sharpe Ratio of all the portfolios evaluated in this study. Overall, the HRP portfolio provides a higher Sharpe ratio but when transaction costs are considered, it is outperformed by MHRP portfolios.

Table 6.11: Risk measures for HRP and MHRP with various linkages

	HRP				**MHRP**			
	Single	Ward	Average	Complete	Single	Ward	Average	Complete
Sharpe Ratio	0.609	0.617	0.621	0.612	0.608	0.617	0.619	0.610
Adjusted SR	0.580	0.584	0.589	0.579	0.581	0.588	0.591	0.580
VaR (5%)	-1.51%	-1.51%	-1.52%	-1.51%	-1.52%	-1.51%	-1.52%	-1.51%
ES (5%)	-2.47%	-2.45%	-2.46%	-2.45%	-2.48%	-2.46%	-2.47%	-2.46%
Max Drawdown	61.46%	60.77%	60.69%	60.56%	61.87%	61.06%	61.19%	61.29%

6.5 Summary of the Results

This section outlines the key takeaways from the results, highlighting the weight characteristics in portfolios along with their associated return and risk metrics. Please note that, as explained in section 4.3, the data these results are based on is not flawless due to the data inconsistency caused by the data sources and the removal of highly inactive securities, which may result in slight survivorship bias.

Table 6.12 depicts the best performing portfolios by each measure. In every column, portfolios are sorted from the best to worst "performing", depending on the meaning of the metrics. For mean returns, Sharpe ratio, and adjusted Sharpe ratio, portfolios are sorted from highest to lowest values, while the opposite approach is taken for the risk and turnover metrics. Note that mean returns here demonstrate the mean values of monthly returns for each portfolio, which are the cumulative returns for every testing period, and volatility means the annualized volatility of the portfolios. The same table for the HRP and MHRP portfolios across linkages can be found in Appendix D.

Table 6.12: Portfolios ranked by key measures

Mean Returns	Sharpe Ratio	Adjusted SR	Volatility	VaR (5%)	ES (5%)	MDD	SSPW	ATR
EW	HRP	IVAR	MinVar	MinVar	MinVar	MinVar	EW	EW
IVOL	MHRP	MHRP	HRP	HRP	HRP	HRP	IVOL	IVOL
MaxSharpe	MinVar	HRP	MHRP	MHRP	MHRP	MHRP	ERC	ERC
IVAR	IVAR	IVOL	IVAR	IVAR	IVAR	IVAR	IVAR	IVAR
ERC	ERC	ERC	ERC	ERC	ERC	ERC	MHRP	MHRP
MHRP	IVOL	MinVar	IVOL	IVOL	IVOL	IVOL	HRP	HRP
HRP	EW	EW	EW	EW	EW	EW	MinVar	MinVar
MinVar	MaxSharpe	MaxSharpe	MaxSharpe	MaxSharpe	MaxSharpe	MaxSharpe	MaxSharpe	MaxSharpe

Weights

Regarding weights, MaxSharpe allocated the largest percentage to a single asset at 23.86%, followed by MinVar at 9.57%, while the largest weights for other portfolios remained below 1.00%. Similarly, MaxSharpe and MinVar exhibited the highest SSPWs, at 46.54% and 6.65%, respectively, proving once again the tendency of these two portfolios to diversify poorly. Across all linkages, MHRP consistently displayed lower ATR, SSPW, minimum, and maximum values than HRP, indicating slightly lower weight concentration and transaction costs.

Performance

In terms of returns, the EW portfolio achieved the highest cumulative returns at 2730%, while MaxSharpe and MinVar had the lowest two, at 226% and 1224%, respectively, with the remaining portfolios ranging from 2417% to 2674%. MaxSharpe also recorded the lowest minimum monthly return of -50.32% but also exhibited the largest monthly rise at 64.14%, indicating unreliable performance and high volatility in the long term. Excluding MinVar's lowest monthly return of -10.73%, the minimum monthly returns for the other portfolios varied from -19.82% to -17.28%. Excluding MaxSharpe, the maximum monthly increases for the remaining portfolios ranged from 10.45% to 18.30%. Across all linkages for HRP and MHRP, MHRP with average linkage generated the highest cumulative returns at 2490%, while HRP

with single linkage recorded the lowest at 2365%. All linkages maintained almost the same range of monthly average returns, highest monthly gains, and losses. All portfolios demonstrated negative skewness and leptokurtic distributions, with MaxSharpe having much lower skewness and excess kurtosis than the other portfolios.

The volatility of the MaxSharpe portfolio was once displayed by the consistently exhibited highest annualized volatility throughout the entire period and its highest VaR and ES, at 4.24% and -6.57%, respectively. In contrast, MinVar demonstrated the lowest potential losses, with a VaR of -1.09% and an ES of -1.77%. MaxSharpe also experienced the greatest maximum drawdown at 96.90%, while HRP and MHRP had the second lowest at 60.77% and 61.06%, respectively, following MinVar. Although MinVar had the highest Sharpe ratio at 0.619, closely followed by HRP and MHRP at 0.617, IVAR achieved the highest adjusted Sharpe ratio of 0.600 after adjusting for anticipated transaction costs. Using different linkages across HRP and MHRP portfolios negligibly altered the results.

Overall, in terms of returns, HRP and MHRP fall short of traditional and other risk-based portfolios with the EW portfolio outperforming others. However, the robust allocation strategy allows them to have the lowest risk figures following MinVar and hold the top positions based on risk-adjusted returns. Only the transaction costs affect HRP and MHRP more negatively than other portfolios, resulting in improved results for the IVAR and IVOL portfolios. This, in turn, causes HRP and MHRP lose their top positions slightly to IVAR.

7 Discussion

In this section, we discuss the rationale of including the MaxSharpe portfolio in our analysis, review our rolling forward analysis methodology, and assess the real-world efficacy of the HRP and MHRP approaches. Furthermore, we explain how we slightly adjusted for the transaction costs and conclude with further research possibilities that could sharpen the results.

7.1 MaxSharpe Portfolio and Its Limitations

The initial part of our discussion centers on the selection of portfolios for analysis. With this thesis primarily examining HRP and MHRP, we aimed to compare these with a broad array of portfolios to evaluate their performance. The decision was prompted by López De Prado's (2016) critique of covariance instability, making it particularly intriguing to compare HRP to Markowitz's mean-variance portfolios (1952, 1959). Our research focused on the MinVar and MaxSharpe portfolios among the various mean-variance portfolios, which also included the target return, target risk, leveraged, and inverse portfolios. We excluded target return and target risk portfolios from our analysis because their specific emphasis on returns or risks makes them less comparable to the larger range of portfolios we investigated. Furthermore, leveraged and inverse portfolios were excluded owing to limits that prevent short selling. Pedersen et al. (2021) highlighted multiple issues with the MaxSharpe portfolio, as briefly mentioned in subsection 3.2.1. The MaxSharpe portfolio relies primarily on historical returns to anticipate future performance, which may be unreliable due to changing market dynamics, resulting in estimation mistakes. Such dependence can also lead to overfitting since portfolios may favor assets that have recently performed well, jeopardizing diversity and stability. Furthermore, the MaxSharpe strategy's success is very susceptible to market movements, which might undermine its performance if market circumstances deviate greatly from the historical data used. These challenges highlight the hazards and difficulty of developing portfolio strategies based primarily on historical data. It is difficult to identify which of these factors has the most influence on the findings given in our study. However, as illustrated in Table 6.12, among the eight portfolios analyzed, the MaxSharpe portfolio was the least effective in numerous categories. Furthermore, while the MaxSharpe portfolio had the best cumulative returns during the first five and ten years, it underperformed severely throughout the duration of the study. These statistics highlight the tremendous volatility of the

MaxSharpe approach. Due to the high volatility and, therefore, fatter tails, the MaxSharpe portfolio also displayed relatively low skewness and kurtosis as the distribution more closely approximated the normal distribution.

7.2 Rolling Walk-Forward Analysis

Our analysis focuses solely on the S&P 500 index, which accounts for around 37% of global equity market capitalization as of the end of 2023 (Securities Industry and Financial Markets Association, 2024). While the S&P 500 index is rebalanced quarterly, we used a monthly rebalancing strategy for our study. This choice is based on the index's dynamic nature, where over a dozen firms can join or leave the S&P 500 index each year. We want to appropriately represent these frequent changes in index composition through monthly rebalancing. Although quarterly rebalancing could potentially reduce our roughly estimated transaction costs, we find that the monthly approach better aligns with the index's evolving composition due to the priority of this thesis being the comparison based on performance and robustness. In our analysis, we observed that the MHRP method with average linkage achieved the highest Sharpe ratio. Notably, both HRP and MHRP portfolios across all linkages performed well in terms of the Sharpe ratio, as evidenced in Table 6.10 and Table 6.11. Consequently, it is worth noting that had we opted for quarterly rebalancing, it is likely that one of the HRP or MHRP portfolios would have attained the highest adjusted Sharpe ratio.

Recognizing the approach used in the data cleaning stage, we strived to maintain a high level of integrity and impartiality throughout the process. Nevertheless, adopting alternative approaches to data cleaning could potentially yield divergent results.

Finally, the walk-forward test, which only allows for one backtest, limits the generalizability of the results. Despite using a rolling-window strategy, this methodology produces only one historical path and may not accurately predict future performance.

7.3 Real-World Efficacy of HRP and MHRP Strategies

Although theoretical data simulations can be useful for evaluating the HRP strategy, they do not fully reflect historical data realities. Despite López De Prado's (2016) study reporting a

Sharpe ratio increase of over 30% in out-of-sample data and Molyboga's (2020) modified HRP version demonstrating a 50% improvement, our analysis does not replicate these enhancements. Notably, Lopez de Prado's simulated data, adjusted by adding shocks of identical magnitude across all simulations, can yield unrealistic results. For example, due to homogenous shocks, the EW portfolio consistently captures and weighs these shocks equally in each simulation.

Additionally, it is noteworthy that Lopez de Prado's simulated portfolios only consisted of 10 assets. Therefore, it is anticipated that the instability of the covariance matrix will escalate with an increase in the number of assets, and HRP will likely outperform portfolios that rely on the inversion of the covariance matrix. It must be noted that our analysis confirms this part with HRP and MHRP portfolios yielding annualized volatilities only higher than the MinVar portfolio. Generally, uniform simulations provide results that differ from the behavior of actual financial data, which is naturally leptokurtic (Albuquerque, 2012) and has a larger range of extreme values. Real financial data complexities are challenging to model, even with advanced machine learning (Chen & Huang, 2008). Examining Molyboga's methodology (2020), the research primarily focuses on Commodity Trading Advisors (CTAs), utilizing data specifically sourced from CTAs that mostly manage derivatives. Given Molyboga's reported 50% increase in the Sharpe ratio, our study reveals no meaningful change from HRP to MHRP. The Sharpe ratio across all linkages remained steady at 0.61 to 0.62. Similarly, after employing the MHRP approach, the results persisted within the same range. These findings call into question the effectiveness of the MHRP technique when applied to S&P 500 data, underscoring the need for further research into the methodology's suitability across various datasets.

7.4 Transaction Costs

To account for transaction costs, our methodology uses a simple approach that estimates these costs based on the average turnover per rebalance, treating them as a fixed percentage of the portfolio's turnover (Lassance, 2021). However, real-world transaction costs can vary widely due to factors such as market liquidity, trading volumes, and brokerage fees, which may have a greater impact on portfolio performance than our model predicts. Portfolios like MinVar, MaxSharpe, HRP, and MHRP are rebalanced monthly but may still incur higher transaction costs. These strategies respond to changes in market conditions or asset volatility, sometimes

requiring significant adjustments to holdings. If these costs are underestimated, they could reduce expected returns. This underscores the importance of including realistic trading expenses when assessing the viability of these investment strategies. Moreover, both the HRP and MHRP strategies demand significant computational resources for managing complex calculations in hierarchical clustering and rebalancing. This could hinder practical use, especially for smaller investors who lack advanced computing access.

Nevertheless, our research has shown that no single strategy consistently outperforms across multiple metrics, and the selection should primarily be driven by the investment objective. From this perspective, the ideal portfolio to implement is contingent upon the investor's specific goals, including factors such as risk tolerance, the need for diversification, and transaction costs.

7.5 Further Research

The results of this thesis offer a significant platform for additional research into hierarchical risk parity (HRP) and the modified HRP (MHRP) strategies, especially in relation to how they perform in comparison to other portfolio strategies under different circumstances, considering the additional costs overlooked in this thesis.

A topic that is ready for more research is the comparative performance of these strategies against a large range of benchmarks, including an S&P 500 value-weighted portfolio constructed with the same components. With the addition of a value-weighted benchmark, more detailed knowledge of the performance of these hierarchical clustering-based methods relative to the traditional market-cap weighted indices would be possible. The lack of access to a complete dataset for market capitalizations did not allow us to benchmark against a truly representative S&P 500 value-weighted portfolio. Future research should aim to close this gap by gaining access to comprehensive market cap data, which would make it possible to gain a detailed comparative analysis and enhance our understanding of portfolio performance across different strategies.

An intriguing expansion of our research may be to investigate the integration of the Global Industry Classification Standard (GICS). GICS clustering might provide a distinct viewpoint on portfolio performance, especially when comparing sector-specific behaviors under methods

such as HRP and MHRP. This methodology may give insights into how these portfolios perform within certain economic sectors and may help mitigate some challenges identified with methods based purely on historical data. By categorizing assets based on industry characteristics, we could potentially reduce the influence of past performance biases and increase the robustness of portfolios to market changes. Investigating the application of GICS in portfolio construction might yield useful insights into the efficacy and flexibility of various portfolio techniques, making it a compelling subject for future research.

Furthermore, one important aspect that needs further investigation is the actual transaction costs of various portfolio strategies, including HRP and MHRP. Future studies could delve into a more extensive assessment of these costs, considering factors such as market liquidity, trading volume, and the frequency of trades. This would explain the net benefits of each strategy in a better way, considering the reduction in returns caused by transaction costs.

Moreover, including different asset classes and a larger pool of investment vehicles could provide insights into the construction strategies. This involves assessing the performance of these strategies in varying market conditions and across different market regimes, thereby testing their robustness and reliability in diverse financial situations.

Lastly, investigating how technology is being applied to portfolio management, specifically how blockchain and artificial intelligence are being implemented, could bring significant benefits. AI can save costs and improve portfolio performance by optimizing trade execution and managing large datasets more effectively.

By pursuing these inputs, future research can improve our understanding of hierarchical clustering in asset allocation and its practical implications, thereby helping investors make more informed decisions that align with their financial goals and market conditions.

8 Conclusion

This thesis seeks to enhance understanding and provide further evidence regarding the effectiveness of the HRP[1] method compared to traditional and heuristic allocation strategies, as outlined in section 1.1. Additional sub-questions were introduced to investigate the primary research question fully. These sub-questions assist in breaking down the fundamental research topic into more manageable, targeted areas of exploration. The concluding chapter begins with responses to these sub-questions, leading to the answering of the main research question.

How do the portfolios' performances vary across different optimization methods from both return and risk perspectives?

In terms of returns, traditional and other risk-based portfolio allocation strategies had slightly superior results in comparison to the HRP and MHRP portfolios, with the only exception of the MinVar portfolio. This is understandable since the pure target of the MinVar portfolio usually hinders it from achieving high returns. One noticeable fact is that although the MaxSharpe portfolio yielded one of the highest means of monthly returns, it had the lowest cumulative and annualized returns due to its fragile nature against the jumps and drops of the returns.

Regarding overall riskiness and the tail-risks of the portfolios, HRP and MHRP portfolios ranked second and third lowest, respectively, and managed to reach their robustness target by only falling slightly behind the MinVar portfolio, which was again expected. The same pattern shows itself in risk-adjusted returns, with HRP and MHRP outperforming the EW and the MinVar portfolios; however, when the transaction costs are considered, the HRP and MHRP portfolios lost their top positions, slipping to second place after IVAR. Importantly, our findings provide compelling evidence for investors who aim to reduce variance but prefer alternatives to the MinVar portfolio, implying that considering the HRP or MHRP portfolio may be worthwhile.

Generally, the IVAR, IVOL, and ERC portfolios provided similar results in terms of both returns and risk, with the EW portfolio yielding higher returns. However, since the EW portfolio is exposed to higher volatility, it provides lower risk-adjusted returns than the

[1] If not stated differently, the HRP and MHRP portfolios depict results using Ward's method.

traditional risk-based portfolios. The IVAR portfolio, on the other hand, utilizes the relatively lower risks and transaction costs and performs the best in terms of risk and cost-adjusted returns.

What degree of concentration is observed in the portfolio weights?

Except for the MinVar and MaxSharpe portfolios, all portfolios kept the average weights of the assets below 1% and provided low SSPW with similar values, which shows almost equally high diversification. The MaxSharpe and the MinVar portfolios, due to their own optimization objectives, provided significantly high maximum average weights and SSPW.

How do different linkage methods affect the results of the HRP and MHRP portfolio optimization?

In our analysis, we examined the effects of employing various linkage methods, specifically single, average, complete linkages, and Ward's method, on the performance of both the HRP and MHRP strategies. The results demonstrated negligible differentiation between the outcomes for all eight portfolios in terms of all metrics, namely return, risk, and risk-adjusted returns. However, MHRP consistently registers lower turnover and SSPW than HRP across all linkages. These differences are attributed to the distinct methodologies each strategy employs: HRP is based on variance, which enhances the weight differences, whereas MHRP utilizes standard deviation, as detailed by Molyboga (2020).

Our results do not confirm a significant 'chaining' effect prominent to single linkage, despite previous indications by Hastie et al. (2009) and (Raffinot, 2018). Furthermore, although Ward's method is preferred amongst many researchers (Raffinot, 2017), it failed to show superior performance across any of the evaluated metrics.

Research aim: How does the HRP method perform compared to traditional and heuristic allocation strategies?

The HRP method, in comparison to heuristic and traditional allocation algorithms, demonstrated a good performance profile, especially in risk management, by displaying lower tail-risks and overall riskiness. While HRP and MHRP are not the top performers in terms of simple returns, they do provide significant stability in risk-adjusted returns. This makes them an acceptable choice for investors who prioritize risk control over maximizing yield. The effectiveness of these strategies also appears in their diversification advantages, as they

maintain lower concentrations in portfolio weights. This, in turn, enhances portfolio stability in comparison to concentrated methods such as the MinVar and MaxSharpe portfolios. Interestingly, the analysis of various linkage methods shows negligible impact on their performance, indicating the robustness of the HRP approach irrespective of the linkage technique used. However, when the transaction costs are included, both HRP and MHRP perform slightly behind, indicating the necessity for optimization in real-life situations. In summary, the HRP method offers a compelling and competitive alternative to standard portfolio allocation strategies for investors who are looking for a conservative investment approach that combines return with risk mitigation.

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Appendix

A Dendrograms for Complete and Average Linkages

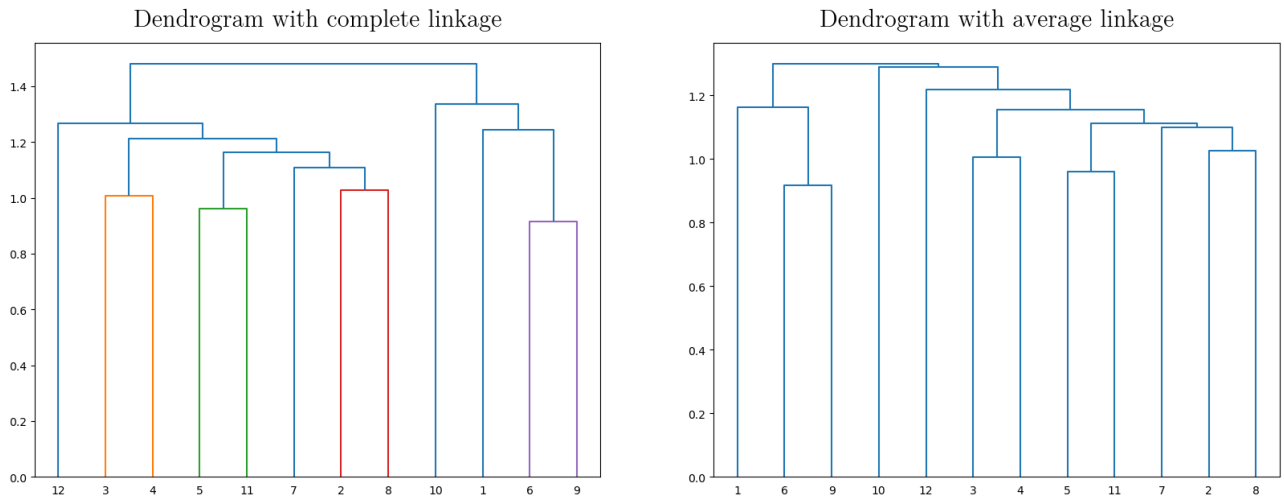

Figure A.1: Dendrogram example using complete (left) and average (right) linkages

B Algorithms for constructing HRP

Algorithm 1 Hierarchical Clustering using HRP

1. Estimate $N \times N$ correlation matrix $\boldsymbol{\rho} = \{\rho_{i,j}\}_{i,j=1,\dots,N}$
2. Convert the correlation matrix into a distance matrix $\mathbf{D} = \{d\}_{i,j=1,\dots,N}$ where $d_{i,j} = \sqrt{\frac{1}{2}(1 - \rho_{i,j})}$
3. Compute a Euclidean distance matrix $\widetilde{\mathbf{D}} = \{\tilde{d}_{i,j}\}_{i,j=1,\dots,N}$ where $\tilde{d}_{i,j} = \tilde{d}(X_i, X_j) = \sqrt{\sum_{n=1}^{N}(d_{n,i} - d_{n,j})^2}$
4. Let us denote the set of clusters as U:
   (a) Construct the first cluster (i^*, j^*) as $U[1] = \underset{i,j}{\operatorname{argmin}}\, \widetilde{D}(i,j)$
   (b) Using single linkage, update the distance matrix by calculating the pairwise distances of the newly formed cluster and other items
   (c) Continue combining assets recursively until we are left with only one single cluster
5. Summarize the information in linkage matrix $\mathbf{Y} \in \mathbb{R}^{(N-1)\times 4}$

Algorithm 2 Naïve Recursive Bisection

1. The algorithm is initialized by:
   (a) Setting the list of items: $L = \{L_0\}$, with $L_0 = \{n\}_{n=1,\dots,N}$
   (b) Assigning a unit weight to all items: $W_n = 1$, $\forall n = 1, \dots, N$
2. If $|L_i| = 1$, $\forall Li \in L$, then stop.
3. For each $L_i \in L$ such that $|L_i| > 1$:
   (a) Bisect L_i into two subsets, $L_i^{(1)} \cup L_i^{(2)} = L_i$, where $|L_i^{(1)}| = int[\frac{1}{2}|L_i|]$, and the order is preserved
   (b) Define the variance of $L_i^{(j)}, j = 1, 2$, as the quadratic form $Var\left(L_i^{(j)}\right) = w_i^{(j)^\top} \mathbf{\Sigma}_j^{\mathrm{S}} w_i^{(j)}$, where $\mathbf{\Sigma}_j^{\mathrm{S}}$ is the covariance matrix between the constituents of the subset $L_i^{(j)}$, and $w_i^{(j)} = \frac{diag\left[\mathbf{\Sigma}_j^S\right]^{-1}}{tr(diag\left[\mathbf{\Sigma}_j^S\right]^{-1})}$, where $diag[.]$ and $tr[.]$ are the diagonal and trace operators
   (c) Compute the split factor: $\alpha_i = 1 - \frac{Var(L_i^{(1)})}{Var\left(L_i^{(1)}\right) + Var(L_i^{(2)})}$, so that $0 \leq \alpha_i \leq 1$
   (d) Re-scale allocations W_n by a factor of $\alpha_i, \forall n \in L_i^{(1)}$
   (e) Re-scale allocations W_n by a factor of $(1 - \alpha_i), \forall n \in L_i^{(2)}$
4. Loop to step 2

C Excluded stocks in the data cleaning

First step	
Issue	**Reason to exclude**
Comcast Special Ord Class A	Merged into Comcast Ord Class A
Constellation Energy Group	Merged into Constellation Energy Ord
Mallinckrodt Pub Ord	Merged into Mallinckrodt Common
Manor Care Ord	Merged into Manor Care Inc Common
Seagate Technology Holdings Ord	Merged into Seagate Tech Common
Time Warner Inc Common	Merged into Time Warner Ord
Waste Management Ord	Merged into Waste Management Common
ITT Corp Ord	Merged into ITT Ord
Allergan Ord	IRL domicile removed, USA domicile kept
Alphabet Ord Class C	Less liquid than class A

AT&T Ord (issue by Corp)	AT and T Ord removed, AT&T Corp kept
Chubb Ord	CHE domicile removed, USA domicile kept
Envision Healthcare Holdings Ord	Shorter activity (1 day) in SP500 than the main issue
Fluor Ord	Shorter activity (11 days) in SP500 than the main issue
Discovery Ord Series C	Shorter activity in SP500 than series A
Fox Ord Class B	Less liquid and shorter activity in SP500 than class A
MCI Group Common	Shorter activity in SP500 than MCI
Morgan Stanley Common	Shorter activity in SP500 than Morgan Stanley Ord
News Ord Class B	Less liquid and shorter activity in SP500 than class A
Safety Kleen Ord	Shorter activity in SP500 than Safety Kleen Common
Santa Fe Pac Common	Shorter activity in SP500 than Sante Fe Pac Gold
21st Century Fox Ord Class B	Less liquid and shorter activity in SP500 than class A
Under Armour Ord Class C	Less liquid and shorter activity in SP500 than class A
Wells Fargo & Co Common	Shorter activity in SP500 than Wells Fargo Ord

Second Step	
Issue	**Reason to exclude**
Amcor Ord	Data not found
Brunos Common	Data not found
Calpine Ord	Data not found
Ceridian Corp Ord	Data not found
Conseco Inc Common	Data not found
Global Crossing Preferred	Data not found
Laidlaw Ord	Data not found
Silicon Graphics Ord	Data not found
Vivendi Exchangeco Series A Ord	Data not found
WR Grace Ord	Data not found
Delta Air Lines Ord	Data not found for one of S&P periods
Dun & Bradstreet Ord	Data not found for one of S&P periods
Sears Holdings Ord	Data not found for one of S&P periods

Third Step (all excluded stocks)	
Adient Ord	Kontoor Brands Ord
AdvanSix Ord	Kraft Heinz Ord
Airtouch Comm Common	Kyndryl Holdings Ord
Alcoa Ord	Linde Ord
Apartment Income REIT Ord	Lockheed Martin Ord
Baxalta Ord	Marriott International Ord Class A
Bioverativ Ord	MasterBrand Ord
Brighthouse Financial Ord	Mobil Corp Common
Cars.com Ord	Motorola Mobility Holdings Ord
ChampionX Ord	nVent Electric Ord
Columbia Pipeline Group Ord	Organon Ord
Comcast MO Ord	Orion Office REIT Ord
Conduent Ord	Perspecta Ord
Covetrus Ord	Pharmacia&Upjohn Ord
COXET 0.177 11/15/29 CVT CALd	Phinia Ord
Dell Technologies Ord Class C	Quality Care Properties Ord
Delphi Technologies Ord	Realogy Ord
DT Midstream Ord	Resideo Technologies Ord
DuPont de Nemours Ord	Sylvamo Ord
Embecta Ord	TechnipFMC Ord
Fort James Common	Travelers Property Ord
Fortrea Holdings Ord	Varex Imaging Ord
Four Corners Property Ord	Veralto Ord
Garrett Motion Ord	Versum Materials Ord
GE HealthCare Technologies Ord	Victoria s Secret Ord
Hartford Financial Services Group Ord	Vontier Ord
Ingevity Ord	WK Kellogg Ord
ITT Ord	Yum China Holdings Ord
JBG SMITH Properties Ord	ZimVie Ord
Kenvue Ord	

D Ranking of HRP and MHRP portfolios across all linkages

Table D.1: HRP and MHRP portfolios across linkages ranked by key measures

Mean Returns	Sharpe Ratio	Adjusted SR	Volatility	VaR (5%)	ES (5%)	MDD	SSPW	ATR
MHRP Average	HRP Average	MHRP Average	HRP Ward	HRP Ward	HRP Ward	HRP Ward	MHRP Average	MHRP Single
HRP Average	MHRP Average	HRP Average	HRP Complete	HRP Complete	HRP Complete	HRP Complete	MHRP Complete	MHRP Average
MHRP Ward	MHRP Ward	MHRP Ward	HRP Average	HRP Average	HRP Average	HRP Average	MHRP Ward	HRP Single
HRP Ward	HRP Ward	HRP Ward	MHRP Ward	MHRP Ward	MHRP Ward	MHRP Ward	MHRP Single	MHRP Ward
MHRP Single	HRP Complete	MHRP Single	MHRP Complete	MHRP Complete	MHRP Complete	MHRP Complete	HRP Average	MHRP Complete
HRP Complete	MHRP Complete	MHRP Complete	HRP Single	HRP Single	HRP Single	HRP Single	HRP Complete	HRP Average
MHRP Complete	HRP Single	HRP Single	MHRP Single	MHRP Single	MHRP Single	MHRP Single	HRP Ward	HRP Ward
HRP Single	MHRP Single	HRP Complete	MHRP Average	MHRP Average	MHRP Average	MHRP Average	HRP Single	HRP Complete